York Region
DISTRICT SCHOOL BOARD

COURSE PACK

Richmond Green S.S.

Functions
Grade 11

MCR 3U1

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We
unite in our purpose to inspire and prepare learners for life in our changing world community

BIOGAS ENERGY

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Polynomial Review

Expressions like x + 2y – 4z +7 are called polynomials. The terms of a polynomial are separated by the _____ or _____ signs. If there is only one term, eg. $2k$, it is specifically called a _____; two terms, eg. 2x + 3w, is specifically called a _____; three terms, eg. 2x + 3y-5z, is called a _____.

Adding and Subtracting Polynomials

Recall that terms such as 8x and 5x, which have the same variable factors, are known as *like terms*. To simplify an expression containing *like terms,* add or subtract their coefficients.

Examples of like terms:

Examples of unlike terms:

1. Simplify the following:

a) $(8x^2 - 7x + 6) + (2x^2 - 5x - 4)$

b) $(6s^2 + 3st - 2t^2) - (4s^2 - 5st + 9t^2)$

c) $(2xy^2 + 3x^2y^2 - 4x^2y) - (5yx^2 + 6y^2x^2 - 7y^2x)$

2. Expand then simplify.

a) $7x(3x^2 - 4x + 1)$

b) $5y(3y + 2) - 6y(8y - 9)$

c) $(2a - 7)(5a - 4)$

d) $(3x - 10)^2$

e) $(x + 4)(x^2 - 3x + 6)$

f) $2(x - 4y) - (x - y)^2 + (x + 2)(x - 7)$

Multiplying Polynomials & Special Products

1. Expand and simplify:

a) $(3x + 5)(2 – 10x)$

b) $(2x + 7)(2x – 7)$

c) $(5x – 1)^2$

d) $(x + 3)(x – 1) + (x + 2)^2$

e) $6x(2x + 9) – (4x + 3)(4x – 3)$

Expanding Practice

Answer the questions on a separate piece of paper. Be sure to show all your work.
Reminder: follow BEDMAS!

Simplify.

a) $(2x^4 - 3x^2 - 6) + (6x^4 - x^3 + 4x^2 + 5)$

b) $(x^2 - 4)(2x^2 + 5x - 2)$

c) $-7x(x^2 + x - 1) - 3x(2x^2 - 5x + 6)$

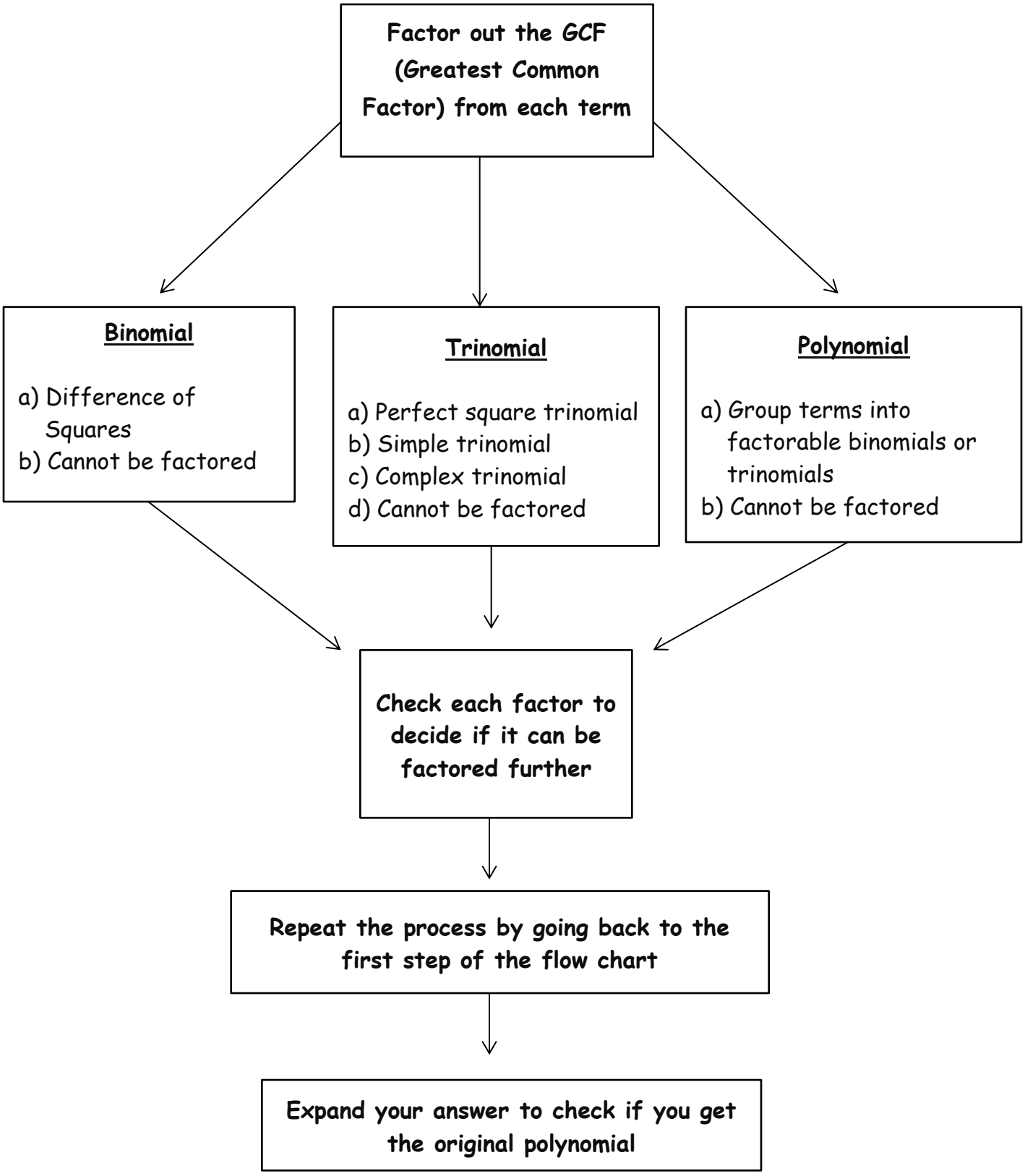
d) $-2x^2(3x^3 - 7x + 2) - x^3(5x^3 + 2x - 8)$

e) $-2x[5x - (2x - 7)] + 6x[3x - (1 + 2x)]$

f) $(x + 2)^2(x - 1)^2 - (x - 4)^2(x + 4)^2$

g) $(x^2 + 5x - 3)^2$

Factoring Polynomials Flowchart



Factoring Practice

Common Factoring

Remember, the greatest common factor will have the greatest common factor of each coefficient, and the lowest power of the common variable!

1) $35x^3y^7 - 14xy^5 + 42x^2y^3$

2) $5x(x+2) + 3(x+2)$

3) $(x+3)^2 - 5(x+3)$

4) $3a^2c^{10}d^3 + \frac{1}{2}a^5bc^8$ Do not leave any fractions in the brackets!

Trinomial Factoring

Simple trinomials of the form $x^2 + bx + c$ can be factored by determining two numbers that _____ to c and _____ to b.

Complex trinomials of the form $ax^2 + bx + c$ can be factored using cross-multiplication or decomposition.

Perfect square trinomials take the form $a^2 + 2ab + b^2$ or $a^2 - 2ab + b^2$.

5) $x^2 + 2x - 15$

6) $8y^2 - 30y + 27$

7) $4x^2 - 20xy + 25y^2$ This is a perfect square trinomial…why?

8) $9(7x - 2)^4 - 42(7x - 2)^2 + 49$

Difference of Squares Factoring

Difference of squares factoring requires determining the square root of the first term and the last term. They take the form $a^2 - b^2$.

9) $25x^2 - 64$

10) $(m + 2)^2 - 9$

11) $625x^4 - 81y^8$

Remember to factor fully!

Factoring by Grouping

Polynomials may be grouped into smaller binomials or trinomials that are factorable.

12) $2y^2 - 6y + 4my - 12m$

13) $3y^5 - 12y^3 - y^2 + 4$

14) $w^2 - 6w + 9 - y^2 + 2xy - x^2$

Factoring Review Sheet

Factor fully:

1) $4x + 6x^2$

2) $5x^3 + 25x^2 + 30x$

3) $12ax + 3ay + 15az$

4) $x^2 + 3x + 2$

5) $y^2 + y - 30$

6) $a^2 - 17a + 60$

7) $2x^2 - 6x + 4$

8) $3k^2 + 15k + 18$

9) $5x^2 + 5x - 30$

10) $3x^2 - 7x + 4$

11) $2x^2 + 9x + 10$

12) $2b^2 - b - 15$

13) $4c^2 - 8c - 21$

14) $4x^2 - 17x + 15$

15) $5w^2 + 11w + 2$

16) $36x^2 - 48x - 20$

17) $48x^2 - 200x + 200$

18) $24n^3 + 68n^2 + 48n$

19) $am - an + bm - bn$

20) $10ax + 4ay - 15x - 6y$

21) $9am + 3bm + 6an + 2bn$

22) $x^3 + x^2 + x + 1$

23) $1 + ab + a + b$

24) $(x+y)^2 - x - y$

25) $9x^2 - 30xy + 25y^2$

26) $16a^2 + 24ab + 9b^2$

27) $50a^2 - 120ab + 72b^2$

28) $x^2 - y^2$

29) $49x^2 - 64y^2$

30) $18a^3 - 50ab^2$

31) $(2x - 1)^2 - y^2$

32) $9k^2 - (5m - 1)^2$

33) $4a^2 - 9 - 2ab + 3b$

34) $(5x)^2 + 7(5x) + 12$

35) $(x^2 + 3x)^2 - 2(x^2 + 3x) - 8$

36) $6(a+b)^2 + 17(a+b) + 5$

37) $m^2 - 9 + 6n - n^2$

38) $49 - (2y - w)^2$

39) $(x+y)^2 - (a+b)^2$

40) $a^2 + 2ab + b^2 - c^2 + 6c - 9$

41) $25y^2 - 30y + 9 - 4c^2 - 4cd - d^2$

MCR 3U

FACTORING REVIEW

SOLUTIONS

1. $2x(2+3x)$
2. $5x(x+3)(x+2)$
3. $3a(4x+y+5z)$
4. $(x+2)(x+1)$
5. $(y+6)(y-5)$
6. $(a-12)(a-5)$
7. $2(x-2)(x-1)$
8. $3(k+3)(k+2)$
9. $5(x+3)(x-2)$
10. $(3x-4)(x-1)$
11. $(2x+5)(x+2)$
12. $(2b+5)(b-3)$
13. $(2c+3)(2c-7)$
14. $(4x-5)(x-3)$
15. $(5w+1)(w+2)$
16. $4(3x+1)(3x-5)$
17. $8(3x-5)(2x-5)$
18. $4n(3n+4)(2n+3)$
19. $(a+b)(m-n)$
20. $(5x+2y)(2a-3)$
21. $(3a+b)(3m+2n)$
22. $(x+1)(x^2+1)$
23. $(a+1)(b+1)$
24. $(x+y)(x+y-1)$
25. $(3x-5y)^2$
26. $(4a+3b)^2$
27. $2(5a-6b)^2$
28. $(x-y)(x+y)$
29. $(7x-8y)(7x+8y)$
30. $2a(3a-5b)(3a+5b)$
31. $(2x-1-y)(2x-1+y)$
32. $(3k-5m+1)(3k+5m-1)$
33. $(2a-3)(2a+3-b)$
34. $(5x+4)(5x+3)$
35. $(x+4)(x-1)(x+2)(x+1)$
36. $(3a+3b+1)(2a+2b+5)$
37. $(m-3+n)(m+3-n)$
38. $(7-2y+w)(7+2y-w)$
40. $(a+b-c+3)(a+b+c-3)$
39. $(x+y-a-b)(x+y+a+b)$
41. $(5y-3-2c-d)(5y-3+2c+d)$

Factoring Practice

I. Greatest Common Factor (GCF)

Find the GCF of the numbers.

$$18, 30$$
$$18 = 2 \cdot 3 \cdot 3$$
$$30 = 2 \cdot 3 \cdot 5$$
$$2 \cdot 3 = 6$$
$$6 = GCF$$

1. 12, 18
2. 10, 35
3. 8, 30
4. 16, 24
5. 28, 49
6. 27, 63
7. 30, 45
8. 48, 72

II. Greatest Common Monomial Factor

Factor, write prime if prime.

$$12a^3b + 15ab^3 = 3ab(4a^2 + 5b^2)$$

1. $6x + 3$
2. $24x^2 - 8x$
3. $6x - 12$
4. $2x^2 + 8x$
5. $4x + 10$
6. $10x^2 + 35x$
7. $10x^2y - 15xy^2$
8. $12x^2 - 9x + 15$
9. $3n^3 - 12n^2 - 30n$
10. $9m^2 - 4n + 12$
11. $2x^3 - 3x^2 + 5x$
12. $13m + 26m^2 - 39m^3$
13. $17x^2 + 34x + 51$
14. $18m^2n^4 - 12m^2n^3 + 24m^2n^2$

III. Factoring the Difference of Two Squares

$$a^2 - 36 = (a + 6)(a - 6)$$
$$3x^2 - 48 = 3(x^2 - 16) = 3(x + 4)(x - 4)$$

Factor, write prime if prime.

1. $x^2 - 1$
2. $x^2 - 9$
3. $x^2 + 4$
4. $x^2 - 25$
5. $9y^2 - 16$
6. $4x^2 - 25$
7. $9x^2 - 1$
8. $a^2 - x^2$
9. $25 - m^2$
10. $x^2 - 16y^2$
11. $25m^2 - n^2$
12. $-x^2 + 16$
13. $36m^2 - 121$
14. $2x^2 - 8$
15. $25 + 4x^2$
16. $4a^2 - 81b^2$
17. $12x^2 - 75$
18. $a^2b - b^3$
19. $-98 + 2x^2$
20. $5x^2 - 45y^2$
21. $9x^4 - 4$
22. $16x^4 - y^2$

IV. Factoring Perfect Square Trinomials

$$\mathbf{x^2 - 14x + 49 = (x - 7)^2}$$

Factor, write prime if prime.

1. $x^2 + 8x + 16$
2. $x^2 - 16x + 64$
3. $y^2 + 12y + 36$
4. $a^2 - 10a + 25$
5. $16y^2 + 8y + 1$
11. $25a^2 + 60a + 36$
12. $16 + 40x + 25x^2$
13. $16x^2 + 24x + 9$
14. $49x^2 - 14x + 1$
15. $9y^2 - 30y + 25$
6. $9x^2 - 6x + 1$
7. $25x^2 + 10x + 1$
8. $n^2 - 14n + 49$
9. $81x^2 - 90x + 25$
10. $4y^2 - 20y + 25$
16. $n^2 + 2n + 4$
17. $b^2 + 2b + 1$
18. $36x^2 + 84x + 49$
19. $81 - 18x + x^2$
20. $4 - 12y + 9y^2$

V. Special Factoring – Challenge

Factor, write prime if prime.

1. $a^2 - 36$
2. $9x^2 - 49$
3. $169m^2 - 4u^2$
4. $x^2y^2 - 9z^4$
5. $\frac{1}{4}x^2 - 25y^2$
6. $\frac{1}{9}x^2 - 16$
7. $64 - a^4b^4$
8. $y^6 - 100$
9. $\frac{4}{9}x^2y^2 - \frac{25}{36}z^2$
10. $y^8 - 81$
11. $1 - 8u + 16u^2$
12. $a^2b^2 + 6ab + 9$
13. $x^2 + 2xy + y^2$
14. $4x^2 + 12xy + 9y^2$
15. $100h^2 + 20h + 1$
16. $9a^2 - 24a + 16$
17. $4a^3 + 8a^2 + 4a$
18. $5c + 20c^2 + 20c^3$
19. $(x + 4)^2 - (y + 1)^2$
20. $(x - 1)^2 - 10(x - 1) + 25$

VI. Factoring Trinomials: $x^2 + bx + c$

$$\mathbf{x^2 + 7x + 10 = (x)^2 + (2 + 5)x + (2)(5) = (x + 2)(x + 5)}$$

Factor, write prime if prime.

1. $x^2 + 6x + 8$
2. $c^2 + 5c + 6$
3. $y^2 - 9y + 14$
4. $x^2 - 10x + 16$
5. $a^2 + 12a + 27$
6. $x^2 - 14x + 24$
7. $x^2 - 15x + 36$
8. $y^2 + 21y + 54$
9. $m^2 + 13m - 36$
10. $x^2 - 8x + 15$
11. $y^2 - 4y - 32$
12. $x^2 - x - 6$
13. $y^2 + 3y - 18$
14. $b^2 + 7b - 18$
15. $a^2 + a - 56$
16. $c^2 - 4c - 12$
17. $x^2 - 9x - 36$
18. $y^2 + 4y - 21$
19. $x^2 - 22x - 75$
20. $x^2 - 3x - 40$
21. $45 + 14y + y^2$
22. $x^2 - 13x + 36$

VII. …More Factoring Trinomials: $x^2 + bx + c$

$$\mathbf{k^2 - k - 20 = (k)^2 + (4 + -5)k + (4)(-5) = (k + 4)(k - 5)}$$

Factor, write prime if prime.

1. $x^2 + 7x + 12$
2. $m^2 + 10m + 21$
3. $y^2 - 7y - 8$
4. $x^2 - 6x + 5$
5. $x^2 + 4x - 32$
6. $x^2 - 2x - 15$
7. $x^2 - 6x + 8$
8. $y^2 + 9y + 18$
9. $3 - 4t + t^2$
10. $v^2 + 12v + 20$
11. $51 - 20k + k^2$
12. $a^2 - 14ab + 24b^2$
13. $y^2 + 6y - 72$
14. $x^2 - 11xy - 60y^2$
15. $15r^2 + 2rs - s^2$
16. $3x^2 + 21xy - 54y^2$ (Hint: Check for GCF)
17. $x^2 - 5xy - 6y^2$
18. $x^2 + 8xy + 12y^2$
19. $y^2 - 7xy + 10x^2$
20. $a^2 - 11ab - 60b^2$

VIII. Factoring Trinomials: $ax^2 + bx + c$

$$\mathbf{2x^2 - 5x - 3 = (2x + 1)(x - 3)}$$

Factor, write prime if prime.

1. $2x^2 - 5x - 3$
2. $3x^2 + 10x - 8$
3. $2y^2 + 15y + 7$
4. $7a^2 - 11a + 4$
5. $5n^2 + 17n + 6$
6. $4y^2 + 8y + 3$
7. $3x^2 + 4x - 7$
8. $2x^2 + 13x + 15$
9. $9y^2 + 6y - 8$
10. $6x^2 - 7x - 20$
11. $2n^2 - 3n - 14$
12. $5n^2 + 2n + 7$
13. $10x^2 + 13x - 30$
14. $12y^2 + 7y + 1$
15. $2n^2 + 9n - 5$
16. $2x^2 + 7x + 6$
17. $5a^2 - 42a - 27$
18. $15x^2 - 28x - 32$
19. $8a^2 - 10a + 3$
20. $2y^2 - 3y - 20$

IX. …More Factoring Trinomials: $ax^2 + bx + c$

Factor, write prime if prime.

1. $3x^2 + 4x + x$
2. $5z^2 + 7z + 2$
3. $2n^2 - 11n + 5$
4. $3z^2 + z - 2$
5. $5h^2 - 2h - 7$
6. $8s^2 - 10st + 3t^2$
7. $6x^2 + 19x + 15$
8. $28a^2 + 5ab - 12b^2$
9. $2a^2 + 7ab - 15b^2$
10. $12x^2 + 17x + 6$
11. $4a^2 - 4ab - 5b^2$
12. $56y^2 + 15y - 56$
13. $12x^2 - 29xy + 14y^2$
14. $64x^2 + 32xy - 21y^2$
15. $16x^2 + 56xy + 49y^2$
16. $18x^2 - 57x + 35$

X. Factoring: Putting It All Together

$$\mathbf{5x^2 + 20x - 60 = 5\left(x^2 + 4x - 12\right) = 5(x + 6)(x - 2)}$$

Factor Completely, write prime if prime.

1. $2x^2 - 8$
2. $2x^2 + 8x + 6$
3. $3n^2 + 9n - 30$
4. $6x^2 - 26x - 20$
5. $2x^2 + 12x - 80$
6. $5t^2 + 15t + 10$
7. $8n^2 - 18$
8. $14x^2 + 7x - 21$
9. $4x^2 + 16x + 16$
10. $18x + 12x^2 + 2x^3$
11. $2x - 2xy^2$
12. $3t^3 - 27t$
13. $24a^2 - 30a + 9$
14. $10x^2 + 15x - 10$
15. $3x^2 - 42x + 147$
16. $4x^4 - 4x^2$

XI. …More Factoring: Putting It All Together

1. $16x^2 - 40x - 24$
2. $27x^2 - 36x + 12$
3. $5x^2 - 60x - 140$
4. $6m^3 + 54m^2 - 6m$
5. $5k^4 + 8k^3 - 4k^2$
6. $x^2y^4 - x^6$
7. $y^4 - 6y^2 - 16$
8. $x^4 - 3x^2 - 4$
9. $h^2 - (a^2 - 6a + 9)$
10. $81x^4 - 16y^4$
11. $4mn^2 - 4m^2n^2 + m^3n^2$
12. $(2a + 3)^2 - (a - 1)^2$
13. $16d^8 - 8d^4 + 1$
14. $x^2(x^2 - 4) + 4x(x^2 - 4) + 4(x^2 - 4)$

XII. Extra: Factoring by Grouping

$$\begin{aligned}\mathbf{6ax - 2b - 3a + 4bx} &= \mathbf{6ax - 3a + 4bx - 2b} \\ &= \mathbf{3a(2x - 1) + 2b(2x - 1)} \\ &= \mathbf{(2x - 1)(3a + 2b)}\end{aligned}$$

1. $x^2 + 2x + xy + 2y$
2. $3a^2 - 2b - 6a + ab$
3. $t^3 - t^2 + t - 1$ Hint: $t - 1 = 1(t - 1)$
4. $10 + 2t - 5s - st$
5. $\frac{2}{3}bc - \frac{14}{3}b + c - 7$
6. $4u^2 + v + 2uv + 2u$
7. $ad + 3a - d^2 - 3d$
8. $n^2 + 2n + 3mn + 6m$
9. $2ax^2 + bx^2 - 2ay^2 - by^2$
10. $yz^2 - y^3 + z^3 - y^2z$
11. $y^3 - y^2 - 4y + 4$
12. $x^2a + x^2b - 16a - 16b$
13. $x^3 + x^2 - x - 1$
14. $a^3 - a^2 - 8a + 8$

Factoring Practice Key

I. Greatest Common Factor

1. 6
2. 5
3. 2
4. 8
5. 7
6. 9
7. 15
8. 24

II. Greatest Common Monomial Factor

1. $3(2x+1)$
2. $8x(3x-1)$
3. $6(x-2)$
4. $2x(x+4)$
5. $2(2x+5)$
6. $5x(2x+7)$
7. $5xy(2x-3y)$
8. $3(4x^2-3x+5)$
9. $3n(n^2-4n-10)$
10. prime
11. $x(2x^2-3x+5)$
12. $13m(1+2m-3m^2)$
13. $17(x^2+2x+3)$
14. $6m^2n^2(3n^2-2n+4)$

III. Factoring the Difference of Two Squares

1. $(x+1)(x-1)$
2. $(x+3)(x-3)$
3. prime
4. $(x+5)(x-5)$
5. $(3y+4)(3y-4)$
6. $(2x+5)(2x-5)$
7. $(3x+1)(3x-1)$
8. $(a+x)(a-x)$
9. $(5+m)(5-m)$
10. $(x+4y)(x-4y)$
11. $(5m+n)(5m-n)$
12. $(4+x)(4-x)$
13. $(6m+11)(6m-11)$
14. $2(x+2)(x-2)$
15. prime
16. $(2a+9b)(2a-9b)$
17. $3(2x+5)(2x-5)$
18. $b(a+b)(a-b)$
19. $-2(7+x)(7-x)$ *or* $2(x+7)(x-7)$
20. $5(x+3y)(x-3y)$
21. $(3x^2+2)(3x^2-2)$
22. $(4x^2+y)(4x^2-y)$

IV. Factoring Perfect Square Trinomials

1. $(x+4)^2$
2. $(x-8)^2$
3. $(y+6)^2$
4. $(a-5)^2$
5. $(4y+1)^2$
6. $(3x-1)^2$
7. $(5x+1)^2$
8. $(n-7)^2$
9. $(9x-5)^2$
10. $(2y-5)^2$
11. $(5a+6)^2$
12. $(4+5x)^2$
13. $(4x+3)^2$
14. $(7x-1)^2$
15. $(3y-5)^2$
16. prime
17. $(b+1)^2$
18. $(6x+7)^2$
19. $(x-9)^2$
20. $(3y-2)^2$

V. Special Factoring – Challenge

1. $(a+6)(a-6)$
2. $(3x+7)(3x-7)$
3. $(13m+2u)(13m-2u)$
4. $(xy+3z^2)(xy-3z^2)$
5. $\left(\frac{1}{2}x+5y\right)\left(\frac{1}{2}x-5y\right)$
6. $\left(\frac{1}{3}x+4\right)\left(\frac{1}{3}x-4\right)$
7. $(8+a^2b^2)(8-a^2b^2)$
8. $(y^3+10)(y^3-10)$
9. $\left(\frac{2}{3}xy+\frac{5}{6}z\right)\left(\frac{2}{3}xy-\frac{5}{6}z\right)$
10. $(y^4+9)(y^2+3)(y^2-3)$
11. $(1-4u)^2$ *or* $(4u-1)^2$
12. $(ab+3)^2$
13. $(x+y)^2$
14. $(2x+3y)^2$
15. $(10h+1)^2$
16. $(3a-4)^2$
17. $4a(a+1)^2$
18. $5c(2c+1)^2$
19. $[(x+4)+(y+1)][(x+4)-(y+1)]$ *or* $(x+y+5)(x-y+3)$
20. $[(x-1)-5]^2$ *or* $(x-6)^2$

VI. Factoring Trinomials: $x^2 + bx + c$

1. $(x+4)(x+2)$
2. $(c+2)(c+3)$
3. $(y-7)(y-2)$
4. $(x-8)(x-2)$
5. $(a+9)(a+3)$
6. $(x-12)(x-2)$
7. $(x-12)(x-3)$
8. $(y+18)(y+3)$
9. prime
10. $(x-5)(x-3)$
11. $(y-8)(y+4)$
12. $(x-3)(x+2)$
13. $(y+6)(y-3)$
14. $(b+9)(b-2)$
15. $(a+8)(a-7)$
16. $(c-6)(c+2)$
17. $(x-12)(x+3)$
18. $(y+7)(y-3)$
19. $(x-25)(x+3)$
20. $(x-8)(x+5)$

21. $(y+9)(y+5)$
22. $(x-9)(x-4)$

VII. …More Factoring Trinomials: $x^2 + bx + c$

1. $(x+4)(x+3)$
2. $(m+3)(m+7)$
3. $(y-8)(y+1)$
4. $(x-1)(x-5)$
5. $(x+8)(x-4)$
6. $(x-5)(x+3)$
7. $(x-2)(x-4)$
8. $(y+3)(y+6)$
9. $(t-1)(t-3)$
10. $(v+2)(v+10)$
11. $(3-k)(17-k)$ *or* $(k-3)(k-17)$
12. $(a-2b)(a-12b)$
13. $(y-6)(y+12)$
14. $(x-15y)(x+4y)$
15. $(5r-s)(3r+s)$
16. $3(x-2y)(x+9y)$
17. $(x-6y)(x+y)$
18. $(x+6y)(x+2y)$
19. $(y-5x)(y-2x)$
20. $(a-15b)(a+4b)$

VIII. Factoring Trinomials: $ax^2 + bx + c$

1. $(2x+1)(x-3)$
2. $(3x-2)(x+4)$
3. $(2y+1)(y+7)$
4. $(7a-4)(a-1)$
5. $(5n+2)(n+3)$
6. $(2y+3)(2y+1)$
7. $(3x+7)(x-1)$
8. $(2x+3)(x+5)$
9. $(3y-2)(3y+4)$
10. $(3x+4)(2x-5)$
11. $(2n-7)(n+2)$
12. prime
13. $(2x+5)(5x-6)$
14. $(3y+1)(4y+1)$
15. $(2n-1)(n+5)$
16. $(2x+3)(x+2)$
17. $(5a+3)(a-9)$
18. $(3x-8)(5x+4)$
19. $(2a-1)(4a-3)$
20. $(2y+5)(y-4)$

IX. …More Factoring Trinomials: $ax^2 + bx + c$

1. $(3x+1)(x+1)$
2. $(5z+2)(z+1)$
3. $(2n-1)(n-5)$
4. $(3z-2)(z+1)$
5. $(5h-7)(h+1)$
6. $(4s-3t)(2s-t)$
7. $(2x+3)(3x+5)$
8. $(7a-4b)(4a+3b)$
9. $(2a-3b)(a+5b)$
10. $(3x+2)(4x+3)$
11. prime
12. $(7y+8)(8y-7)$
13. $(4x-7y)(3x-2y)$
14. $(8x+7y)(8x-3y)$
15. $(4x+7y)^2$
16. $(6x-5)(3x-7)$

X. Factoring: Putting It All Together

1. $2(x+2)(x-2)$
2. $2(x+3)(x+1)$
3. $3(n+5)(n-2)$
4. $2(3x+2)(x-5)$
5. $2(x+10)(x-4)$
6. $5(t+1)(t+2)$
7. $2(2n+3)(2n-3)$
8. $7(2x+3)(x-1)$
9. $4(x+2)^2$
10. $2x(3+x)^2$ *or* $2x(x+3)^2$
11. $2x(1+y)(1-y)$
12. $3t(t+3)(t-3)$
13. $3(2a-1)(4a-3)$
14. $5(2x-1)(x+2)$
15. $3(x-7)^2$
16. $4x^2(x+1)(x-1)$

XI. …More Factoring: Putting It All Together

1. $8(2x+1)(x-3)$
2. $3(3x-2)^2$
3. $5(x-14)(x+2)$
4. $6m(m^2+9m-1)$
5. $k^2(5k-2)(k+2)$
6. $x^2(y^2+x^2)(y+x)(y-x)$
7. $(y^2-8)(y^2+2)$
8. $(x^2+1)(x+2)(x-2)$
9. $[h+(a-3)][h-(a-3)]$ *or* $(h+a-3)(h-a+3)$
10. $(9x^2+4y^2)(3x+2y)(3x-2y)$
11. $mn^2(2-m)^2$ *or* $mn^2(m-2)^2$
12. $[(2a+3)+(a-1)][(2a+3)-(a-1)]$ *or* $(3a+2)(a+4)$
13. $(2d^2+1)^2(2d^2-1)^2$
14. $(x+2)^3(x-2)$

XII. Extra: Factoring by Grouping

1. $(x+2)(x+y)$
2. $(a-2)(3a+b)$
3. $(t-1)(t^2+2)$
4. $(5+t)(2-s)$
5. $(c+7)\left(\frac{2}{3}b+1\right)$
6. $(2u+1)(2u+v)$
7. $(d+3)(a-d)$
8. $(n+2)(n+3m)$
9. $(x-y)(x+y)(2a+b)$
10. $(z-y)(z+y)^2$
11. $(y-1)(y+2)(y-2)$
12. $(a+b)(x+4)(x-4)$
13. $(x+1)^2(x-1)$
14. $(a-1)(a^2-8)$

Review: Solving Equations

To solve an equation means to find the values of the variable(s) so that the statement is TRUE; that is, find x so that _____ equals _____.

e.g. Determine whether or not $x = 2$ is a solution to:

a) $4x - 3 = 5$ b) $3x - 8 = 10 - x$

In order to solve linear equations:

1. Clear any fractions by multiplying all terms by the LCD
2. Collect all like terms (variable terms on one side, constant terms on the other side)
3. Take the steps necessary to isolate for the variable by applying _____ BEDMAS operations in _____ order.
   Opposite operations:

Practice: Solve the following equations.

a) $6x - 5 = 4(x + 3)$ b) $2x - 3 + 7x = 5(2 - x)$ c) $x - 3 = x + 5$

d) $\frac{x}{3} - 4 = \frac{x+1}{6}$ e) $\frac{x-3}{6} - \frac{2x+1}{4} = 3$

In grade 10, you also learned how to solve quadratic equations.

1. Equations of the form $ab = 0$ mean you have two solutions: $a = 0$ and $b = 0$.

   e.g. Solve $(x - 3)(x + 4) = 0$

2. When you have an equation with x^2, remember that the inverse operation is _____ and you will have two solutions.

   e.g. Solve $2(4x + 1)^2 - 50 = 0$

Practice: Solve the following equations.

a) $(2x - 1)(x + 5) = 0$

b) $6x(3x + 2) = 0$

c) $2x^2 - 18 = 0$

d) $7(3x + 2)^2 = 28$

Review: Solving Linear Systems

To solve a linear system means to find the _____ of two lines.

1. Solve the following linear systems by substitution.

a) $y = 3x - 4$
$4x + 2y = 3$

b) $y = 2 + x$
$3x - 3y + 4 = 0$

2. Solve the following linear systems by elimination.

a) $5x - 3y = 1$
$3y + 2x = -4$

b) $4x + 2y = 10$
$2x + y - 5 = 0$

Practice:

1. For more practice with solving linear equations, refer to the review package.

2. Solve the following quadratic equations:

a) $4x(x+1)=0$ b) $3x(2x-5)=0$ c) $(x-3)(x+7)=0$

d) $(3x-4)(5x+1)=0$ e) $2x^2-32=0$ f) $-3x^2+27=0$

g) $(x+4)^2-9=0$ h) $(3x-1)^2-25=0$ i) $4(2x+1)^2-9=0$

3. Solve the following linear

a) $y=11-x$
$2x-3y-2=0$

b) $4x+2y=3$
$x-2y-1=0$

c) $3x+2y=3$
$12x+8y=5$

Answers:

2. a) $x=0,-1$ b) $x=0,\frac{5}{2}$ c) $x=3,-7$ d) $x=\frac{4}{3},-\frac{1}{5}$ e) $x=\pm 4$

f) $x=\pm 3$ g) $x=-1,-7$ h) $x=2,-\frac{4}{3}$ i) $x=\frac{1}{4},-\frac{5}{4}$

3. a) $(7,4)$ b) $\left(\frac{4}{5},-\frac{1}{10}\right)$ c) No solution

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Solving Equations (1)

Date:

Solve each of the following equations:

1. a) $x-3=7$ b) $a-2=3$ c) $x-2=2$ d) $x-3=6$
   e) $x+2=5$ f) $y+6=6$ g) $1+x=4$ h) $2+a=-2$
2. a) $4x=12$ b) $6x=24$ c) $3a=3$ d) $2x=16$
   e) $-5t=45$ f) $-3y=-9$ g) $-p=16$ h) $-y=-12$
3. a) $\frac{x}{2}=3$ b) $\frac{a}{3}=4$ c) $\frac{1}{2}x=7$ d) $\frac{1}{3}y=8$
   e) $\frac{a}{3}=-2$ f) $\frac{1}{4}b=-3$ g) $-\frac{y}{4}=-4$ h) $\frac{-d}{5}=-2$
4. a) $2x+3=13$ b) $5x+7=22$ c) $6y+5=41$ d) $4x-5=-17$
   e) $6y-4=20$ f) $7n+6=6$ g) $12-y=16$ h) $11-6p=-49$
5. a) $7x=6$ b) $-5x=7$ c) $10x=-6$ d) $-9m=-12$
   e) $5x+4=7$ f) $4x-3=7$ g) $3+10x=-22$ h) $1-8a=-5$
6. a) $\frac{2x}{3}=12$ b) $\frac{5x}{-4}=-65$ c) $\frac{-6x}{7}=-18$ d) $\frac{-7y}{-6}=-28$
   e) $\frac{7x}{2}=3$ f) $\frac{3x}{4}=2$ g) $\frac{21x}{-2}=7$ h) $\frac{8a}{3}=-6$
7. a) $\frac{x+3}{7}=6$ b) $\frac{x-5}{3}=8$ c) $\frac{a-4}{9}=-3$ d) $\frac{x+1}{-7}=1$
   e) $\frac{5-x}{3}=-7$ f) $\frac{7x-6}{4}=9$ g) $\frac{4x+5}{-3}=13$ h) $\frac{7-2x}{-5}=-13$
8. a) $3x-5=-26$ b) $5(x-6)=-15$ c) $-3(n+2)=12$
   d) $5(x-2)-3=42$ e) $3(2x+1)=-9$ f) $3(2t+8)=0$
9. a) $\frac{x-1}{3}+5=14$ b) $\frac{3x+5}{4}+1=12$ c) $\frac{7x-10}{9}-1=-6$
   d) $\frac{4-x}{5}+12=0$ e) $\frac{11-5x}{6}+1=7$ f) $\frac{2(x-3)}{4}+5=5$
10. Solve ***and check.***
    a) $3x-4=23$ b) $2m+5=-21$ c) $-5a-7=-62$
    d) $8y+3=-37$ e) $2(3x-5)=32$ f) $5(3x+4)=-10$
    g) $9x-(4x+7)=28$ h) $4x-7(7x-8)=-34$ i) $4(x-3)+9x=-38$

Answers:

1. 10, 5, 4, 9, 3, 0, 3, –4
2. 3, 4, 1, 8, –9, 3, –16, 12
3. 6, 12, 14, 24, –6, –12, 16, 10
4. 5, 3, 6, –3, 4, 0, –4, 10
5. $\frac{6}{7}$, $-\frac{7}{5}$, $-\frac{3}{5}$, $\frac{4}{3}$, $\frac{3}{5}$, $\frac{5}{2}$, $-\frac{5}{2}$, $\frac{3}{4}$
6. 18, 52, 21, –24, $\frac{6}{7}$, $\frac{8}{3}$, $-\frac{2}{3}$, $-\frac{9}{4}$
7. 39, 29, –23, –8, 26, 6, –11, –29
8. –7, 3, –6, 11, –2, –4
9. 28, 13, –5, 64, –5, 3

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Solving Equations (2)

Date:

Solve each of the following equations.

1. a) $14+5x-2=4x-15-2x$ b) $10-24m+6=6m+4$ c) $y^2-5y+8=y^2+7y-6$

2. a) $5(p+2)=4(p+1)$ b) $5(k-3)+3(8-k)=21-7(k-6)$
   c) $3(2x-5)-4(x-2)=11-2(x-5)$ d) $3(x-5)-2(x-4)=3(x-3)-6(x-2)$
   e) $3(4x-8)+2(x-7)=4(3x-8)$ f) $6(x-2)+4(x-3)+3(x-4)=120$
   g) $2(x-2)-8(x-1)=2(3-x)+4(x+4)$ h) $2x-4(x-9)=4(5x+9)$

3. a) $\frac{x}{3}=\frac{7}{8}$ b) $\frac{x}{4}=\frac{5}{6}$ c) $\frac{11}{70}=\frac{x}{7}$ d) $\frac{15}{66}=\frac{x}{11}$
   e) $\frac{9}{x}=\frac{12}{7}$ f) $\frac{-4}{-9}=\frac{-3}{x}$ g) $\frac{-10}{x}=\frac{15}{-18}$ h) $\frac{100}{-3}=\frac{-24}{x}$

4. a) $\frac{5}{x}=7$ b) $-8=\frac{-4}{x}$ c) $\frac{6}{x}=\frac{1}{5}$ d) $9=\frac{1}{x}$
   e) $\frac{3x}{4}=\frac{-9}{8}$ f) $\frac{2}{25x}=\frac{6}{35}$ g) $\frac{-5}{-4}=\frac{5x}{-7}$ h) $\frac{-9}{20}=\frac{6x}{-5}$

5. a) $\frac{-x}{2}=\frac{4}{3}$ b) $\frac{-x}{3}=\frac{5}{-6}$ c) $\frac{5}{-7}=\frac{-3}{-x}$ d) $\frac{8}{-x}=-3$
   e) $\frac{3}{x}+\frac{4}{x}=\frac{5}{6}$ f) $\frac{4}{x}=\frac{4}{3}+\frac{7}{3}$ g) $\frac{6}{x}=\frac{4}{x}+\frac{3}{2}$ h) $\frac{4}{5}-\frac{4}{x}=\frac{7}{10}$

6. a) $2(x^2-3x+2)-2(x^2+8x+7)=1$ b) $5(r^2+3r+7)-3(2r^2+5r-6)=8+3r-r^2$
   c) $5(2y^2-8y+3)+2(y^2+10y-4)=12y^2-9$ d) $s^2-3(s^2+s)=-2(s^2-4s+5)$
   e) $3(x^3+2x^2+3x-2)-2(x^3+3x^2-4x-6)=x^3-x$

7. a) $26-15+10a=a+4a+36$ b) $5x-3(2x-11)=x-1$
   c) $2-(3-2m)=3(5-6m)-2(1+4m)$ d) $4(t^2-3t+1)-2(2t^2+3t-9)=-3t-8$

Answers:

1. a) -9 b) $\frac{2}{5}$ c) $\frac{7}{6}$
2. a) -6 b) 6 c) 7 d) $\frac{5}{2}$ e) 3 f) 12 g) $-\frac{9}{4}$ h) 0
3. a) $\frac{21}{8}$ b) $\frac{10}{3}$ c) $\frac{11}{10}$ d) $\frac{5}{2}$ e) $\frac{21}{4}$ f) $-\frac{27}{4}$ g) 12 h) $\frac{18}{25}$
4. a) $\frac{5}{7}$ b) $\frac{1}{2}$ c) 30 d) $\frac{1}{9}$ e) $-\frac{3}{2}$ f) $\frac{7}{15}$ g) $-\frac{7}{4}$ h) $\frac{3}{8}$
5. a) $-\frac{8}{3}$ b) $\frac{5}{2}$ c) $-\frac{21}{5}$ d) $\frac{8}{3}$ e) $\frac{42}{5}$ f) $\frac{12}{11}$ g) $\frac{4}{3}$ h) 40
6. a) $-\frac{1}{2}$ b) 15 c) $\frac{4}{5}$ d) $\frac{10}{11}$ e) $-\frac{1}{3}$
7. a) 5 b) 17 c) $\frac{1}{2}$ d) 2

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Solving Equations (3)

Date:

Solve each of the following.

1. a) $\frac{x}{2}+\frac{x}{3}=5$ b) $\frac{r}{5}+\frac{r}{2}=2$ c) $\frac{m}{6}-\frac{m}{2}=5$

   d) $\frac{a}{5}-a=\frac{1}{2}$ e) $\frac{x}{2}+\frac{2x}{3}=\frac{7}{2}$ f) $\frac{3x}{4}=\frac{x}{3}+\frac{5}{6}$

   g) $\frac{x}{6}+\frac{x}{5}=\frac{5}{12}$ h) $\frac{4x}{5}-\frac{3}{2}=\frac{2}{3}+\frac{1}{3}x$ i) $-\frac{3}{4}x-\frac{4x}{5}+\frac{7x}{10}=\frac{-1}{20}$

   j) $-\frac{x}{3}+\frac{x}{4}-\frac{x}{6}=\frac{1}{10}$ k) $\frac{x}{5}+\frac{2}{5}=\frac{5x}{6}-\frac{3}{2}$ l) $\frac{3x}{10}-\frac{2x}{5}=\frac{3x}{2}+\frac{1}{2}$

2. a) $\frac{x+1}{3}=\frac{x-2}{2}$ b) $\frac{a+5}{3}=\frac{3-a}{7}$ c) $\frac{x-3}{4}+\frac{3+x}{3}=\frac{1}{2}$

   d) $\frac{m+3}{2}-4=\frac{4-m}{3}+7$ e) $\frac{8y+9}{4}-17=\frac{8-y}{2}$ f) $\frac{2-x}{2}+\frac{1}{4}=\frac{4-2x}{5}$

   g) $\frac{3}{7}=\frac{5-6x}{14}+\frac{2x}{7}$ h) $\frac{11s-6}{2}=-\frac{3s}{4}+\frac{9-7s}{2}$ i) $\frac{7(-x-6)}{8}+\frac{3(2x+3)}{4}=-\frac{3}{8}$

Solve and Check your Solution.

3. a) $\frac{a}{2}+3=\frac{2a}{3}$ b) $\frac{3k}{4}+\frac{1}{2}=\frac{2k}{3}$ c) $\frac{5x-3}{6}=\frac{10x+6}{3}$

   d) $\frac{y+3}{5}-\frac{5}{4}=\frac{y-1}{4}$ e) $\frac{5-2a}{3}+\frac{6a-15}{4}=\frac{25}{4}$

Answers:

1. a) 6 b) $\frac{20}{7}$ c) -15 d) $-\frac{5}{8}$ e) 3 f) 2
   g) $\frac{25}{22}$ h) $\frac{65}{14}$ i) $\frac{1}{17}$ j) $-\frac{2}{5}$ k) 3 l) $-\frac{5}{16}$
2. a) 8 b) $-\frac{13}{5}$ c) $\frac{3}{7}$ d) 13 e) $\frac{15}{2}$ f) $\frac{9}{2}$
   g) $-\frac{1}{2}$ h) $\frac{10}{13}$ i) $\frac{21}{5}$
3. a) 18 b) –6 c) –1 d) –8 e) 10

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Unit 1:
Rational Expressions

Simplifying Rational Expressions

A **rational expression** is the quotient of two polynomials. e.g. $\frac{5y^2}{6x^3}$ or $\frac{a^2+b+c^3}{2a+b}$

Simplifying rational expressions means to determine an equivalent expression that will yield the same output.

Steps for simplifying rational expressions:

1) **Factor** the numerator and denominator
2) State any **restrictions** on the variable(s)
   - ➔ Restrictions: value(s) of the variable(s) that make the _____ equal to _____
   - ➔ These value(s) are not permitted because they will make the entire expression _____
   - ➔ To determine restriction(s), set each factor in the denominator to not equal 0 and then isolate for each variable in the denominator
3) Divide by any common factors ("**cancel out**")

Example 1 → Simplify the following to determine an equivalent expression.
State any restrictions on the variables.

a) $\frac{8y^2-10xy}{4y}$

Therefore, $\frac{8y^2-10xy}{4y}$ =

for any values of x and y, except

b) $\frac{x^2}{3x^3-6x^2}$

Therefore, $\frac{x^2}{3x^3-6x^2}$ =

for any values of x, except

c) $\frac{6x-10}{5-3x}$

d) $\frac{3x^2-6xy}{3xy}$

e) $\frac{x^2-4}{x^2-5x+6}$

f) $\frac{2m^2+m-3}{3m^2+2m-5}$

Example 2: Determine if $\frac{2x^2-4x-6}{x+1}$ and $8x^2-2x(4x-1)-6$ are equivalent.

Applications of Simplifying Rational Expressions

Warm-Up: Simplify $\frac{6x^2-5x-4}{3x^2+11x-20}$ and state restrictions.

Example 1: A rectangular piece of cardboard measures 400cm by 600cm. Squares measuring *x* cm by *x* cm will be cut from each corner in order to create an open-topped box.

Determine a simplified expression for the ratio of the volume to the surface area.

Lesson 1.2 Applications of Simplifying Rational Expressions

Example 2: The dimensions of a rectangular prism are $3x + 3$ by $2x + 2$ by $4x - 2$.
Determine the simplified expression for the ratio of the volume to the surface area of the prism.

Example 3: The height of a cylinder is 5 times as long as the radius.
Determine the simplified expression for the ratio of the volume to the surface area of the cylinder.

Communicate Your Understanding

1. Explain why $x \neq 3$ is a restriction on the variable for the expression $\frac{x+4}{x-3}$.
2. Describe how you would simplify $\frac{x^2-x}{x}$.
3. a) Describe how you would simplify $\frac{x^2+3x+2}{x^2-x-2}$.

b) Describe how you would determine the restrictions on the variable.

4. Write an expression in one variable for the denominator of a rational expression, if the restrictions on the variable are $x \neq 2, -3$.

Practise

In each of the following, state any restrictions on the variables.

A

1. Simplify.

a) $\frac{3t^3+6t^2-15t}{3t}$ b) $\frac{6a^2+9a}{12a^2}$

c) $\frac{10y^4+5y^3-15y^2}{5y}$

d) $\frac{14n^4-4n^3+6n^2+8n}{2n^2}$

e) $\frac{4m^2-8mn}{4mn}$ f) $\frac{-6x^2y^3}{-18x^3y}$

g) $\frac{16a^2bc}{4a^2b^2c^2}$ h) $\frac{-4x^4y^2z}{20x^3y^3z}$

i) $\frac{21m(m-4)}{7m^2}$

2. Express in simplest form.

a) $\frac{5x}{5(x+4)}$ b) $\frac{8t^2(t+5)}{4t(t-5)}$

c) $\frac{7x(x-3)}{14x^2(x-3)}$ d) $\frac{(m-1)(m+2)}{(m+4)(m-1)}$

e) $\frac{2x}{2x+8}$ f) $\frac{y^2}{y^2+2y}$

g) $\frac{10x}{5x^2-15x}$ h) $\frac{4x}{16x^3-12x}$

i) $\frac{3xy}{6x^2y-12xy^2}$

3. Simplify.

a) $\frac{6t-36}{t-6}$ b) $\frac{4m+24}{8m-24}$

c) $\frac{5x-10}{3x-6}$ d) $\frac{a^2+2a}{a^2-3a}$

e) $\frac{8x^2+4x}{6x^2+3x}$ f) $\frac{2x^2-2x}{2x^2+2x}$

g) $\frac{4x+4y}{5x+5y}$ h) $\frac{4a^2b+8ab}{6a^2-6a}$

i) $\frac{5xy+10x}{2y^2+4y}$

4. Express in simplest equivalent form.

a) $\frac{m-2}{m^2-5m+6}$ b) $\frac{y^2+10y+25}{y+5}$

c) $\frac{2x+6}{x^2-6x-27}$ d) $\frac{r^2-4}{5r+10}$

e) $\frac{d^2+a}{d^2+2a+1}$

f) $\frac{x^2-9}{2x^2y-6xy}$

g) $\frac{2w+2}{2w^2+3w+1}$

h) $\frac{3t^2-8t+4}{6t^2-4t}$

i) $\frac{8z+6z^2}{9z^2-16}$

j) $\frac{5x^2+3xy-2y^2}{3x^2+3xy}$

5. Simplify.

a) $\frac{y-2}{2-y}$

b) $\frac{3-x}{x-3}$

c) $\frac{2t-1}{4-8t}$

d) $\frac{6-10w}{15w-9}$

e) $\frac{x^2-1}{1-x^2}$

f) $\frac{1-4y^2}{8y^2-2}$

6. Simplify.

a) $\frac{x^2+4x+4}{x^2+5x+6}$

b) $\frac{d^2-a-12}{d^2-9a+20}$

c) $\frac{m^2-5m+6}{m^2+2m-15}$

d) $\frac{y^2-8y+15}{y^2-25}$

e) $\frac{x^2-10x+24}{x^2-12x+36}$

f) $\frac{n^2-n-2}{n^2+n-6}$

g) $\frac{p^2+8p+16}{p^2-16}$

h) $\frac{2t^2-t-1}{t^2-3t+2}$

i) $\frac{6v^2+11v+3}{4v^2+8v+3}$

j) $\frac{6x^2-13x+6}{8x^2-6x-9}$

k) $\frac{3z^2-7z+2}{9z^2-6z+1}$

l) $\frac{2m^2-mn-n^2}{4m^2-4mn-3n^2}$

Apply, Solve, Communicate

7. Saskatchewan flag The area of a Saskatchewan flag can be represented by the polynomial x^2+3x+2 and its width by $x+1$.

a) Write a rational expression that represents the length.

b) Write the expression in simplest form.

c) If x represents 1 unit of length, what is the ratio length:width for a Saskatchewan flag?

B

8. Simplify, if possible.

a) $\frac{1-x}{x-1}$

b) $\frac{x-1}{x+1}$

c) $\frac{y^2+1}{y^2-1}$

d) $\frac{3t-7}{3t-7}$

e) $\frac{t^2-s^2}{(s+t)^2}$

f) $\frac{x^3-2x^2+3x}{2x^2-4x+6}$

9. For which values of x are the following rational expressions not defined?

a) $\frac{2x-y}{x-y}$

b) $\frac{4x}{3x+y}$

c) $\frac{3}{x^3}$

d) $\frac{x^2}{x^3-8}$

e) $\frac{x^2+3x-11}{x^2-1}$

f) $\frac{3x^2+5xy+2y^2}{4x^2-9y^2}$

10. Communication State whether each of the following rational expressions is equivalent to the expression $\frac{x+1}{x-1}$. Explain.

a) $\frac{x+2}{x-2}$ b) $\frac{x^2+x}{x^2-x}$ c) $\frac{4+4x}{4x-4}$

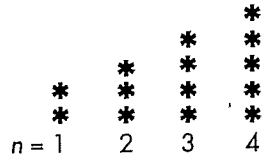
d) $\frac{3x+1}{3x-1}$ e) $\frac{(x+1)^2}{(x-1)^2}$ f) $\frac{1+x}{1-x}$

11. Cube For a cube of edge length x, find the ratio of the volume to the surface area. Simplify, if possible.

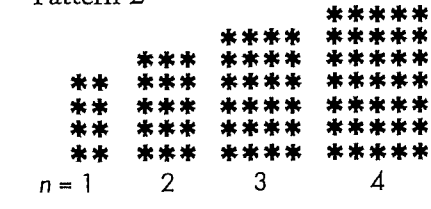
12. Application For a sphere of radius r, find the ratio of the volume to the surface area. Simplify, if possible.

13. Pattern The first 4 diagrams of two patterns are shown.

Pattern 1



Pattern 2



a) For pattern 1, express the number of asterisks in the nth diagram in terms of n.

b) For pattern 2, the number of asterisks in the nth diagram is given by the binomial product $(n + \blacktriangle)(n + \blacksquare)$, where $\blacktriangle$ and $\blacksquare$ represent whole numbers. Replace $\blacktriangle$ and $\blacksquare$ in the binomial product by their correct values.

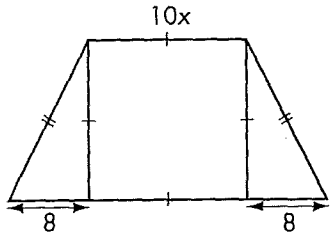
c) Divide your polynomial from part b) by your expression from part a).

d) Use your result from part c) to calculate how many times as many asterisks there are in the 10th diagram of pattern 2 as there are in the 10th diagram of pattern 1.

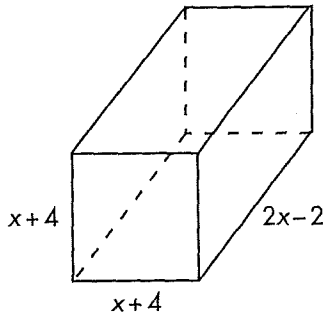
e) If a diagram in pattern 1 has 20 asterisks, how many asterisks are in the corresponding diagram of pattern 2?

f) If a diagram in pattern 2 has 1295 asterisks, how many asterisks are in the corresponding diagram in pattern 1?

14. Measurement Find the ratio of the area of the square to the area of the trapezoid. Simplify, if possible.



15. Rectangular prism Find the ratio of the volume to the surface area for the rectangular prism shown. Simplify, if possible.



C

16. Write rational expressions in one variable so that the restrictions on the variables are as follows.

a) $x \neq 1$ b) $y \neq 0, -3$ c) $a \neq \frac{1}{2}, -\frac{3}{4}$ d) $t \neq -1, \pm\sqrt{3}$

17. Technology a) Use a graphing calculator to graph the equations $y = \frac{2x^2 + 3x}{x}$ and $y = 2x + 3$ in the same standard viewing window. Explain your observations.

b) Display the tables of values for the two equations. Compare and explain the values of y when $x = 0$.

18. Inquiry/Problem Solving a) For a solid cone with radius r, height h, and slant height s, find the ratio of the volume to the surface area. Simplify, if possible.

b) Determine whole-number values of r, h, and s that give the ratio in part a) a numerical value of 1.

WORD *Power*

Lewis Carroll invented a word game called doublets. The object of the game is to change one word to another by changing one letter at a time. You must form a real word each time you change a letter. The best solution has the fewest steps. Change the word RING to the word BELL by changing one letter at a time.

Multiplying & Dividing Rational Expressions

When multiplying fractions, we often reduce first by dividing out common factors and then multiply to get our final answer.

$$\frac{14}{30} \times \frac{15}{21} =$$

Steps for multiplying rational expressions:

1) Factor the numerator and denominator
2) Divide by any common factors ("cancel out")
3) Express product as a single rational expression
   ➔ Multiply numerators together and multiply denominators together
4) State any restrictions on the variable(s) in the denominators

Example 1: Simplify and state any restrictions on the variables.

a) $\frac{12a^2b}{15} \times \frac{5ab}{8b^2}$

b) $\frac{2x+6}{x^2+7x+10} \times \frac{x^2+3x-10}{x^2-4}$

Steps for <u>dividing</u> rational expressions:

1) Multiply by the reciprocal of the divisor
2) Factor the numerator and denominator
3) Divide by any common factors ("cancel out")
4) Express product as a single rational expression
   ➔ Multiply numerators together and multiply denominators together
5) State any restrictions on the variable(s) in the denominators AND the numerator of the divisor
   ➔ Divisor's numerator and denominator switched when we changed division into multiplication and changed the divisor into its reciprocal

Example 2: Simplify and state any restrictions on the variables.

a) $\frac{6m^5}{5n^3} \div \frac{8m^6}{15mn}$

b) $\frac{a^2+5a-14}{a^2+8a+7} \div \frac{a^2-4a+4}{3a^2-6a}$

c) $\frac{nm-2m-3n+6}{2-n} \div \frac{m^2-6m+9}{6-2m}$

 Name_____

Rational Expressions

Date_____ Period____

State the excluded values for each.

1) $\frac{60x^3}{12x}$

2) $\frac{70v^2}{100v}$

3) $\frac{m+7}{m^2+4m-21}$

4) $\frac{n^2+6n+5}{n+1}$

5) $\frac{35x-35}{25x-40}$

6) $\frac{-n^2+16n-63}{n^2-2n-35}$

Simplify each and state the excluded values.

7) $\frac{p+4}{p^2+6p+8}$

8) $\frac{9}{15a-15}$

9) $\frac{2a^2+10a}{3a^2+15a}$

10) $\frac{p^2-3p-10}{p^2+p-2}$

11) $\frac{x^2+x-6}{x^2+8x+15}$

12) $\frac{a^2+5a+4}{a^2+9a+20}$

13) $\dfrac{x^2-2x-15}{x^2-6x+5}$

14) $\dfrac{10x-6}{10x-6}$

15) $\dfrac{(v-7)(v+8)}{(v+8)(v-10)} \div \dfrac{1}{v-10}$

16) $\dfrac{n+3}{n+2} \div \dfrac{(n-1)(n+3)}{(n-1)^2}$

17) $\dfrac{x+3}{4} \cdot \dfrac{3(x-6)}{3(x+3)}$

18) $\dfrac{x-8}{(x+6)(x-8)} \cdot \dfrac{4x(x+10)}{x+10}$

19) $\dfrac{2b^2-12b}{b+5} \div \dfrac{b-6}{b+5}$

20) $\dfrac{1}{n+9} \div \dfrac{6-n}{3n-18}$

21) $\dfrac{28-7b}{b-4} \cdot \dfrac{1}{b+10}$

22) $\dfrac{2}{v^2-12v+27} \cdot \dfrac{v^2-12v+27}{3}$

23) $\dfrac{1}{5p^2} \div \dfrac{9p-36}{5p^3-35p^2}$

24) $\dfrac{8-7x-x^2}{x+8} \cdot \dfrac{x+5}{9x-9}$

25) $\dfrac{x^2-16}{9-x} \cdot \dfrac{x^2+x-90}{x^2+14x+40}$

26) $\dfrac{10x^2-20x}{40x^3-80x^2} \cdot \dfrac{16x^3+80x^2}{6x+30}$

Name_____

Rational Expressions

Date_____ Period____

State the excluded values for each.

1) $\frac{60x^3}{12x}$

$\{0\}$

2) $\frac{70v^2}{100v}$

$\{0\}$

3) $\frac{m+7}{m^2+4m-21}$

$\{-7, 3\}$

4) $\frac{n^2+6n+5}{n+1}$

$\{-1\}$

5) $\frac{35x-35}{25x-40}$

$\left\{\frac{8}{5}\right\}$

6) $\frac{-n^2+16n-63}{n^2-2n-35}$

$\{-5, 7\}$

Simplify each and state the excluded values.

7) $\frac{p+4}{p^2+6p+8}$

$\frac{1}{p+2}$; $\{-2, -4\}$

8) $\frac{9}{15a-15}$

$\frac{3}{5(a-1)}$; $\{1\}$

9) $\frac{2a^2+10a}{3a^2+15a}$

$\frac{2}{3}$; $\{0, -5\}$

10) $\frac{p^2-3p-10}{p^2+p-2}$

$\frac{p-5}{p-1}$; $\{-2, 1\}$

11) $\frac{x^2+x-6}{x^2+8x+15}$

$\frac{x-2}{x+5}$; $\{-3, -5\}$

12) $\frac{a^2+5a+4}{a^2+9a+20}$

$\frac{a+1}{a+5}$; $\{-4, -5\}$

13) $\dfrac{x^2 - 2x - 15}{x^2 - 6x + 5}$

$\dfrac{x+3}{x-1}$; $\{1, 5\}$

14) $\dfrac{10x - 6}{10x - 6}$

1 ; $\left\{\dfrac{3}{5}\right\}$

15) $\dfrac{(v-7)(v+8)}{(v+8)(v-10)} \div \dfrac{1}{v-10}$

$v - 7$; $\{-8, 10\}$

16) $\dfrac{n+3}{n+2} \div \dfrac{(n-1)(n+3)}{(n-1)^2}$

$\dfrac{n-1}{n+2}$; $\{-2, 1, -3\}$

17) $\dfrac{x+3}{4} \cdot \dfrac{3(x-6)}{3(x+3)}$

$\dfrac{x-6}{4}$; $\{-3\}$

18) $\dfrac{x-8}{(x+6)(x-8)} \cdot \dfrac{4x(x+10)}{x+10}$

$\dfrac{4x}{x+6}$; $\{-6, 8, -10\}$

19) $\dfrac{2b^2 - 12b}{b+5} \div \dfrac{b-6}{b+5}$

$2b$; $\{-5, 6\}$

20) $\dfrac{1}{n+9} \div \dfrac{6-n}{3n-18}$

$-\dfrac{3}{n+9}$; $\{-9, 6\}$

21) $\dfrac{28 - 7b}{b-4} \cdot \dfrac{1}{b+10}$

$-\dfrac{7}{b+10}$; $\{4, -10\}$

22) $\dfrac{2}{v^2 - 12v + 27} \cdot \dfrac{v^2 - 12v + 27}{3}$

$\dfrac{2}{3}$; $\{3, 9\}$

23) $\dfrac{1}{5p^2} \div \dfrac{9p - 36}{5p^3 - 35p^2}$

$\dfrac{p-7}{9(p-4)}$; $\{0, 7, 4\}$

24) $\dfrac{8 - 7x - x^2}{x+8} \cdot \dfrac{x+5}{9x-9}$

$-\dfrac{(x+5)}{9}$; $\{-8, 1\}$

25) $\dfrac{x^2 - 16}{9 - x} \cdot \dfrac{x^2 + x - 90}{x^2 + 14x + 40}$

$-(x-4)$; $\{9, -4, -10\}$

26) $\dfrac{10x^2 - 20x}{40x^3 - 80x^2} \cdot \dfrac{16x^3 + 80x^2}{6x + 30}$

$\dfrac{2x}{3}$; $\{0, 2, -5\}$

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Warm-Up

1. Simplify $\frac{9x^2-16}{x^2+2x-15} \div \frac{6x^2+5x-4}{x^2+5x}$.

2. Simplify the ratio of the volume to the surface area of a cylinder with radius r and height $3r$.

Adding and Subtracting Rational Expressions

Steps for Adding/Subtracting Rational Expressions	Example: $\frac{x+5}{x^2+x-6} - \frac{4x}{x^2-9}$
1. Factor the numerators and denominators. 2. Cancel any common factors within one rational expression. 3. Determine the Lowest Common Denominator. ➔ Monomials: LCM of the coefficients, highest power of the variables ➔ Binomials: multiply factors 4. Rewrite each expression using the LCD. ➔ Multiply the numerator and denominator by the missing factor. 5. Expand and simplify the numerators. 6. State restrictions on the denominators.	

Example 1: Simplify and state restrictions.

a) $\frac{3}{x^2} - \frac{1}{3x} + \frac{7}{5x^3}$

b) $\frac{3}{x^2+9x+14} - \frac{5}{x^2-4}$

c) $\frac{3x}{(x+2)}+\frac{6x}{(x+2)}-\frac{5}{(x+2)}$

d) $\frac{a-2}{2a-3}+\frac{a+3}{3-2a}$

e) $m+\frac{2m}{n}-\frac{3+n}{m}$

f) $\frac{2x-1}{2x^2+3x+1}+\frac{2x+1}{3x^2+4x+1}$

g) $\frac{3x}{(8x+4)}+\frac{4}{(6x-6)}+2$

h) $\frac{2x^2+5x}{6x^2+11x-10}-\frac{3x+1}{9x^2-4}$

HOMEWORK – Lesson 1.4 (Day 1)

2. Find the LCM.

a) $10a^2b,\ 4ab^3$ **b)** $3m^2n,\ 2mn^2,\ 6mn$

c) $2x^3,\ 6xy^2,\ 4y$ **d)** $10s^2t^2,\ 20s^2t,\ 15st^2$

3. Simplify.

a) $\dfrac{3}{2x}+\dfrac{4}{5x}$

b) $\dfrac{2}{4y}+\dfrac{3}{3y}-\dfrac{1}{2y}$

c) $\dfrac{1}{2x^2}+\dfrac{3}{3x}-\dfrac{2}{x^3}$

d) $\dfrac{3}{2m^2n}-\dfrac{1}{m^2n^3}+\dfrac{4}{5mn}$

e) $x-\dfrac{2}{x}+5$

f) $\dfrac{3m+4}{mn}-\dfrac{1}{m}-2$

g) $\dfrac{4x-1}{3x^2}-\dfrac{2x+3}{x}+\dfrac{5x+2}{5x^2}$

h) $\dfrac{x-2y}{x}-\dfrac{4x+y}{xy}-\dfrac{3x-4y}{y}$

6. Simplify.

a) $\dfrac{2}{x+1}+\dfrac{3}{x+2}$ **b)** $\dfrac{m}{m-3}-\dfrac{5}{m+2}$

c) $\dfrac{3}{x}+\dfrac{5}{x-1}$ **d)** $\dfrac{2}{t-1}+\dfrac{1}{5}+2$

e) $\dfrac{2x}{x-2}-\dfrac{3x}{x+2}$ **f)** $\dfrac{4}{3n-1}-\dfrac{3}{2n+3}$

g) $\dfrac{1}{2x-2}+\dfrac{3}{4x-8}$ **h)** $\dfrac{t}{3t+15}-\dfrac{1}{6t-24}$

i) $\dfrac{4}{2s-12}-\dfrac{s}{5s-5}$ **j)** $\dfrac{2m}{3m-15}+\dfrac{m}{4m-8}$

7. State the LCM in factored form.

a) $x+2,\ x^2+4x+4$

b) $y^2+6y+8,\ y^2-4$

c) $t^2-t-12,\ t^2-3t-4$

d) $2x-4,\ x^2-3x-4$

e) $m^2+6m+9,\ m^2-2m-15$

8. Simplify.

a) $\dfrac{2}{x+3}+\dfrac{3}{x^2+5x+6}$

b) $\dfrac{y}{y^2-16}-\dfrac{4}{y+4}$

c) $\dfrac{3x}{x-5}+\dfrac{2x}{x^2-4x-5}$

d) $\dfrac{a}{a^2-7a+12}-\dfrac{2a}{a-3}$

e) $\dfrac{4}{2x^2+3x+1}+\dfrac{2}{2x+1}$

f) $\dfrac{6}{2n-1}-\dfrac{3}{6n^2-5n+1}$

ANSWERS:

2. a) $20a^2b^3$ **b)** $6m^2n^2$ **c)** $12x^3y^2$ **d)** $60s^2t^2$

3. a) $\dfrac{23}{10x},\ x\neq 0$ **b)** $\dfrac{1}{y},\ y\neq 0$ **c)** $\dfrac{2x^2+x-4}{2x^3},\ x\neq 0$

d) $\dfrac{15n^2+8mn^2-10}{10m^2n^3},\ m,\ n\neq 0$ **e)** $\dfrac{x^2+5x-2}{x},\ x\neq 0$

f) $\dfrac{3m-2mn-n+4}{mn},\ m,\ n\neq 0$ **g)** $\dfrac{-30x^2-10x+1}{15x^2},\ x\neq 0$

h) $\dfrac{-3x^2-2y^2+5xy-4x-y}{xy},\ x,y\neq 0$

6. a) $\dfrac{5x+7}{(x+1)(x+2)},\ x\neq -1,-2$ **b)** $\dfrac{m^2-3m+15}{(m-3)(m+2)},\ m\neq -2,3$

c) $\dfrac{8x-3}{x(x-1)},\ x\neq 0,1$ **d)** $\dfrac{11t-1}{5(t-1)},\ t\neq 1$ **e)** $\dfrac{10x-x^2}{(x-2)(x+2)},$

$x\neq \pm 2$ **f)** $\dfrac{15-n}{(3n-1)(2n+3)},\ n\neq -\dfrac{3}{2},\dfrac{1}{3}$ **g)** $\dfrac{5x-7}{4(x-1)(x-2)},$ $x\neq 1,2$ **h)** $\dfrac{2t^2-9t-5}{6(t+5)(t-4)},\ t\neq -5,4$ **i)** $\dfrac{-s^2+16s-10}{5(s-6)(s-1)},\ s\neq 1,6$

j) $\dfrac{11m^2-31m}{12(m-5)(m-2)},\ m\neq 2,5$ **7. a)** $(x+2)^2$

b) $(y-2)(y+2)(y+4)$ **c)** $(t+3)(t-4)(t+1)$

d) $2(x-2)(x+1)(x-4)$ **e)** $(m+3)^2(m-5)$

8. a) $\dfrac{2x+7}{(x+3)(x+2)},\ x\neq -3,-2$ **b)** $\dfrac{-3y+16}{(y-4)(y+4)},\ y\neq \pm 4$

c) $\dfrac{3x^2+5x}{(x-5)(x+1)},\ x\neq -1,5$ **d)** $\dfrac{9a-2a^2}{(a-3)(a-4)},\ a\neq 3,4$

e) $\dfrac{2x+6}{(2x+1)(x+1)},\ x\neq -1,-\dfrac{1}{2}$ **f)** $\dfrac{18n-9}{(2n-1)(3n-1)},$

$n\neq \dfrac{1}{3},\dfrac{1}{2}$

HOMEWORK – Lesson 1.4 (Day 2)

9. Simplify.

a) $\frac{2}{m^2+4m+3}+\frac{1}{m^2+7m+12}$

b) $\frac{1}{x^2+4x+4}-\frac{3}{x^2-4}$

c) $\frac{a}{a^2-25}-\frac{2}{a^2-9a+20}$

d) $\frac{4m}{m^2-9m+18}+\frac{2m}{m^2-11m+30}$

e) $\frac{5}{3x^2+4x+1}+\frac{2}{3x^2-2x-1}$

f) $\frac{3y}{4y^2-9}-\frac{2y}{4y^2-12y+9}$

10. Simplify.

a) $\frac{t+1}{t-1}+\frac{2}{t^2-5t+4}$

b) $\frac{y+1}{y-1}+\frac{y-1}{y^2+y-2}$

c) $\frac{x-2}{x^2+4x+3}-\frac{2x+1}{x+3}$

d) $\frac{n^2+4n-3}{n^2-16}+\frac{4-3n}{3n-12}$

e) $\frac{m+4}{m^2-m-12}-\frac{m}{m^2-5m+4}$

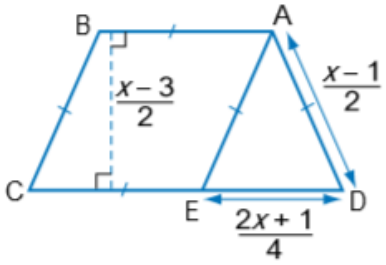
f) $\frac{a+2}{a^2-1}-\frac{a-1}{a^2+2a+1}$

g) $\frac{3w-4}{w^2+5w+4}+\frac{2w-3}{w^2+2w-8}$

h) $\frac{2x-1}{2x^2+3x+1}+\frac{2x+1}{3x^2+4x+1}$

i) $\frac{2z-1}{4z^2-25}-\frac{2z+5}{4z^2-8z-5}$

12. Measurement The diagram shows trapezoid ABCD divided into rhombus ABCE and isosceles triangle ADE.



a) Write an expression that represents the area of the triangle in terms of x.

b) Write an expression that represents the area of the rhombus in terms of x.

c) Add and simplify the expressions you wrote in parts a) and b).

d) Write and simplify an expression that represents the longer base of the trapezoid in terms of x.

e) Use the formula for the area of a trapezoid to write and simplify an expression that represents the area of the trapezoid in terms of x.

f) Compare your expressions from parts c) and e).

14. Simplify.

a) $\frac{m+3}{m+2} \times \frac{m+2}{m+1} + \frac{5m}{m+1}$

b) $\frac{2x^2-2x}{x^2+4x-5} - \frac{4x}{x+5}$

c) $\frac{x^2+2x-15}{x^2-7x+12} + \frac{x^2-6x-7}{x^2-3x-4}$

d) $\frac{3y-1}{y-4} - \frac{y^2+4y-12}{y^2-6y+8}$

e) $\frac{2z^2-5z+3}{z^2-1} + \frac{4z^2-9}{4z+6}$

f) $\frac{x^2+5x+6}{x^2-3x+2} \div \frac{x+3}{x-1} - \frac{6}{x+3}$

C

15. Write two rational expressions with binomial denominators and with each of the following sums. Compare your answers with a classmate's.

a) $\frac{5x+8}{(x+1)(x+2)}$

b) $\frac{5x-5}{6x^2-13x+6}$

c) $\frac{x^2-3}{(x-1)(x-3)}$

d) $\frac{4x^2}{4x^2-9}$

Answers:

9. a) $\frac{3}{(m+1)(m+4)}$, $m \neq -4, -3, -1$

b) $\frac{-2x-8}{(x+2)^2(x-2)}$, $x \neq \pm 2$ **c)** $\frac{a^2-6a-10}{(a-5)(a+5)(a-4)}$, $a \neq \pm 5, 4$

d) $\frac{6m^2-26m}{(m-3)(m-6)(m-5)}$, $m \neq 3, 5, 6$

e) $\frac{7x-3}{(3x+1)(x+1)(x-1)}$, $x \neq -\frac{1}{3}, \pm 1$

f) $\frac{2y^2-15y}{(2y-3)^2(2y+3)}$, $y \neq \pm\frac{3}{2}$ **10. a)** $\frac{t^2-3t-2}{(t-1)(t-4)}$, $t \neq 1, 4$

b) $\frac{y^2+4y+1}{(y-1)(y+2)}$, $y \neq -2, 1$ **c)** $\frac{-2x^2-2x-3}{(x+3)(x+1)}$, $x \neq -3, -1$

d) $\frac{4n+7}{3(n+4)(n-4)}$, $n \neq \pm 4$ **e)** $\frac{-4}{(m+3)(m-4)(m-1)}$, $m \neq -3, 1, 4$ **f)** $\frac{5a+1}{(a-1)(a+1)^2}$, $a \neq \pm 1$

g) $\frac{5w^2-11w+5}{(w+1)(w+4)(w-2)}$, $w \neq -4, -1, 2$

h) $\frac{10x^2+3x}{(2x+1)(x+1)(3x+1)}$, $x \neq -1, -\frac{1}{2}, -\frac{1}{3}$

i) $\frac{-20z-26}{(2z-5)(2z+5)(2z+1)}$, $z \neq \pm\frac{5}{2}, -\frac{1}{2}$

12. a) $\frac{(2x+1)(x-3)}{16}$ **b)** $\frac{(x-1)(x-3)}{4}$ **c)** $\frac{3(x-3)(2x-1)}{16}$

d) $\frac{4x-1}{4}$ **e)** $\frac{3(x-3)(2x-1)}{16}$ **f)** They are equal.

14. a) $\frac{6m+3}{m+1}$, $m \neq -1, -2$

b) $\frac{-2x}{x+5}$, $x \neq -5, 1$ **c)** $\frac{2x-2}{x-4}$, $x \neq 3, 4, -1$ **d)** $\frac{2y-7}{y-4}$, $y \neq 2, 4$

e) $\frac{2z^2+3z-9}{2(z+1)}$, $z \neq \pm 1, -\frac{3}{2}$ **f)** $\frac{x^2-x+18}{(x-2)(x+3)}$, $x \neq 1, 2, -3$

15. Answers may vary. **a)** $\frac{2}{x+2} + \frac{3}{x+1}$ **b)** $\frac{1}{2x-3} + \frac{1}{3x-2}$

c) $\frac{x}{x-3} + \frac{1}{x-1}$ **d)** $\frac{4x+3}{2(2x+3)} + \frac{3}{2(2x-3)}$

Adding + Subtracting Rational Expressions

Date_____ Period____

Simplify each expression.

1) $\frac{u+5v}{8v^2u^2} - \frac{u-6v}{8v^2u^2}$

2) $\frac{5n}{30m} + \frac{2m+4n}{30m}$

3) $\frac{a+2b}{6a^3} - \frac{5a+4b}{6a^3}$

4) $\frac{x+y}{18xy} - \frac{6x+y}{18xy}$

5) $\frac{4a-5}{6a^2+30a} + \frac{a-1}{6a^2+30a}$

6) $\frac{5x-4}{9x^3+27x^2} - \frac{x+6}{9x^3+27x^2}$

7) $\frac{b-3}{12b+18} + \frac{4b}{12b+18}$

8) $\frac{n-4}{n^2-n-20} + \frac{n+1}{n^2-n-20}$

9) $\frac{7x}{2x} - \frac{x-2}{20x+16}$

10) $\frac{8}{7v-6} + \frac{4}{3v^2}$

11) $\frac{7v}{8} - \frac{8v-4}{5v-2}$

12) $\frac{4}{n+7} - \frac{7}{n-2}$

13) $\frac{7}{3n^2+24n} - \frac{7}{2n}$

14) $\frac{6}{v-2} - \frac{7}{2v+7}$

15) $\frac{6x}{3} + \frac{7}{15x+3}$

16) $\frac{5v}{v-3} + \frac{5}{v+6}$

17) $\frac{4x}{x^2+4x-5} - \frac{5}{4}$

18) $\frac{2}{x+3} - \frac{6x}{2x+1}$

19) $\frac{4x}{x+3} - \frac{4x}{x+6}$

20) $\frac{2x}{3x+3} - \frac{2}{x+5}$

21) $\frac{6}{x-2} + \frac{6}{x+1}$

22) $\frac{v-2}{3v^4-15v^3-18v^2} + 3v$

Adding + Subtracting Rational Expressions

Date_____ Period____

Simplify each expression.

1) $\frac{u+5v}{8v^2u^2} - \frac{u-6v}{8v^2u^2}$

$\frac{11}{8vu^2}$

2) $\frac{5n}{30m} + \frac{2m+4n}{30m}$

$\frac{9n+2m}{30m}$

3) $\frac{a+2b}{6a^3} - \frac{5a+4b}{6a^3}$

$\frac{-2a-b}{3a^3}$

4) $\frac{x+y}{18xy} - \frac{6x+y}{18xy}$

$-\frac{5}{18y}$

5) $\frac{4a-5}{6a^2+30a} + \frac{a-1}{6a^2+30a}$

$\frac{5a-6}{6a^2+30a}$

6) $\frac{5x-4}{9x^3+27x^2} - \frac{x+6}{9x^3+27x^2}$

$\frac{4x-10}{9x^3+27x^2}$

7) $\frac{b-3}{12b+18} + \frac{4b}{12b+18}$

$\frac{5b-3}{12b+18}$

8) $\frac{n-4}{n^2-n-20} + \frac{n+1}{n^2-n-20}$

$\frac{2n-3}{n^2-n-20}$

9) $\frac{7x}{2x} - \frac{x-2}{20x+16}$

$\frac{69x+58}{4(5x+4)}$

10) $\frac{8}{7v-6} + \frac{4}{3v^2}$

$\frac{24v^2+28v-24}{3v^2(7v-6)}$

11) $\frac{7v}{8} - \frac{8v-4}{5v-2}$

$\frac{35v^2 - 78v + 32}{8(5v-2)}$

12) $\frac{4}{n+7} - \frac{7}{n-2}$

$\frac{-3n - 57}{(n+7)(n-2)}$

13) $\frac{7}{3n^2 + 24n} - \frac{7}{2n}$

$\frac{-154 - 21n}{6n(n+8)}$

14) $\frac{6}{v-2} - \frac{7}{2v+7}$

$\frac{5v + 56}{(2v+7)(v-2)}$

15) $\frac{6x}{3} + \frac{7}{15x+3}$

$\frac{30x^2 + 6x + 7}{3(5x+1)}$

16) $\frac{5v}{v-3} + \frac{5}{v+6}$

$\frac{5v^2 + 35v - 15}{(v+6)(v-3)}$

17) $\frac{4x}{x^2 + 4x - 5} - \frac{5}{4}$

$\frac{-4x - 5x^2 + 25}{4(x+5)(x-1)}$

18) $\frac{2}{x+3} - \frac{6x}{2x+1}$

$\frac{-14x + 2 - 6x^2}{(2x+1)(x+3)}$

19) $\frac{4x}{x+3} - \frac{4x}{x+6}$

$\frac{12x}{(x+3)(x+6)}$

20) $\frac{2x}{3x+3} - \frac{2}{x+5}$

$\frac{2x^2 + 4x - 6}{3(x+1)(x+5)}$

21) $\frac{6}{x-2} + \frac{6}{x+1}$

$\frac{12x - 6}{(x+1)(x-2)}$

22) $\frac{v-2}{3v^4 - 15v^3 - 18v^2} + 3v$

$\frac{9v^5 - 45v^4 - 54v^3 + v - 2}{3v^2(v+1)(v-6)}$

Create your own worksheets like this one with **Infinite Algebra 1**. Free trial available at KutaSoftware.com

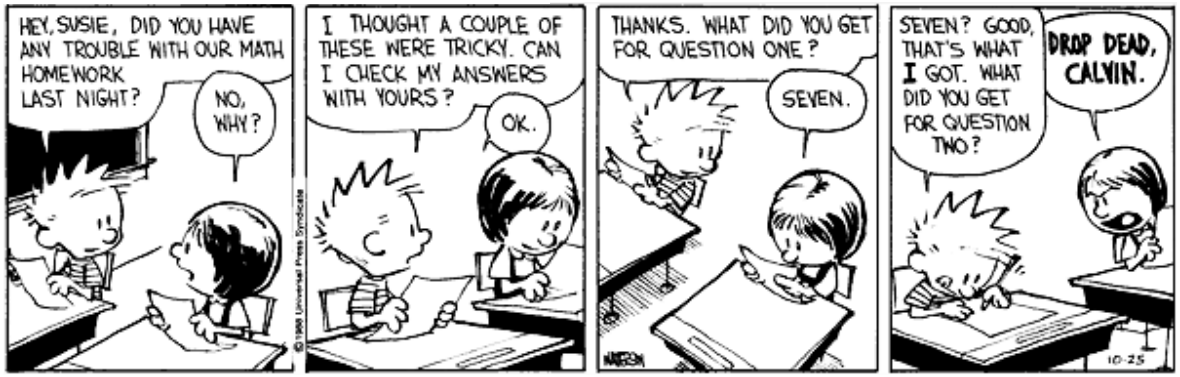
Name: _____ Period: _____

Riddle me this!

Directions: Simplify each expression below by adding, subtracting, multiplying and dividing.

Show all your work to receive full credit.

Color each time you find your answer below!



"IF YOU'RE ALWAYS RIGHT, WHAT IS YOUR SINE?"

Can't simplify	$x+8$	$\frac{1}{x+4}$	$\frac{2x-3}{x+2}$	$\frac{36x}{(x+6)(x+3)}$	1	$\frac{2x^2+4x+6}{3(x+1)(x+5)}$	$\frac{2x}{3}$	$x(x+5)$	$\frac{2x^2+4x+6}{3(x+1)(x+5)}$
$\frac{2x-3}{x+2}$	$x(x+5)$	$\frac{1}{(x-3)(x-2)}$	$\frac{12x}{(x+3)(x+6)}$	$x+8$	$\frac{x+8}{(x+4)^2}$	$\frac{2x-3}{x+2}$	$\frac{1}{x+2}$	$\frac{x+4}{(x+2)^2}$	$\frac{36x}{(x+6)(x+3)}$
$x(x+5)$	$\frac{2x-3}{x+2}$	$\frac{x-10}{x^2-4}$	$\frac{2x}{3}$	$\frac{2x}{3}$	Can't simplify	$\frac{1}{x+2}$	1	$\frac{2x^2+4x-6}{3(x+1)(x+5)}$	0
1	$\frac{2x-3}{x+2}$	Can't simplify	$\frac{2x^2+4x-6}{3(x+1)(x+5)}$	0	$\frac{1}{x+4}$	$\frac{36x}{(x+6)(x+3)}$	$\frac{x-2}{5}$	$\frac{2x-3}{x+2}$	$\frac{x+8}{(x+4)^2}$
$\frac{2x}{3}$	$x+8$	$\frac{2x^2+4x-6}{3(x+1)(x+5)}$	$\frac{2x-3}{x+2}$	$x(x+5)$	1	$\frac{2x^2+4x+6}{3(x+1)(x+5)}$	$\frac{2}{3x}$	$\frac{1}{(x-3)(x-2)}$	Can't simplify
$\frac{x+4}{(x+2)^2}$	$\frac{2x-3}{x+2}$	$x(x+5)$	$\frac{2x}{3}$	$\frac{x-10}{x^2-4}$	$\frac{1}{x+2}$	$x(x+5)$	$\frac{2x}{3}$	$\frac{1}{(x-3)(x-2)}$	$x+8$
0	$\frac{x-10}{x^2-4}$	$\frac{1}{x+4}$	$x+8$	Can't simplify	$\frac{2}{3x}$	$\frac{x+8}{(x+4)^2}$	$\frac{1}{x+4}$	$\frac{36x}{(x+6)(x+3)}$	1

1. $\frac{x+2}{x^2-9} \cdot \frac{x+3}{x^2-4}$

2. $\frac{x-3}{2x+4} + \frac{3x-3}{2x+4}$

3. $\frac{x+4}{x^2+6x+8} \div \frac{x+2}{x+4}$

4. $\frac{3x-6}{x^2-4} - \frac{2}{(x-2)}$

5. $\frac{x^2+10x+16}{x^2+6x+8} \div \frac{1}{x+4}$

6. $\frac{2x}{3x+3} - \frac{2}{x+5}$

7. $\frac{4x}{x+3} - \frac{4x}{x+6}$

8. $\frac{10x^2-20x}{40x^3-80x^2} \cdot \frac{16x^3+80x^2}{6x+30}$

9. $\left(x^2-2x-35\right) \div \frac{x^2-6x+8}{x^2-x-12} \cdot \frac{x^2-2x}{x^2-4x-21}$

Radicals

A radical is an expression that contains a root. e.g. $\sqrt{24}$ $3\sqrt{10}$

Radical expressions allow us to keep a value in its exact form rather than round to a decimal place.

The default **index** of a radical expression is 2 (this means we are taking the _____ root). We will deal with higher indices later in the course.

The **radicand** is _____

The **coefficient** is _____

$\sqrt{24}$ is an example of an _____ radical.

$3\sqrt{10}$ is an example of a _____ radical.

The following properties are used to simplify radicals:	Note that we **cannot** do the following:
$\sqrt{ab} =$ _____ x _____, $a \geq 0$, $b \geq 0$ $\sqrt{\frac{a}{b}} =$ _____, $a \geq 0, b > 0$	$\sqrt{a+b} \neq \sqrt{a} + \sqrt{b}$ e.g.

We **simplify radicals** in order to create an equivalent expression. **A radical is in simplest form when**:

Condition 1: There aren't any _____ (other than 1) in the radicand

- ➔ Rewrite the radical as a product of two factors, one of which is a perfect square
- ➔ Square root the perfect square factor to turn it into the coefficient of the mixed radical

e.g. $\sqrt{24}$ is not fully simplified; $\sqrt{24} =$

Condition 2: There aren't any _____ in the radicand

- ➔ Square root the numerator and denominator separately

e.g. $\sqrt{\frac{5}{9}}$ is not fully simplified; $\sqrt{\frac{5}{9}} =$

Condition 3: There aren't any _____ in the denominator of a rational expression

- ➔ Multiply the numerator and denominator both by the _____
- ➔ This means we are multiplying by _____; changing form of radical and not its value
- ➔ This is called **rationalizing the denominator**

e.g. $\frac{1}{\sqrt{3}}$ is not fully simplified; $\frac{1}{\sqrt{3}} =$

To multiply or divide radical expressions, be sure to multiply/divide _____ and _____.

e.g. $4\sqrt{3} \times 5\sqrt{2}$

e.g. $\frac{64\sqrt{10}-30\sqrt{20}}{12\sqrt{2}}$

Practice: Simplify the following radical expressions.

1) $\sqrt{288} =$

2) $5\sqrt{128} =$

3) $3\sqrt{6} \times 5\sqrt{12} =$

4) $\frac{\sqrt{12}}{\sqrt{24}} =$

5) $\frac{\sqrt{180}}{\sqrt{15}} =$

6) $\frac{12+\sqrt{72}}{4} =$

7) $\frac{\sqrt{12-28}}{2} =$

*Notice that the solution to #7 does not exist in the real number system.

Application Problems Involving Radicals

Recall the **Pythagorean theorem**, which helps us find the hypotenuse in right-angle triangles:

Ex 1: Determine the exact hypotenuse length of a right triangle whose legs measure 20cm and 30cm.

Ex 2: Determine the area of a right triangle whose hypotenuse measures 16cm and one leg measures 10cm.

Application Problems Involving Radicals

Recall the **Pythagorean theorem**, which helps us find the hypotenuse in right-angle triangles:

Ex 1: Determine the exact hypotenuse length of a right triangle whose legs measure 20cm and 30cm.

Ex 2: Determine the area of a right triangle whose hypotenuse measures 16cm and one leg measures 10cm.

Warm-Up

1. Simplify $\frac{x^2+4x-5}{2x^2-x-1} + \frac{3x^2+6x}{2x^2-7x-4}$.

2. Simplify $\frac{80\sqrt{12}+\sqrt{24}}{14\sqrt{3}}$.

3. The diagonal length of a computer screen is 20 inches. If the screen is 15 inches wide, determine the exact height of the screen.

Operations with Radical Expressions

Steps for adding/subtracting radical expressions:

1) Express the radicals in simplest radical form

2) Combine **like radicals** → same _____ in simplest form
   → Add/subtract coefficients of like radicals

e.g. $\sqrt{18}+\sqrt{48}-\sqrt{8}+\sqrt{27}$

Example 1: Simplify.

a. $4\sqrt{48}-7\sqrt{12}$

b. $3\sqrt{20}-4\sqrt{8}+\sqrt{50}-\sqrt{45}$

Steps for simplifying radical expressions involving mixed operations:

→ Use the distributive property (multiply coefficients together and radicands together)
→ Add or subtract coefficients of like radicals

e.g. $(3\sqrt{2}+4\sqrt{5})(4\sqrt{2}-3\sqrt{5})$

Example 2: Determine the perimeter of a rectangle with length $20+\sqrt{8}$ cm and width $14+\sqrt{18}$ cm.

Example 3: Simplify fully.

a) $(2\sqrt{7}+\sqrt{5})(3\sqrt{7}+2\sqrt{3})$

b) $(4+2\sqrt{3})(4-2\sqrt{3})$

What do you notice about the question and answer for example b) above?

Recall that one of the conditions for a radical to be in simplest form is to ensure that **no radical appears in the denominator** of a rational expression.

e.g. (From lesson 1.5) Simplify $\frac{3}{2\sqrt{5}}$

- **Rationalize the monomial denominator** by multiplying both the numerator and denominator by _____
- Changes the denominator from an _____ number to a _____ number

To simplify expressions that have a binomial radical expression in the denominator:

- Multiply the numerator and denominator by the _____ of the denominator
- This means we are multiplying by _____; changing form of radical and not its value
- The product of conjugate binomials is always _____ (see ex. b)

e.g. Simplify $\frac{3}{2\sqrt{3}-\sqrt{6}}$

Practice: Simplify each expression. Leave your answers as exact values.

a) $\sqrt{200} - 2\sqrt{27} + \sqrt{300} - 12\sqrt{18}$

b) $\left(5\sqrt{12} - \sqrt{45}\right)\left(4\sqrt{20} + \sqrt{3}\right)$

c) $\frac{3\sqrt{2}}{2\sqrt{5}+\sqrt{10}}$

Unit 2:
Quadratic Functions

Review of Key Concepts of Quadratic Relations

The graph of a quadratic relation is called a _____.

The _____ of the quadratic relation is $y = ax^2 + bx + c$.

- In this form, a represents _____ and c represents _____.

The _____ of the quadratic relation is $y = a(x - h)^2 + k$.

- a still represents _____,
- h represents _____ or _____ and
- k represents _____ or _____.

The **maximum/minimum value** of a quadratic is _____

and **where it occurs** is _____.

- A quadratic has a maximum value when _____.
- A quadratic has a minimum value when _____.

The _____ of the quadratic relation is $y = a(x - r)(x - s)$.

- a still represents _____,
- r and s represent the _____ or _____.

Remember, there are two types of problems that you will be asked to answer when it comes to quadratics:

Type 1: Finding the optimal or max/min value of the quadratic relation Key words:	Type 2: Solving the quadratic equation (finding the zeros of the quadratic relation) Key words:

Maximum/Minimum of a Quadratic Function by Completing the Square

Standard form of the equation

$y = ax^2 + bx + c$

→

Vertex form of the equation

$y = a(x - h)^2 + k$

How to change from the standard form to the vertex form

$$y = ax^2 + bx + c$$
$$= a(x^2 + \frac{b}{a}x) + c$$
$$= a\left(x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 - \left(\frac{b}{2a}\right)^2\right) + c$$
$$= a\left(x + \frac{b}{2a}\right)^2 - a\left(\frac{b}{2a}\right)^2 + c$$
$$= a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a} + c$$

Then the maximum/minimum value is _____ and it occurs at _____.

- We have a minimum value when _____
- We have a maximum value when _____

Ex. 1) Complete the square for $y = -3x^2 - 4x + 6$. State the maximum/minimum value of the relation and where it occurs.

Ex. 2) Complete the square for $y = \frac{1}{2}x^2 - 3x + 1$. State the maximum/minimum value of the relation and where it occurs.

Ex. 3) A fireworks display travels in a parabolic trajectory. The following formula relates the height of the fireworks shell, h in metres, with respect to time in seconds, t. Determine the maximum height reached by the fireworks shell.

$$h = -4.9t^2 + 39.2t + 1.5$$

Maximum and Minimum Value of a Quadratic Function by Averaging the Zeros

Standard form of the equation $y = ax^2 + bx + c$ → **(Fully) Factored form of the equation** $y = (dx - r)(ex - s)$	**How to change from the standard form to the factored form** $y = ax^2 + bx + c$ Trinomial factor into the form $y = a(x - r)(x - s)$ • The zeros are $x_1 = r$ and $x_2 = s$ or Trinomial factor into the form $y = (dx - r)(ex - s)$ • The zeros are $x_1 = \frac{r}{d}$ and $x_2 = \frac{s}{e}$

- Average the zeros to find $x_v = \frac{x_1 + x_2}{2}$ then substitute x_v into the equation to find y_v
  - ➢ Then the vertex $(h, k) = (x_v, y_v)$

Note: if $y = ax^2 + bx$ (i.e. there is no "c" value), then $x_1 = 0$ and $x_2 = -\frac{b}{a}$

Ex. 1) Find the maximum/minimum value of the relation $y = x^2 + 3x - 10$ using the zeros.

Ex. 2) Find the maximum/minimum value of the relation $y = -4x^2 + 4x + 35$ using the zeros.

Explain why this method will not work every time.

Lesson 2.2 Max and Min Value by Partial Factoring

Warm-Up

1. Determine the maximum/minimum value of the function $y = -4x^2 + 12x - 1$ by completing the square.

2. Determine the maximum/minimum value of the function $y = 2x^2 + 4x - 30$ by averaging the zeros.

Determining the Maximum or Minimum Value of a Quadratic Function by Partial Factoring

Standard form of the equation
$y = ax^2 + bx + c$

→

Partially factored form of the equation
$y = ax\left(x + \frac{b}{a}\right) + c$

How to change from standard form to partially factored form

Factor out ax from the equation $y = ax^2 + bx + c$ → $y = ax\left(x + \frac{b}{a}\right) + c$

Determining the max/min value from partially factored form

1) Set $y = c$ to determine the x-coordinates of the _____ and its _____ – the coordinates of these two points are $(x_1, y_1) =$ and $(x_2, y_2) =$

2) Average the x-coordinates to determine the _____ or the _____

3) Substitute x_v back into the relation to find y_v

Ex. 1) Find the maximum/minimum value of the function $y = 2x^2 - 8x + 5$ using partial factoring.

Lesson 2.2 Max and Min Value by Partial Factoring

Ex. 2) Find the maximum/minimum value of the function $y = -\frac{1}{3}x^2 - 4x - 3$ using partial factoring.

Explain the similarities and the differences between Averaging the Zeros method and Partial Factoring method.

Homework

5. For each quadratic relation,
   - **i.** use partial factoring to find two points that are equidistant from the axis of symmetry
   - **ii.** find the coordinates of the vertex
   - **iii.** express the relation in vertex form
   - **iv.** sketch the graph

   (a) $y = x^2 - 6x + 5$ (b) $y = x^2 - 4x - 11$

   (c) $y = -2x^2 + 12x - 11$ (d) $y = -x^2 - 6x - 13$

   (e) $y = -\frac{1}{2}x^2 + 2x + 3$ (f) $y = 2x^2 - 10x + 11$

6. **Communication:** Write down the steps you would take to find the coordinates of the vertex of the quadratic relation $y = -3x^2 + 12x + 29$. Don't actually find the vertex, just explain the steps in your own words.

5. (a) (0, 5), (6, 5), (3, −4), $y = (x - 3)^2 - 4$
   (b) (0, −11), (4, −11), (2, −15), $y = (x - 2)^2 - 15$
   (c) (0, −11), (6, −11), (3, 7), $y = -2(x - 3)^2 + 7$
   (d) (0, −13), (−6, −13), (−3, −4), $y = -(x + 3)^2 - 4$
   (e) (0, 3), (4, 3), (2, 5), $y = -\frac{1}{2}(x - 2)^2 + 5$
   (f) (0, 11), (5, 11), (2.5, −1.5), $y = 2(x - 2.5)^2 - 1.5$

Warm-Up

1. Determine the maximum/minimum value of the function $y = -\frac{1}{2}x^2 + 3x + 2$ by partial factoring.

Quadratic Relations – Optimization Problems

An overall strategy for answering optimization problems:

1) **Understand the problem.**
   - ➔ Identify the quantities that can change, the _____ variable
   - ➔ Identify the quantity to be optimized, the _____ variable
   - ➔ Introduce variables using 'let' statements
   - ➔ Whenever possible, draw a diagram labeling all quantities.

2) **Write a quadratic relation** using the two variables you introduced.
   - ➔ Use the information in the problem to write the variable to be optimized in terms of only one other variable

3) **Find the maximum or minimum** of this relation using a suitable method
   - ➔ Completing the square and partial factoring always work
   - ➔ Factoring and averaging the zeroes does not always work but is best when...

4) Answer the question → write a **therefore statement** and include appropriate units.

Number Type

Ex. 1) The sum of two numbers is 20. What is the least possible sum of their squares?

#1	#2	Sum of squares

Ex. 2) The sum of two numbers is 10. What are the numbers if the sum of their squares is a minimum?

Area / Perimeter Type * Draw a diagram!

Ex. 1) A rectangular field is to be enclosed with 600m of fencing. What is the maximum area that can be enclosed?

Ex. 2) A farmer wants to enclose a rectangular field against the side of his barn and subdivide it to keep his horses and cows separate. He has only 300m of fencing material. Find the dimensions of the field that he should make to optimize its area.

Revenue / Profit / Cost Type

In business, revenue is the total amount of money taken in from sales. In the case of selling an item,

Revenue = Price per item × Number of items sold

Ex. 1) A group of students are organizing a school dance. From past dance-hosting experiences, they know that if they charge $8 per ticket, they will sell about 200 tickets. They poll the students and discover that for each 25-cent price reduction, 20 more students would attend. What is the maximum revenue?

n	Cost per ticket	# of tickets sold	Revenue

Ex. 2) The student council plans to run the annual talent show to raise money for charity. Last year, tickets sold for $10 each, and 400 people attended. The student council has determined that for every $1 increase in ticket price, attendance decreases by 20 people. What ticket price would maximize the revenue?

Solving Quadratic Equations

Part 1: What's the difference?

	Quadratic relation	Quadratic equation
Standard form	$y = ax^2 + bx + c, \ a \neq 0$	$0 = ax^2 + bx + c, \ a \neq 0$
Definition/ description	x: y:	
What we can do with it	Determine x-intercepts/zeros	Solve for the roots
	Determine the max/min value	
	Determine the y-value for a given x-value (or vice versa)	
	Determine the features of the parabola, such as y-intercept and vertex	

Part 2: Solving a quadratic equation

To solve a quadratic equation means to find the value(s) of the variable that will make the statement true so that the left side of the equation equals the right side.

Method #1: Isolating x

When there is only one x in the equation, use reverse BEDMAS to isolate x.

Ex. 1) Solve the equation $-25(x-3)^2 + 100 = 0$

Method #2: Factoring

If there is more than one term with x:

1) Expand and simplify the equation to write it in standard form, $0 = ax^2 + bx + c$.
2) Then, factor if possible → when $b^2 - 4ac$ is a _____

Ex. 2) Solve the equation $2x^2 + 8 = 5 - 7x$.

Method #3: Using the Quadratic Formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

If there is more than one term with x:

1) Expand and simplify the equation to write it in standard form, $0 = ax^2 + bx + c$.
2) Then, if factoring is not possible, use the quadratic formula to solve the equation.
3) Leave your answers as exact values, in fully simplified radical form.

*The quadratic formula is derived from completing the square on the standard form of the equation and then isolating for x.

Ex. 3) Solve the equation $4x^2 + 8x - 6 = 0$. Leave your answers in exact radical form, fully simplified.

If a question does not specify that you use one of the methods, you can choose which is best under the circumstances.

When would it be best or most efficient to use each of the solving methods?

Isolating x	Factoring	Using the Quadratic Formula

Solving Quadratic Equations Homework

1. $m^2 - 5m - 14 = 0$
2. $7r^2 - 14r = -7$
3. $k^2 - 31 - 2k = -6 - 3k^2 - 2k$
4. $3 - 4x^2 = -85$
5. $5x^2 - 44x + 120 = -30 + 11x$
6. $8n^2 = 4 + 6n$
7. $8b^2 - 7 = 193$
8. $-4p(p + 2) - 3 = -3 - 5p^2$
9. $4(3 - x)^2 - 24 = 0$
10. $15a^2 - 3a = 3 - 7a$

Answers:

1. 7; –2
2. 1
3. $\frac{5}{2}; -\frac{5}{2}$
4. $\pm\sqrt{22}$
5. 6; 5
6. $\frac{3\pm\sqrt{41}}{8}$
7. 5; –5
8. 8; 0
9. $3 \pm \sqrt{6}$
10. $\frac{1}{3}; -\frac{3}{5}$

Applications Involving Quadratic Equations

Review of key concepts:
To solve a quadratic equation, such as $3 = x^2 - 2x$, means to _____

_____.

The three methods we used to solve a quadratic equation were:

1) Use reverse BEDMAS to isolate for x when _____
2) Factor when _____
3) Use the quadratic formula when _____

Part 1: Applications involving quadratic equations

Ex. 1) A building measuring 90m by 60m is to be built. A paved sidewalk of uniform width will surround the building. If the sidewalk is to have an area of 9000m^2, determine how wide it will be.

Problem solving strategy: Use a KWH chart.

What do we know?	What do we want to know?	How can we find we need?

Ex 2) Tickets to a rock concert cost $80 and there are 450 attendees. For every $5 increase in ticket price, there are 15 fewer attendees. If the revenue for a particular concert was $39,000 determine how much each ticket cost and how many people attended the concert.

Problem solving strategy: Using the Problem Solving Framework

1) Understand the problem - *Re-read the question; put it into your own words;* - *use pictures or diagrams;* - *identify the problem; underline/highlight important information;* - *ask clarification questions*	
2) Make a plan - *Make connections to similar problems;* - *use pictures or diagrams;* - *write an algebraic model* - *discuss with a partner*	
3) Carry out the plan - *Apply the strategy(ies) chosen to answer the question*	
4) Look back at the solution - *Check to see if all important information was used;* - *check calculations for errors;* - *check to see if answer is reasonable*	

Part 2: Try it on your own!

Ex 3) Two numbers have a difference of 10. Determine the integers if the sum of their squares is 68.

Part 3: What's the difference?

Fill in the chart below to compare the differences in the questions being asked.

Ex. 4) A DVD production company models their profit by the quadratic relation $P = -2d^2 + 28d - 90$, where d is the number of DVDs sold in hundred thousands and P is the profit in millions of dollars.

Question	a) Determine how many DVDs the company must sell in order to make the maximum profit possible.	b) Determine how many DVDs the company must sell in order to make a profit of 6 million dollars.
What are you being asked to determine? How do you know?		
How will you answer the question? Which lesson(s) can you refer to for similar examples?		
Solution		

Lesson 2.5 Applications Involving Quadratic Equations

What am I trying to find?

Used for Optimization Questions

Used to Solve a Quadratic Equation

Vertex

COMPLETING THE SQUARE

Works when relation is in Standard Form.

Ex: $y = 2x^2 - 4x + 5$

$y = (2x^2 - 4x) + 5$

$y = 2(x^2 - 2x) + 5$

$y = 2(x^2 - 2x + 1 - 1) + 5$

$y = 2(x - 1)^2 - 2 + 5$

$\mathbf{y = 2(x - 1)^2 + 3}$

This is vertex form. The vertex is (+1, +3)

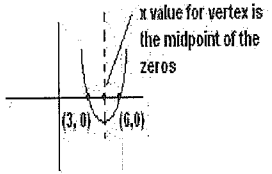
*Converts standard form into vertex form

AVERAGING THE ZEROS

Works when relation can be written in factored form y=a(x-r)(x-s), then x-coordinate of vertex is $\frac{r+s}{2}$

Ex: y = (x-3)(x-6)

0 = (x-3)(x-6)



$x = \frac{3+6}{2}$

Now substitute x into original equation to find y for the vertex.

PARTIAL FACTORING

Works when relation is in standard form. Only the first two terms are factored while y is set to c.

Ex: $y = 2x^2 - 4x + 5$

$5 = 2x(x - 2) + 5$

$0 = 2x(x - 2)$

$2x = 0$ OR $x - 2 = 0$

$x = 0$ $x = 2$

These are x coordinates of the y-intercept and its mirrored point. Average of these two x-coordinates will give x-coordinate of vertex:

$\frac{0+2}{2} = 1$

Substitute x=1 into original equation to find y-coordinate of vertex.

NOTE: We never use partial factoring to find zeros

Roots

FACTORING

Works when you can factor the polynomial.

Always Common Factor first!!!

Ex. $0 = 2x^2 - 2x - 12$

$0 = 2(x^2 - x - 6)$

Simple Trinomial

Ex. $0 = x^2 - 9x + 18$

$0 = (x - 3)(x - 6)$

$x = 3$ or 6

Difference of Squares

Ex. $0 = x^2 - 16$

$0 = (x - 4)(x + 4)$

$x = 4$ or -4

Complex Trinomial

Ex. $0 = 2x^2 + 5x - 3$

$0 = (2x-1)(x+3)$

$x = ½$ or -3

QUADRATIC FORMULA

Works when you can't factor and equation is in standard form.

Leave answer as an exact value in fully simplified radical form

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Ex. $0 = x^2 - 4x - 10$

ISOLATING FOR X

Works when you are given the equation with only one x term

Ex. $0 = 4(5x+6)^2 - 9$

Using the Discriminant to Determine the Number of Zeros

The **zeros** of a quadratic relation are the **x-intercepts** of the quadratic relation.

- To find the zeros, we set y to 0
- This changes our quadratic relation, $y = ax^2 + bx + c$, into a quadratic equation, $0 = ax^2 + bx + c$, which we can then solve using one of the three methods!

Function	Value of the discriminant $b^2 - 4ac$	How many zeros?	Rough sketch of graph
$y = -x^2 - 2x + 9$			
$y = 2x^2 + 5x + 1$			
$y = 2x^2 - 12x + 18$			
$y = -9x^2 + 12x - 4$			
$y = -2x^2 - 4x - 3$			
$y = 3x^2 + 4$			

Part 3: The relationship between the value of the discriminant $b^2 - 4ac$ and the number of zeros

Given a quadratic relation in standard form, $y = ax^2 + bx + c$,

- When $b^2 - 4ac > 0$, there are _____ zeros. → Same as, there will be _____ different values for x when we use the quadratic formula to determine the x-intercepts.

- When $b^2 - 4ac = 0$, there are _____ zeros. → Same as, there will only be _____ value for x when we use the quadratic formula to determine the x-intercept.

- When $b^2 - 4ac < 0$, there are _____ zeros. → Same as, the quadratic formula will yield an _____ answer for x.

Part 4: Let's make some more connections!

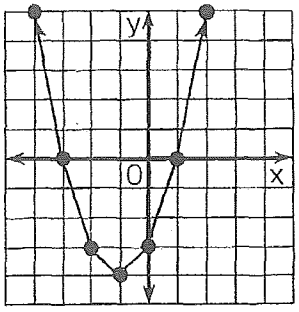
- If a is positive and c is negative, there will always be _____ zeros

- If a is negative and c is positive, there will always be _____ zeros

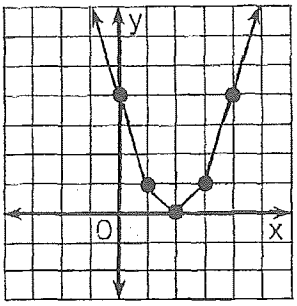
The number of zeros of a quadratic relation is related to the **nature of the roots** of the **quadratic *equation***. We will learn about this in an upcoming lesson.

How Can You Help Control Soil Erosion?

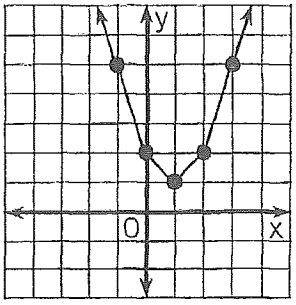
Use the related graph or the discriminant of each equation to determine how many real-number solutions it has. Circle the letter of the correct choice and write this letter in the box containing the exercise number.



(1) $x^2 + 2x - 3 = 0$
(D) two solutions
(E) one solution
(M) no solutions



(2) $x^2 - 4x + 4 = 0$
(C) two solutions
(A) one solution
(W) no solutions



(3) $x^2 - 2x + 2 = 0$
(H) two solutions
(D) one solution
(O) no solutions

	two solutions	one solution	no solutions
(4) $x^2 + 5x + 4 = 0$	K	B	G
(5) $x^2 - 3x = 2$	U	O	A
(6) $y^2 + 10y + 25 = 0$	V	A	I
(7) $2x^2 = 4x - 3$	F	C	H
(8) $4x^2 + 9 = 12x$	S	P	N
(9) $-3n^2 + 5n - 2 = 0$	N	R	S
(10) $\frac{1}{2}x^2 + 3x + 8 = 0$	R	P	L
(11) $\frac{1}{3}t^2 + 3 = 2t$	Y	B	T

7	3	10	1	5	8	2	11	6	9	4

OBJECTIVE 4–f: To use the related graph or the discriminant of a equation to determine how many real-number solutions it has.

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Using the Discriminant to Determine the Number of Zeros and the Nature of the Roots

Part 1: Review of key concepts

- A quadratic relation is _____.
- A quadratic equation is _____.

A quadratic relation may have _____, _____, or _____ zeros, which are the number of x-intercepts of the parabola.

A quadratic equation is a polynomial of degree _____.

➔ This means that there must always be _____ solutions to a quadratic equation.

➔ The nature of the roots of the quadratic equation tells us the types of solution we have.

Recall that we used the discriminant to find the number of zeros/x-intercepts a quadratic relation has. We can also use the discriminant to determine the nature of the roots of a quadratic equation.

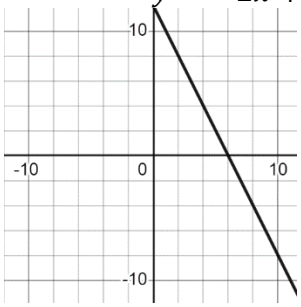
	Quadratic Relation	**Quadratic Equation**
Standard Form		
If the discriminant is…		
Positive $b^2 - 4ac > 0$		
Zero $b^2 - 4ac = 0$		
Negative $b^2 - 4ac < 0$		

Ex. 1)

Determine the number of x-intercepts that $y = 3x^2 - 4x + 5$ has.	Determine the nature of the roots of $0 = 3x^2 - 4x + 5$.

Part 2: Linear Equations vs Linear Inequalities

Inequalities are expressions that include any of the following: ≤ < ≥ >

Solve the following **linear equation**:	Solve the following **linear inequality**:	Do your solutions from 1) and 2) make sense based on the graph?
1) $-2x + 12 = 0$	2) $-2x + 12 > 0$	$y = -2x + 12$ 

Summary: Solving Linear Inequalities

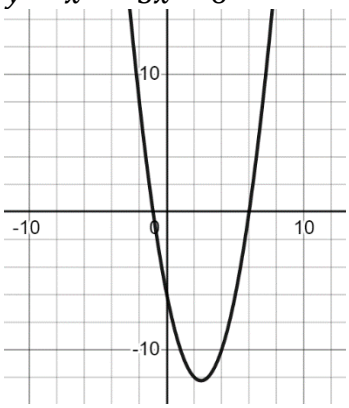
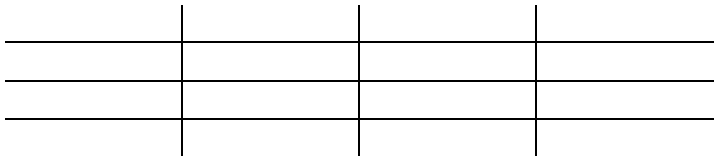
The rules for solving an equation and solving an inequality are the same except that when we multiply or divide both sides of an inequality by a negative number, we must _____ the inequality sign. The solution to an inequality is a range of values.

e.g. Solve the inequality $2x - 4 > 3x + 5$.

Lesson 2.7 – Solving Discriminant Problems

Part 3: Solving Quadratic Inequalities

Solve the inequality $x^2 - 5x - 6 \geq 0$:

Graph of relation $y = x^2 - 5x - 6$  	**Solve $x^2 - 5x - 6 \geq 0$ graphically:** Look at where the graph of the relation is _____ or _____ the x-axis (where the y-values are equal to/greater than 0). State the x-value interval(s):
Solve $x^2 - 5x - 6 \geq 0$ algebraically, using intervals: 1) Factor and solve the equation → determine the roots of the equation 2) Determine whether each factor is positive or negative in the intervals separated by the roots. 3) Determine whether the overall product of the factors in each interval is positive or negative. 4) Answer the inequality given by stating the correct x-value interval(s).	

Practice: Solve the inequality $x^2 - 4 < 0$.

Part 4: Solving problems involving the discriminant

Ex. 1) Determine the value(s) of k for which the quadratic relation $y = x^2 + kx + 9$ will have:
a) two x-intercepts, b) one x-intercept, and c) no-x-intercepts.

Ex. 2) Determine the value(s) of k for which the quadratic equation $2x^2 + kx = 4$ has no real roots.

Recall that given $y = ax^2 + bx + c$, when the signs of a and c are different, we will always have _____ zeroes/x-intercepts.

In the same manner, given $0 = ax^2 + bx + c$, when the signs of a and c are different, we will always have _____ roots.

Classwork/Homework:

7. For what value(s) of k does the function $f(x) = kx^2 - 4x + k$ have no zeros?
8. The function $g(x) = 4x^2 - 4x + m$ has exactly one zero. What is the value of m?
9. For what values of k does the equation $3x^2 + 4x + k = 0$ have no real roots? one real root? two real roots?
10. The graph of function $f(x) = x^2 - kx + k + 8$ touches the x-axis at one point. What is the value of k?

7. $k < -2$ or $k > 2$
8. $m = 1$
9. (a) $k > \frac{4}{3}$ (b) $k = \frac{4}{3}$ (c) $k < \frac{4}{3}$
10. $k = -4$ or $k = 8$

Determining the Algebraic Model of a Quadratic Function

Recall: The algebraic expression of a quadratic function may be written in different forms:

1) Standard form: $y = ax^2 + bx + c$
2) Vertex form: $y = a(x - h)^2 + k$
3) Factored form: $y = a(x - s)(x - t)$

If we are asked to find the algebraic model of a quadratic function we can use any of these forms (unless specifically asked for a particular one). Remember, 'find the algebraic model' means you need to **find the equation**.

Generally, we will be given the zeros of a quadratic function. **All quadratic functions with the same zeros are said to be members of the same _____ of quadratic functions**. Specific family members will pass through a specific third point.

Ex. 1) The zeros of a family of quadratic functions are -3 and 5.

a) Determine the factored form of the algebraic model that represents this family of quadratic functions. Then, sketch a few different parabolas that model this family of quadratic functions.

b) Determine the algebraic model of the specific parabola (which is a member of the above family) that crosses the y-axis at -75.

c) Determine the algebraic model, in standard form, of the specific family member that passes through the point (2, 12).

Ex. 2) Determine the algebraic model of the family of quadratic functions if the roots of the corresponding quadratic equation are $1 \pm \sqrt{5}$. Then, determine the specific family member whose parabola passes through (2, 5).

Ex. 3) Determine the quadratic relation that has zeros of -3 and 11 and a minimum y-value of -35.

Solving Linear-Quadratic Systems

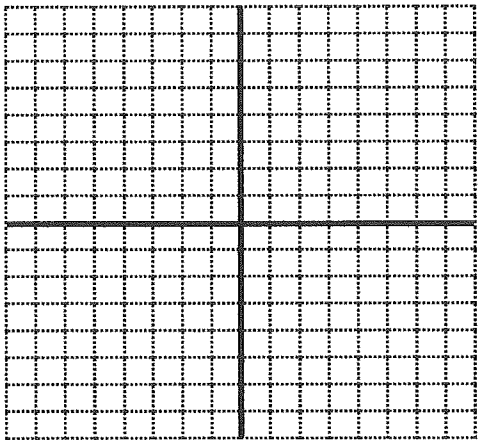
Review: Solving a linear system

Recall that a linear system is _____ linear relations. To solve a linear system means to find the _____ of all the lines.

Ex. 1) Solve the linear system $y = x + 3$
$2x + y = 6$

i) Graphically

ii) Algebraically by Substitution

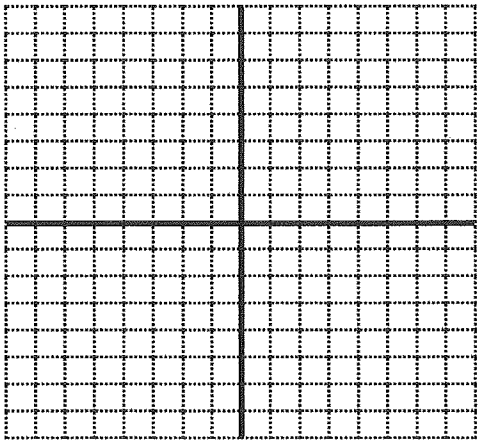


Solving a linear-quadratic system

Now, let us consider a linear-quadratic system. This system is made up of a _____ and a _____. To solve a linear-quadratic system means to find the _____ between the line and the parabola.

Ex. 2) Solve the system $y = x^2 - 6x + 5$
$y = -4x + 5$

a) Graphically.



b) By substitution. $y = x^2 - 6x + 5$
$y = -4x + 5$

Ex. 3) Solve the system algebraically.

$y = 2x - 1$
$y = 5x^2 - 4$

Possible Solutions for a Linear-Quadratic System of Equations

What are the possible arrangements for how a line might intersect a parabola? For example, when solving a linear-quadratic system, is it possible to have 4 solutions? Show using diagrams.

Using the Discriminant to Determine the Number of Solutions

For each of the following, determine the number of solutions to the system; do not solve for the points of intersection.

a) $y = 2x - 1$
$y = 5x^2 - 4$

b) $y = 10$
$y = -2x^2 + 4x + 3$

c) $y = -x - 4$
$y = x^2 + x - 3$

Steps to determining how many solutions (P.O.I.s) a linear-quadratic system will have:

1)

2)

3)

4)

Ex. 4) Determine the value of k such that the parabola modeled by $y = -x^2 + 4x + k$ will intersect with the line $y = 8x - 2$ at a) two points; b) one point, and c) no point.

Ex. 5) *(From textbook p.69 #15)*

A parachutist jumps from an airplane and immediately opens his parachute. His altitude, h_1, in metres, after t seconds is modeled by the linear relation $h_1 = -4t + 300$. A second parachutist jumps 5 seconds later and freefalls for a few seconds. Her altitude during this time, h_2, in metres, is modelled by the quadratic relation $h_2 = -4.9(t - 5)^2 + 300$.

When does the second parachutist reach the same altitude as the first parachutist?

Fill in the KWH chart below to help organize your thoughts. Then solve the problem in the space below the chart.

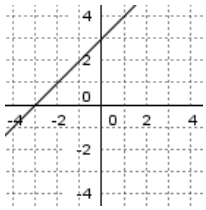
What do we **k**now?	What do we **w**ant to know?	**H**ow can we solve the problem?

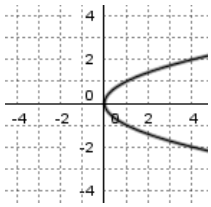
Unit 3: Functions

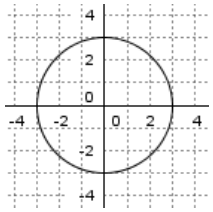
Relations vs Functions

A **relation** is a set of _____.

e.g. ① $y = x + 3$ ② $x = y^2$ ③ $x^2 + y^2 = 9$







④ $\{\ldots(-3,2),(-2,1),(-1,0)\ldots\}$

⑤ $\{(-4,1),(-4,2),(0,6),(5,2)\}$

A **function** is a relation in which ***no*** two ordered pairs have the _____ first coordinate.

For every *x-value* (independent set of numbers), there is ONLY one _____ corresponding *y-value* (dependent set of numbers).

e.g.

This means that **all functions are relations** but some relations are not functions.

There are several **ways of representing functions**.

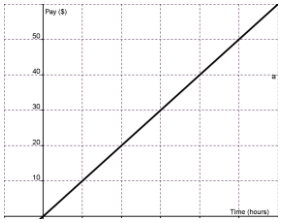
Example: *Kevin has a part-time job that pays $10/hr.*

Table of Values

Time	Pay
0	0
1	10
2	20
3	30
4	40
5	50

Set of Ordered Pairs

$\{(0,0),(1,10),(2,20),(3,30),(4,40),(5,50)\}$



Graph

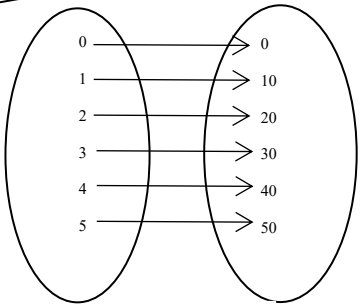
Function

$P = 10t$

Equation

$P(t) = 10t$

FUNCTION NOTATION



Mapping Diagram

Lesson 3.1 Relations vs Functions

Given a relation, how can we tell whether or not it is a function? Here are a couple of ways to **test for functions**:

1. **Algebraic Test:** If a relation is a function, substituting any value for x will result in _____ value of y.

Given an equation, isolate for y. If a value of x can be found which produces more than one value of y, then the equation does not represent a function.

2. **Vertical Line Test:** If a relation is a function, a _____ drawn anywhere on the graph will cross the graph at exactly one point only.

Examples: Which of the following are functions. Justify your answer.

a) $\{(1, 3), (2, 3), (3, 4), (4, 4) \ldots\}$

b) $\{(1, 3), (1, 4), (1, 5), (1, 6)\}$

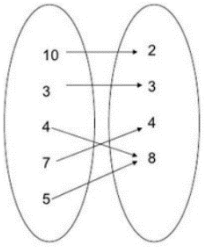
c) $x = y^2$

d) $y = 3x - 8$

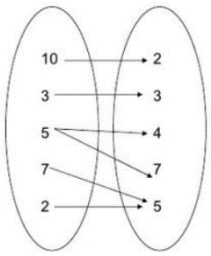
e) $y = x^2$

f) $x^2 + y^2 = 25$

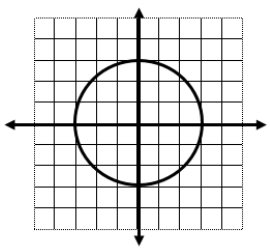
g)



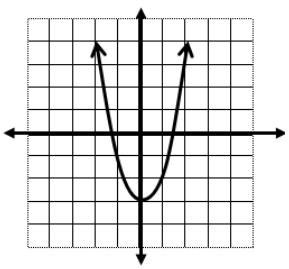
h)



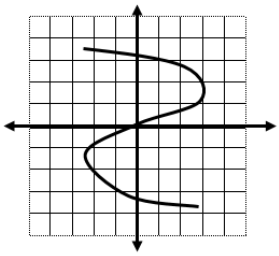
i)



j)



k)



Using Function Notation f(x)

Function notation is just another way of representing functions algebraically. Note that function notation is reserved specifically for functions; so, we should not use f(x) to represent a relation that is not a function.

For example: the function $y = x^2 + 3$ can be written using function notation: $f(x) = x^2 + 3$.

Example: Write $y = 3x^3 + 2x$ in function notation. _____

Example: Given two functions f and g where $f(x) = 2x + 3$ and $g(x) = x^2$, find...

a) f(1)

b) g(-4)

c) f(0)

d) f(3x -1)

e) g(a + 1)

f) g(2) + f(7)

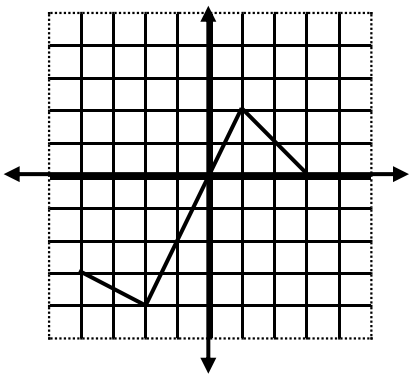
g) f(g(-4))

h) the solution to f(x) = 7

i) the solution to g(x) = 8

Lesson 3.1 Relations vs Functions

Example: Use the graph of $y = f(x)$ below to find



a) f(2)

b) f(–3)

c) f(0)

d) the solution to $f(x) = -4$

e) the solution to $f(x) = 0$

The Domain and Range of a Function

Description/set of numbers	Number set is called	Represented by
$\{1, 2, 3, 4, \dots\}$		
Natural numbers and 0 $\{0, 1, 2, 3, \dots\}$		
Positive and negative whole numbers $\{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$		
Division of any two integers $\frac{a}{b}$, $b \neq 0$ (positive and negative repeating or terminating decimals) e.g. $\frac{3}{7}$, 0.128, $-\frac{4}{5}$		
Numbers that cannot be expressed as a fraction of two integers; any non-repeating, non-terminating decimal value e.g. $\sqrt{3}$, π, $-7\sqrt{5}$		
All rational and irrational numbers		

We need to be familiar with number sets in order to state the domain and range of functions.

Domain is the set of all _____ of a relation; the set of all numbers upon which the relation is defined.

Range is the set of all _____ of a relation; the set of all numbers that are associated with an element of the domain.

We state the domain and range of relations using **formal set notation**. In order to determine the domain and range of a function or relation, determine the smallest and largest possible values of x and y. Then, write the domain and range as

Continuous, Discontinuous and Discrete Functions

Quadratic relations are examples of **continuous functions**, where the domain and range is always the set of real numbers (or a smaller set that does not have gaps, such as $\{x \leq 3 | x \in \mathbb{R}\}$). The graphs of continuous functions do not have any _____.

Another type of functions are **discontinuous functions**, where there are value(s) where the function is not defined; these functions still have a smaller set of real numbers as their domain and range, but there may be a gap.

e.g. a function that has $\{-5 \leq x \leq 3, x \neq 0 | x \in \mathbb{R}\}$ as its domain would be a discontinuous function.

A third type of functions are **discrete functions**. These functions are made up of separate points that are not connected.

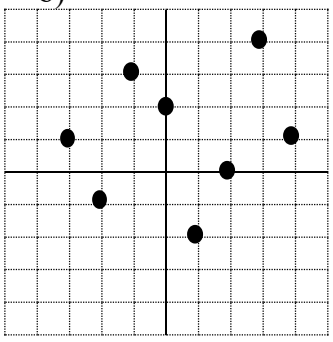
Examples: In each of the following cases, $f(x)$ represents a function.

i. Classify whether the relation is continuous, discontinuous, or discrete.
ii. State the domain and range of $f(x)$ using formal set notation.

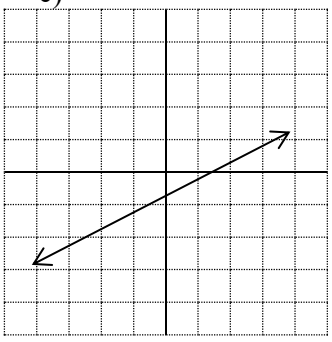
a) $f(x) = \{(-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4)\}$

*Scale: 1 tick = 1 unit

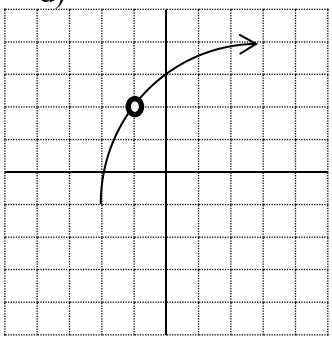
b)



c)



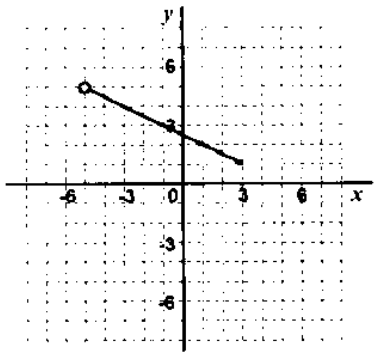
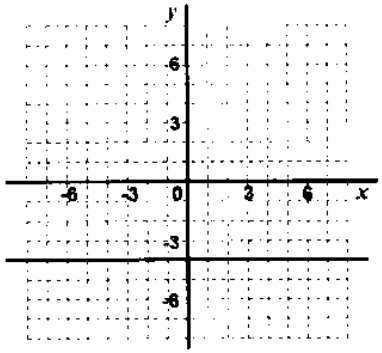
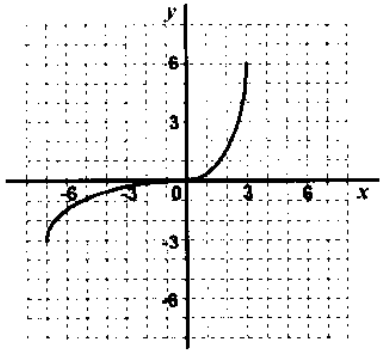
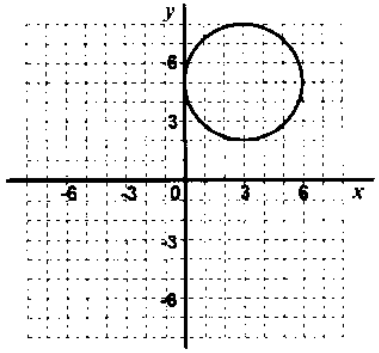
d)



Domain & Range

Date:

State the Domain and Range of each of the given relations in the space provided. ***Assume that graphs drawn to the edge of the grid continue on infinitely.***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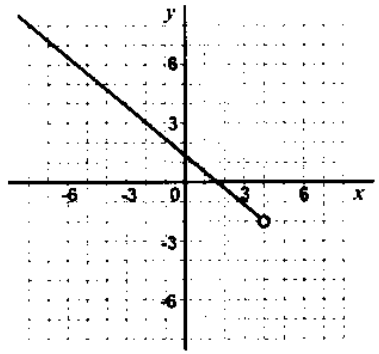
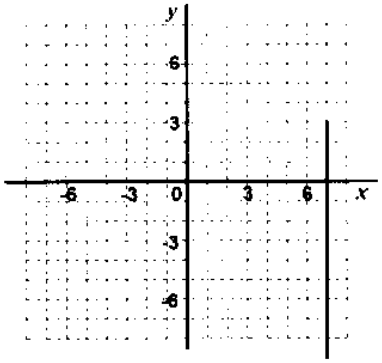
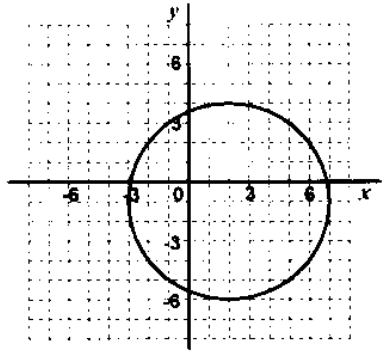
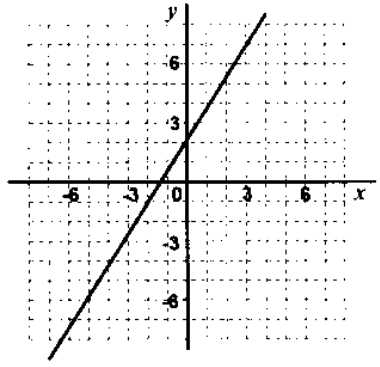
R:	***R:***	***R:***	***R:***
			
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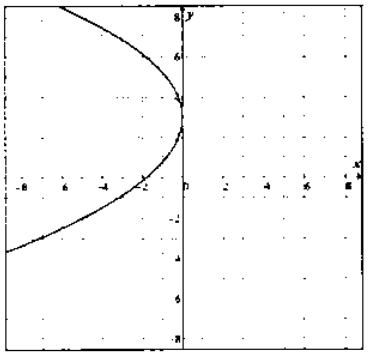	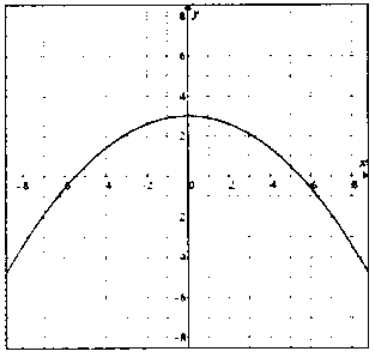	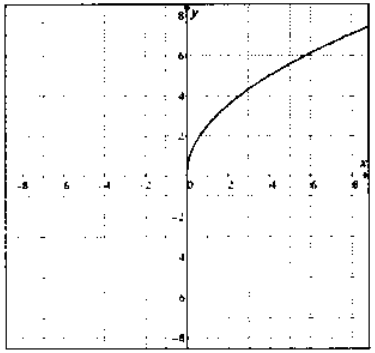	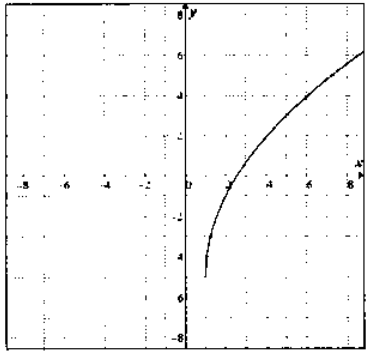
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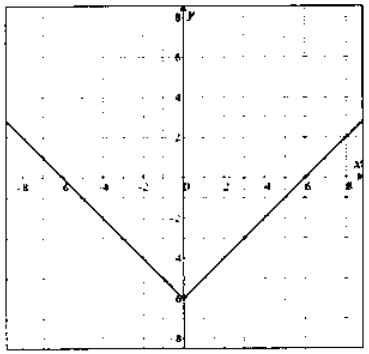	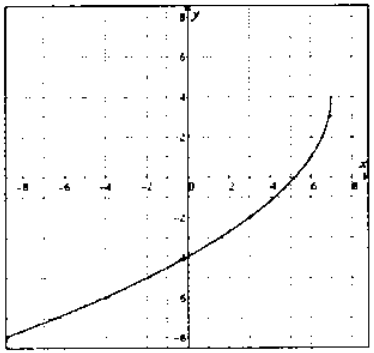	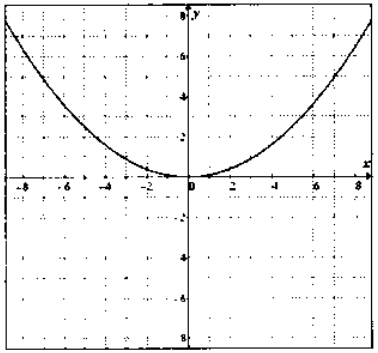	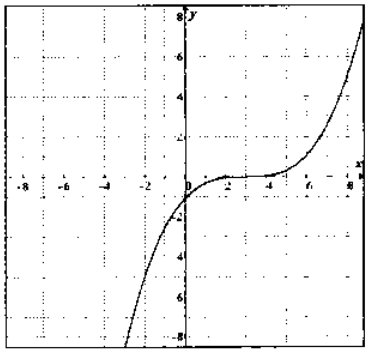
D:	***D:***	***D:***	***D:***
R:	***R:***	***R:***	***R:***

DOMAIN
for **Continuous** Relations

Look at your graph...

Will it keep going to the **LEFT** forever?

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YES

Look at your graph...

Will it keep going to the **RIGHT** forever?

NO

Find the point that's furthest to the right.

What's it's x-value? We'll call it R.

R = _____

The domain is $x \leq R$

YES

The domain is $\mathbb{R}$ (all real numbers)

NO

Find the point that's furthest to the left.

What's it's x-value? We'll call it L.

L = _____

Look at your graph...

Will it keep going to the **RIGHT** forever?

YES

The domain is $x \geq L$

NO

Find the point that's furthest to the right.

What's it's x-value? We'll call it R.

R= _____

Your domain is $L \leq x \leq R$.

Note:

If any of your endpoints have an OPEN POINT (○) or are on a DASHED LINE (- -), then replace the inequality symbols as follows:

$\leq \rightarrow <$ & $\geq \rightarrow >$

RANGE
for **Continuous** Relations

Look at your graph...

Will it keep going **DOWN** forever?

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YES

Look at your graph...

Will it keep going **UP** forever?

NO

Find the point that's furthest to the top.

What's it's y-value? We'll call it T.

T = _____

The range is $y \leq T$

YES

The range is $\mathbb{R}$ (all real numbers)

NO

Find the point that's furthest to the bottom.

What's it's y-value? We'll call it B.

B = _____

Look at your graph...

Will it keep going **UP** forever?

YES

The range is $y \geq B$

NO

Find the point that's furthest to the top.

What's it's y-value? We'll call it T.

T= _____

Your range is $B \leq y \leq T$

Note:

If any of your endpoints have an OPEN POINT (○) or are on a DASHED LINE (- -), then replace the inequality symbols as follows:

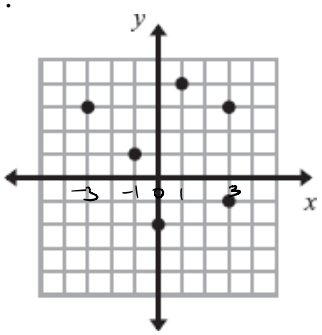
$\leq \rightarrow <$ & $\geq \rightarrow >$

Function Notation and Evaluating Functions Practice Worksheet

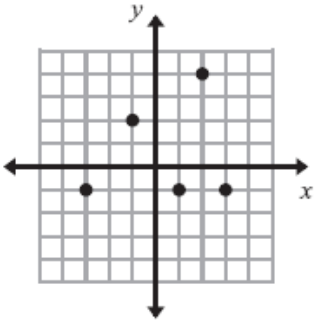
Name_____ Date_____

Decide whether the graph represents y as a function of x. If it is a function, give the domain and range.

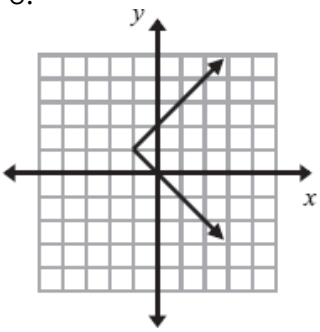
1.



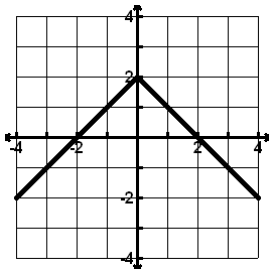
2.



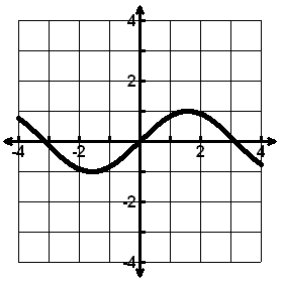
3.



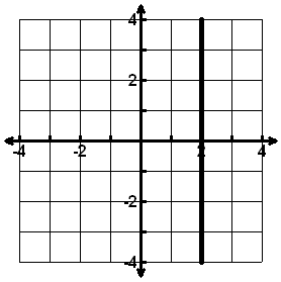
4.



5.



6.



✳ Determine whether each algebraic model represents a function or a relation.

a) $x^2 + y = -x^3 + 4$

b) $x - y^2 = 8$

Decide whether the relation is a function.
If it is a function, give the domain and the range.

7.

Input	Output
1	7
	-7
2	8
	-8

8.

Input	Output
3	2
5	4
7	6

9.

Input	Output
0	-6
2	-4
4	-2
6	0

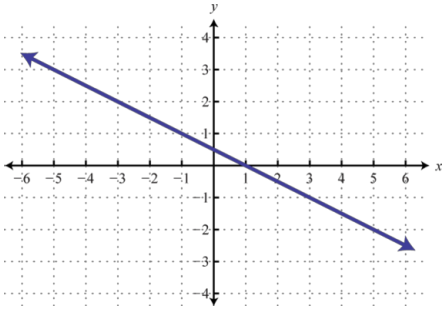
Evaluate the function when $x = 3,\ x = 0,$ and $x = -2$.
(You will have 3 answers for each problem)

10. $f(x) = 2x - 5$

11. $h(x) = 6x + 2$

12. $h(x) = x^3 - 4x$

Evaluate the function using the following graph.



13. **f(– 1) =**_____

14. **f(3)=**_____

15. **f(**_____**)= 0**

16. **f(**_____**)= 3**

If $f(x)=2x-3$, $g(x)=x^3-2$, and $h(x)=x^2-3x+5$, find each of the following:

17. f(4) =

18. h(-3) =

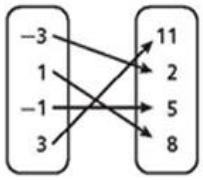
19. g(-2) =

✳ f(g(-2))

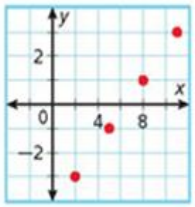
---

20. Which is NOT a correct way to describe the function $\{(-3, 2), (1, 8), (-1, 5), (3, 11)\}$?

Ⓕ



Ⓖ



Ⓗ Domain: {−3, 1, −1, 3}
Range: {2, 8, 5, 11}

Ⓙ

x	y
−3	2
−1	5
1	8
3	11

21. Use the table to answer the following:

x	-3	-1	0	1	3
y	5	7	9	11	13

a. Give the domain and range of the relation.

Domain:

Range:

b. Does the relation represent a function? Explain.

Find the following:

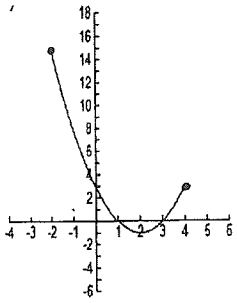
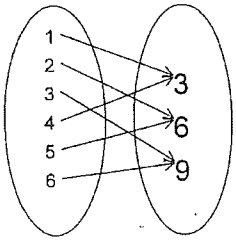
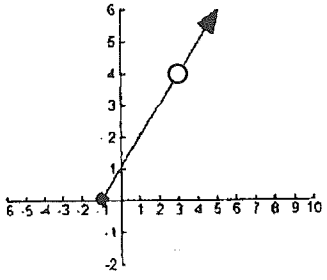
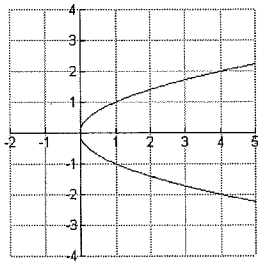
c. f(0)=_____

d. f(3)=_____

e. f(_____)= 7

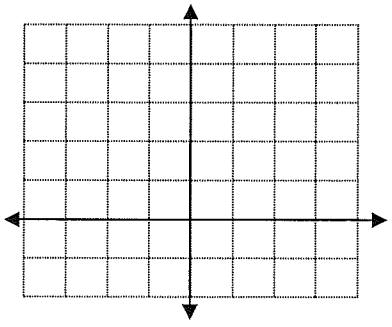
f. f(_____)=5

Warm-Up: Domain and Range

Relation	**Function or just relation? Type?** (Continuous, discontinuous, or discrete)	**Domain** The set of the first coordinates of the ordered pairs in the relation.	**Range** The set of the second coordinates of the ordered pairs in the relation.
a) {(2,-1), (3,-2), (4,-4), (5,-5), (2,-3)}			
b) {..., (-2,5), (-1,6), (0,7), (1,8), (2,9),...}			
c) 			
d) 			
e) 			
f) (curves drawn to edge continue) 			

Graphing Quadratic, Root and Rational Functions

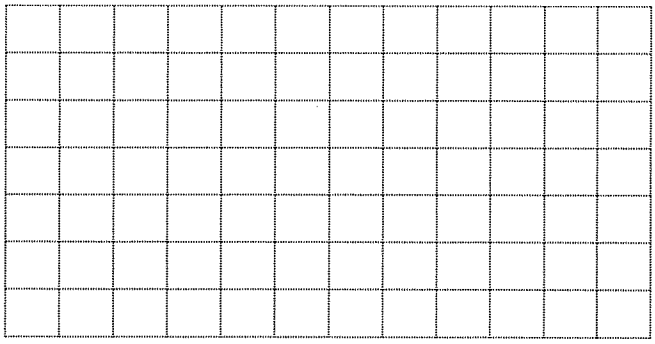
Base Function #1: Quadratic Function

Base Equation / Algebraic Model	Base Points	Base Graph	Domain and Range
$f(x) = x^2$	$(-3, 9)$ $(-2, 4)$ $(-1, 1)$ $(0, 0)$ $(1, 1)$ $(2, 4)$ $(3, 9)$		

Base Function #2: Root Function

1. Complete the table of values then graph $f(x) = \sqrt{x}$. Connect your points with a smooth curve.

x	$f(x) = \sqrt{x}$
0	
1	
4	
9	



2. How do you know $f(x) = \sqrt{x}$ is a function? What type of function is it?

3. Why does the graph of $f(x) = \sqrt{x}$ appear only in the first quadrant?

4. Determine the domain and range of $f(x) = \sqrt{x}$.

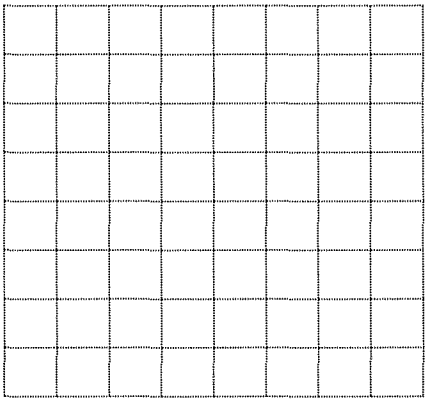
Base Function #3: Rational/Reciprocal Function

1. Complete the table of values then graph $f(x) = \frac{1}{x}$. Connect your points with a smooth curve.

x	$f(x) = \frac{1}{x}$
-2	
-1	
-1/2	
0	
1/2	
1	
2	

2. What happens to the y-values as x decreases, closer to $-\infty$? As x remains negative but gets closer to 0?

3. What happens to the y-values as x increases, closer to $+\infty$? As x remains positive but gets closer to 0?



4. How do you know $f(x) = \frac{1}{x}$ is a function? What type of function is it?

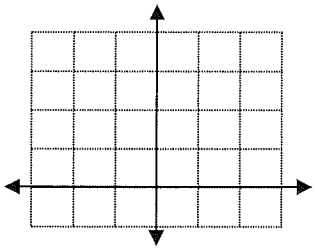
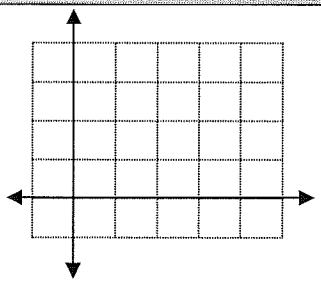
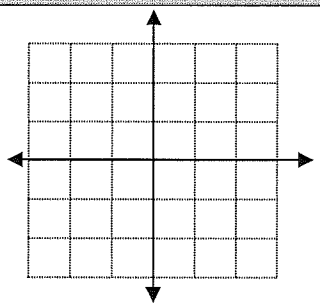
5. Why does the graph of $f(x) = \frac{1}{x}$ have a break in the middle?

6. Determine the domain and range of $f(x) = \frac{1}{x}$. (Think about what the restrictions on x and $f(x)$.)

7. An **asymptote** is a line that a graph will come very close to but never touch. What are the equations of the asymptotes of $y = \frac{1}{x}$?

Lesson 3.3 Graphing Base Functions

Base Graphs Summary:

Function	Quadratic	Root	Rational/Reciprocal
Base graph			
Function type			
Base points			
Domain			
Range			

Warm-Up

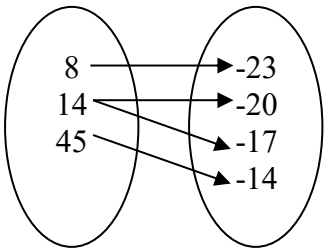
1. Given $f(x) = 3x^2 + 2x - 1$ and $g(x) = 4x + 5$, determine
   a) $f(-3)$ b) $g(4) - f(1)$ c) $f(g(x))$

2. Determine whether the following relations are functions or not, and explain how you know.
   a) $y^2 = x - 4$

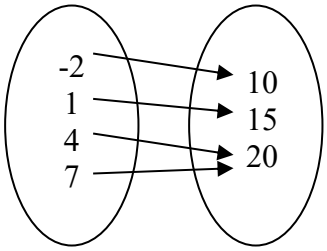
   b)

x	y
2	-3
5	-4
7	-5
8	-5

   c)



3. What type of function (continuous, discontinuous, discrete) is each of the following examples? Explain how you know.
   a) Quadratic b) Rational c)

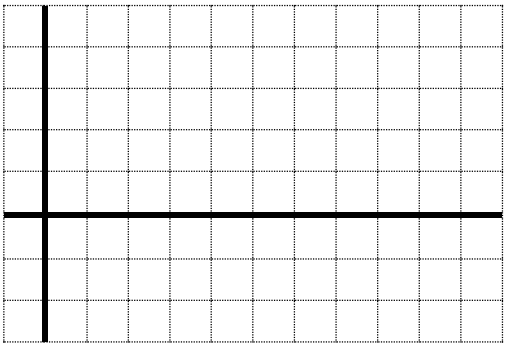


4. State the algebraic model and the domain and range for the three base functions – quadratic, root, and rational.

Translations of Functions and General Function Notation

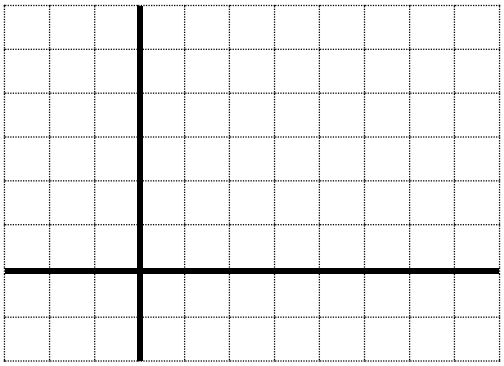
Investigating the Effects of Translating Functions

1. Complete each table of values for the functions shown below using www.desmos.com/calulator. Graph each function on the coordinate axes provided in different colours.



	Base function	$g(x) = f(x) + 3$	$h(x) = f(x) - 2$
Algebraic model	$f(x) = \sqrt{x}$		
Points	x | y 0 | 0 1 | 1 4 | 2 9 | 3	x | y 0 | 1 | 4 | 9 |	x | y 0 | 1 | 4 | 9 |
Transformation from base function			
Changes to x-coordinates from base			
Domain			
Changes to y-coordinates from base			
Range			

2. Complete each table of values for the functions shown below using www.desmos.com/calulator. Graph each function on the coordinate axes provided in different colours.

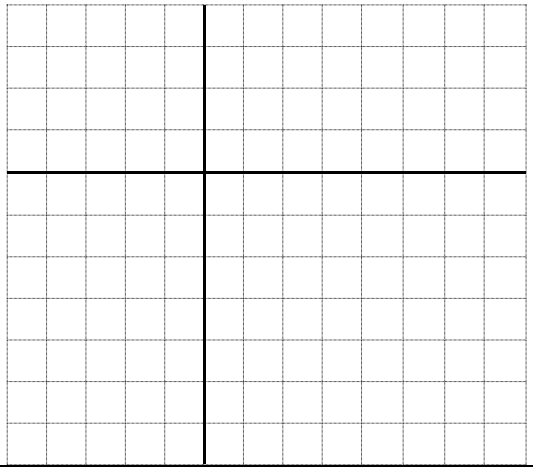
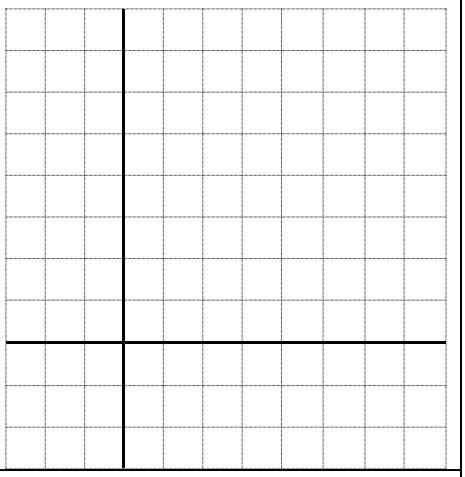


	Base function	$g(x) = f(x-3)$	$h(x) = f(x+1)$
Algebraic model	$f(x) = \sqrt{x}$		
Points	x | y 0 | 0 1 | 1 4 | 2 9 | 3	x | y | 0 | 1 | 2 | 3	x | y | 0 | 1 | 2 | 3
Transformation from base function			
Changes to x-coordinates from base			
Domain			
Changes to y-coordinates from base			
Range			

Summary: Vertical/Horizontal Translations of Functions

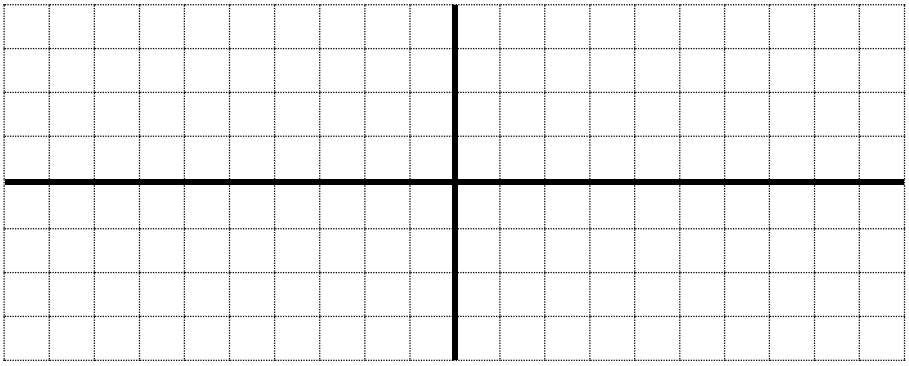
Transformation	General Function Notation	Quadratic Function	Root Function	Rational Function
Translate $\|c\|$ units up/down				
Transformation	**General Function Notation**	**Quadratic Function**	**Root Function**	**Rational Function**
Translate $\|d\|$ units right/left				
	Vertical Translation		**Horizontal Translation**	
Coordinate changes				
Domain and range changes				

Practice:

	$g(x) = \sqrt{x+1} - 4$	$h(x) = \dfrac{1}{x-3} + 2$
Transformations		
Graph		
Domain and Range		

Reflections of functions

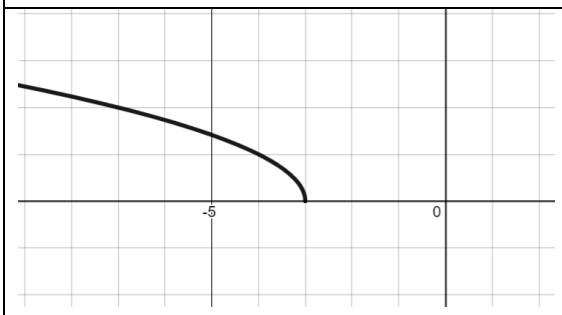
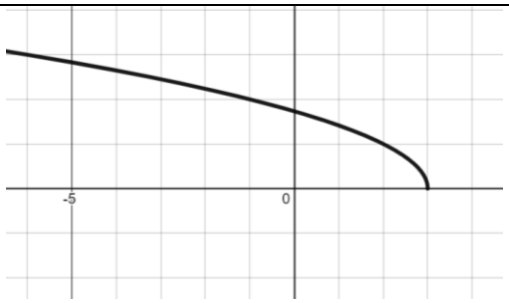
1. Complete each table of values for the functions shown below using www.desmos.com/calculator. Graph each function on the coordinate axes provided in different colours.



	Base function	$g(x) = -f(x)$	$h(x) = f(-x)$
Algebraic model	$f(x) = \sqrt{x}$		
Points	x | y 0 | 0 1 | 1 4 | 2 9 | 3	x | y 0 | 1 | 4 | 9 |	x | y | 0 | 1 | 2 | 3
Transformation from base function			
Changes to x-coordinates from base			
Domain			
Changes to y-coordinates from base			
Range			

Investigation: Which is correct?

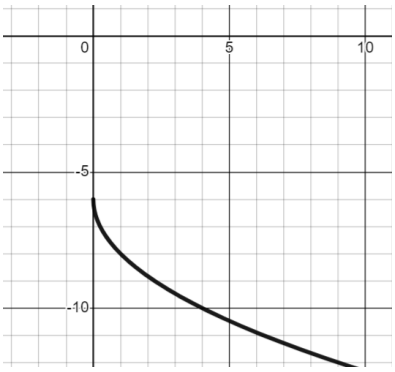
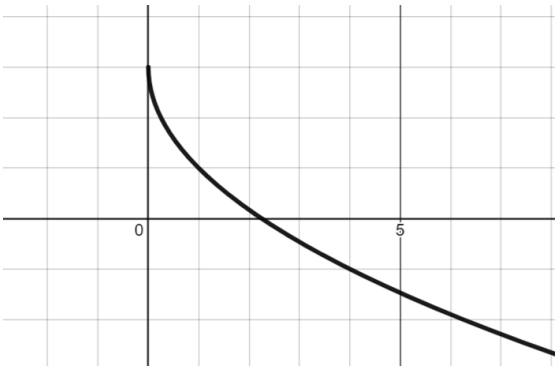
1) Which is the correct graph of the function $f(x) = \sqrt{-x-3}$?

a) Graph of $y = \sqrt{x}$ has been reflected on the y-axis and then translated 3 units left	b) Graph of $y = \sqrt{x}$ has been reflected on the y-axis and then translated 3 units right
	

➔ Try substituting a test value for x.

➔ Write $f(x) = \sqrt{-x-3}$ in factored form…what do you notice?

2) Which is the correct graph of the function $(x) = -2\sqrt{x} + 3$?

Translated 3 units up Reflected on x-axis Vertically stretched, factor of 2	Reflected on x-axis Vertically stretched, factor of 2 Translated 3 units up
	

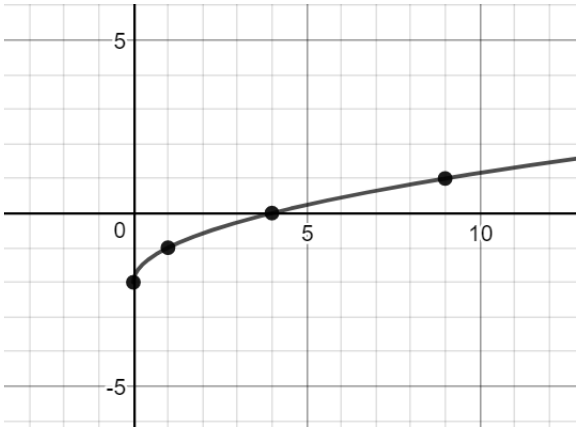
➔ Try substituting a test value for x.

➔ What is the correct order for applying transformations?

Invariant points are point(s) that do not change/move after a transformation.

Ex 1. Given $f(x) = \sqrt{x} - 2$ whose graph is shown below,

a) Graph $-f(x)$.



b) Which point is invariant?

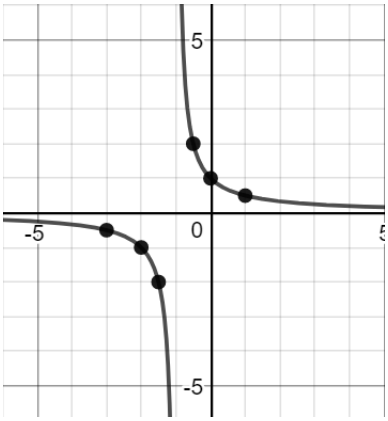
<u>When we reflect a function in the x-axis</u>, the _____ are multiplied by -1.

The only value of $f(x)$ (or y) that will always stay the same after being multiplied by -1 is _____.

Therefore, the _____, if there is one, will be the only possible invariant point after a reflection in the x-axis.

Ex 2. Given $g(x) = \frac{1}{x+1}$ whose graph is shown below,

a) Graph $g(-x)$.



b) Which point is invariant?

<u>When we reflect a function in the y-axis</u>, the _____ are multiplied by -1.

The only value of x that will always stay the same after being multiplied by -1 is _____.

Therefore, the _____, if there is one, will be the only possible invariant point after a reflection in the y-axis.

<u>After reflecting in both axes</u>

The only possible invariant point after reflection in both axes is _____.

Summary: Reflections of Functions

	General Function Notation	Quadratic Function	Root Function	Rational Function
Base				
Reflect in the x-axis				
Reflect in the y-axis				
Reflect in both axes				

Practice:

1) State the transformations in the proper order.
   a) $y = \sqrt{-x-6} - 2$
   b) $y = \frac{1}{-x+4} + 5$

2) A function's graph has been reflected in both axes and translated right 6 units.
   a) Write these transformations in general, using function notation.
   b) State the new transformed function if $f(x) = \sqrt{x}$.

3) Determine the invariant point(s) when
   a) $f(x) = \frac{1}{x-4}$ is reflected in the y-axis.
   b) $g(x) = \sqrt{x} + 3$ is reflected in both axes.

Warm-Up

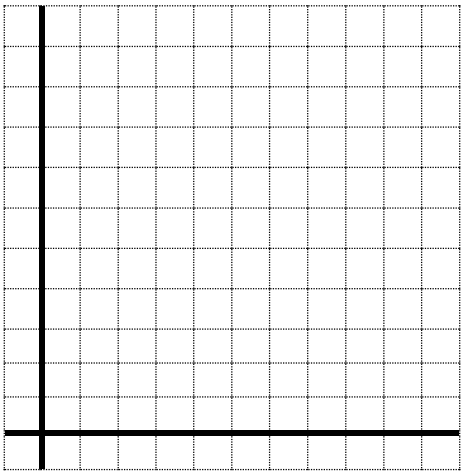
1. Fill in the table.

	$f(x) = -(3x+3)^2 + 2$	$h(x) = \frac{5}{3x-14} + \frac{2}{3}$
Reflect the function in the x-axis $y = -f(x)$		
Reflect the function in the y-axis $y = f(-x)$		
Reflect the function in both axes $y = -f(-x)$		

Vertical Stretches/Compressions of Functions

Part 1: Investigating the Effects of Vertically Scaling Functions

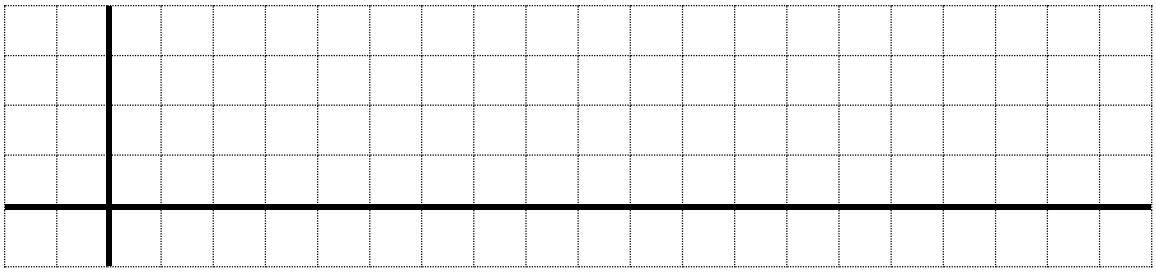
1. Complete each table of values for the functions shown below using www.desmos.com/calulator. Graph each function on the coordinate axes provided in different colours.



	Base function	$g(x) = 3f(x)$	$h(x) = \frac{1}{2}f(x)$
Algebraic model	$f(x) = \sqrt{x}$		
Points	x | y 0 | 0 1 | 1 4 | 2 9 | 3	x | y 0 | 1 | 4 | 9 |	x | y 0 | 1 | 4 | 9 |
Transformation from base function			
Changes to x-coordinates from base			
Domain			
Changes to y-coordinates from base			
Range			

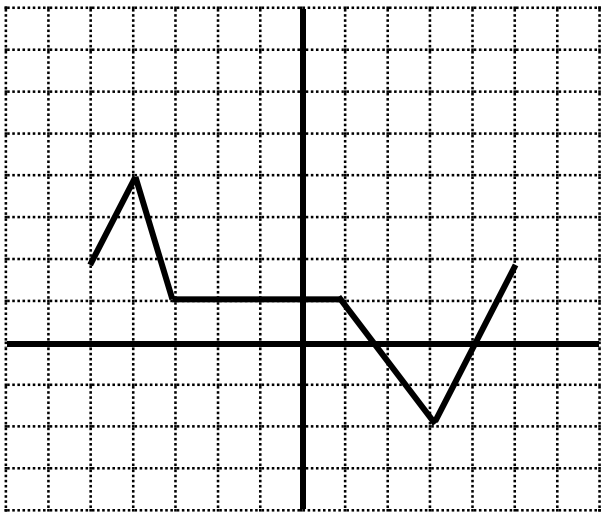
Part 2: Investigating the Effects of Horizontally Scaling Functions

1. Complete each table of values for the functions shown below using www.desmos.com/calulator. Graph each function on the coordinate axes provided in different colours.



	Base function	$g(x) = f(3x)$	$h(x) = f\left(\frac{1}{2}x\right)$
Algebraic model	$f(x) = \sqrt{x}$		
Points	x | y 0 | 0 1 | 1 4 | 2 9 | 3	x | y | 0 | 1 | 2 | 3	x | y | 0 | 1 | 2 | 3
Transformation from base function			
Changes to x-coordinates from base			
Domain			
Changes to y-coordinates from base			
Range			

Ex. 1) Given the graph of $y = f(x)$ on the right:



a) Graph $y = 2f(x)$.

b) Graph $y = f(2x)$.

c) Compare the domain and range of the original function $f(x)$ with the transformed functions, $2f(x)$ and $f(2x)$.

	$\boldsymbol{f(x)}$	$\boldsymbol{2f(x)}$	$\boldsymbol{f(2x)}$
Domain			
Range			

Summary: Vertical Scaling (Stretches/Compressions)

Transformation	General Function Notation	Quadratic Function	Root Function	Rational Function
Vertical Scaling by a factor of $\|a\|$				

When vertically scaling, we multiply the _____by _____.

➔ We also multiply the _____ start/end values by _____.

➔ The _____ and _____ do not change.

Summary: Horizontal Scaling (Stretches/Compressions)

Transformation	General Function Notation	Quadratic Function	Root Function	Rational Function
Horizontal Scaling by a factor of $\left\|\frac{1}{k}\right\|$				

When horizontally scaling, we multiply the _____by _____.

➔ We also multiply the _____ start/end values by _____.

➔ The _____ and _____ do not change.

***** Important reminders about transformations:**

→ Always factor out the k-value
→ Apply and state translations last

Combinations of Transformations

$$y = af[k(x-d)] + c$$

Vertical stretch/compress, factor of	Horizontal stretch/compress, factor of	Horizontal translation	Vertical translation
If $a < 0$	If $k < 0$	If $d < 0$	If $c < 0$
If $\|a\| > 1$	If $\|k\| > 1$	If $d > 0$	If $c > 0$
If $0 < \|a\| < 1$	If $0 < \|k\| < 1$		
Default: $a = 1$	Default: $k = 1$	Default: $d = 0$	Default: $c = 0$
Affects _____ by multiplication	Affects _____ by multiplication	Affects _____ by addition	Affects _____ by addition

Writing Transformations Using Function Notation

	Base Function	Base points	$y = af[k(x-d)] + c$, where $f(x)$ represents any function
Quadratic		$(-2, 4)$ $(-1, 1)$ $(0, 0)$ $(1, 1)$ $(2, 4)$	
Root		$(0, 0)$ $(1, 1)$ $(4, 2)$ $(9, 3)$	
Rational		$\left(-2, -\frac{1}{2}\right)$ $\left(\frac{1}{2}, 2\right)$ $(-1, -1)$ $(1, 1)$ $\left(-\frac{1}{2}, -2\right)$ $\left(2, \frac{1}{2}\right)$ Asymptotes at $y = 0$ and $x = 0$	

We will put together all of our transformations in order to graph functions and state domain and range.

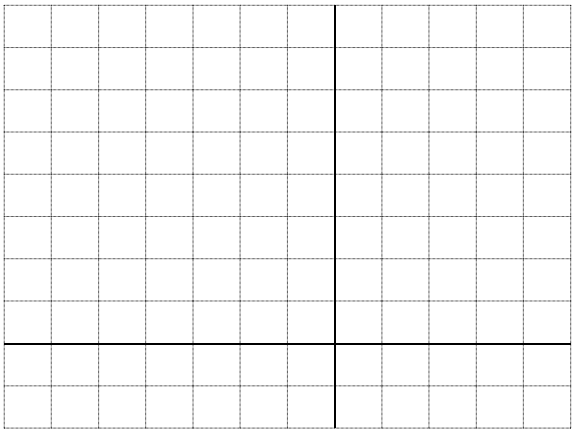
- ➔ Always factor out the k-value
- ➔ It does not matter if you stretch/compress first or reflect first; translations should be applied last

Overview of the Methods for Graphing Transformations of Functions

Method #1: **Multiple Graphs**

1) 1st graph: base function
2) 2nd /3rd graph(s): stretches and reflections
3) Last graph: translations

Ex. 1) Graph $f(x) = 2\sqrt{-x-2}+3$ using the multiple graphs method.



Method #2: **Mapping Notation**

1) Multiply all of the base graph's x-coordinates by $\frac{1}{k}$ then add d
2) Multiply all of the base graph's y-coordinates by a, then add c
3) Plot points from origin (0, 0)

	Base →	**Transformed**
Function	$f(x)$ →	$y = af[k(x-d)]+c$
Points	(x,y) →	$\left(\frac{1}{k}x+d, ay+c\right)$ Plot from origin (0, 0)

Ex. 2) Use mapping notation to determine the transformed points of $f(x) = 2\sqrt{-x-2}+3$.

Base graph points → Transformed
(x,y) →

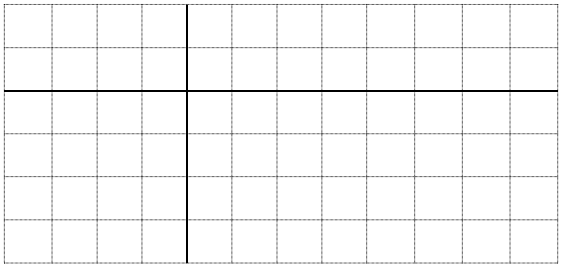
Method #3: **Floating Origin**

1) Move the origin of the base graph to (d,c). Draw in any asymptotes, if applicable.
2) Calculate transformed points using $(x,y) \rightarrow \left(\frac{1}{k}x, ay\right)$
   Plot from the "new origin" (d,c).

*** Note: For Method #2 and #3, you can use the positive base points for a rational, and then graph the other "branch" by symmetry

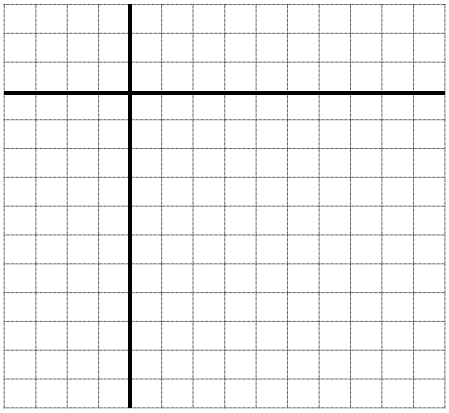
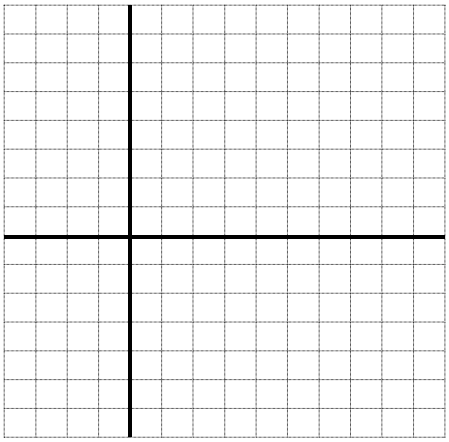
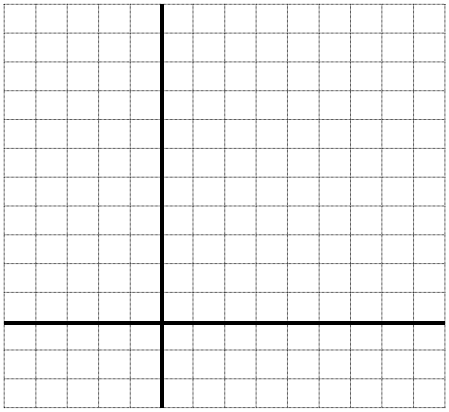
Ex. 3) Graph $f(x) = -\sqrt{\frac{1}{2}x+\frac{3}{2}}-2$ using the floating origin method.

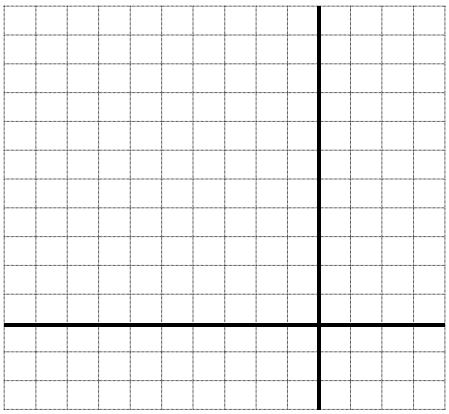
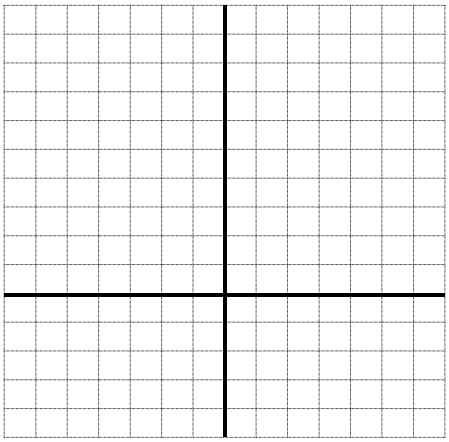
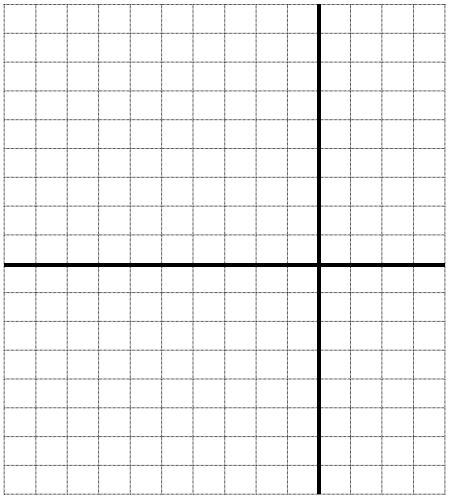
New origin
(x,y) →



Combinations of Transformations Practice

List, in order, the transformations that are applied to the base function's graph to produce each function. Sketch each graph using any method, showing the appropriate amount of work.

FUNCTION	WORK	GRAPH
1. $y = -2(x-5)^2$ Base: Transformations:		
2. $y = \frac{1}{2}(2x-6)^2 + 1$ Base: Transformations:		
3. $y = 3\sqrt{\frac{1}{2}x + 2}$ Base: Transformations:		

FUNCTION	WORK	GRAPH
4. $y = 2\sqrt{-\frac{1}{2}x - 1} + 3$ Base: Transformations:		
5. $y = -\frac{1}{x} + 4$ Base: Transformations:		
6. $y = \frac{2}{3x+6} + 1$ Base: Transformations:		

Domain and Range Summary

	Quadratic $f(x) = x^2$		**Root** $f(x) = \sqrt{x}$		**Rational** $f(x) = \frac{1}{x}$
Algebraic model $y = af[k(x-d)] + c$	$g(x) = a[k(x-d)]^2 + c$		$f(x) = a\sqrt{k(x-d)} + c$		$f(x) = \frac{a}{k(x-d)} + c$
Base Graph Domain	$D: \{x \in \mathbb{R}\}$		$D: \{x \in \mathbb{R} \mid x \geq 0\}$		$D: \{x \in \mathbb{R} \mid x \neq 0\}$
Base Graph Range	$R: \{y \in \mathbb{R} \mid y \geq 0\}$		$R: \{y \in \mathbb{R} \mid y \geq 0\}$		$D: \{y \in \mathbb{R} \mid y \neq 0\}$
Transformed Graph Domain			If $k > 0$	If $k < 0$	
Transformed Graph Range	If $a > 0$	If $a < 0$	If $a > 0$	If $a < 0$	
Example	$f(x) = -3\left(\frac{1}{2}(x+5)\right)^2 - 4$		$f(x) = 2\sqrt{-\frac{1}{3}x + 2} + 5$		$f(x) = \frac{-4}{3x+6} - 5$
Domain					
Range					

Lesson 3.8 Domain and Range Summary and Equivalent Transformations

Practice:

1) State the domain and range of the functions.

Function	Domain	Range
$y = -(x+2)^2 + 7$		
$y = (-x+13)^2 - 9$		
$y = \sqrt{x-8} - 3$		
$y = \sqrt{-4x-12}$		
$y = \dfrac{1}{x+10} - 3$		
$y = \dfrac{1}{-2x+10} + 7$		

Equivalent Transformations

Horizontal stretches and compressions can **have the same effect (produce the same graphs)** as vertical compressions and stretches. However, the **scaling factor may be different.**

A) Comparing $f(x) = \sqrt{kx}$ and $g(x) = a\sqrt{x}$

Graph the following pair of functions on the same set of axes	Describe the transformation for $f(x)$	Describe the transformation for $g(x)$	How can we change $f(x) = \sqrt{kx}$ to $g(x) = a\sqrt{x}$? (Consider the relationship between a and k?)
$f(x) = \sqrt{4x}$ and $g(x) = 2\sqrt{x}$			

B) Comparing $f(x) = (kx)^2$ and $g(x) = ax^2$

Graph the following pair of functions on the same set of axes	Describe the transformation for $f(x)$	Describe the transformation for $g(x)$	How can we change $f(x) = (kx)^2$ to $g(x) = ax^2$? (Consider the relationship between a and k?)
$f(x) = \left(\frac{1}{3}x\right)^2$ and $g(x) = \frac{1}{9}x^2$			

C) Comparing $f(x) = \frac{1}{kx}$ and $g(x) = \frac{a}{x}$?

Graph the following pair of functions on the same set of axes	Describe the transformation for $f(x)$	Describe the transformation for $g(x)$	How can we change $f(x) = \frac{1}{kx}$ to $g(x) = \frac{a}{x}$? (Consider the relationship between a and k?)
$f(x) = \frac{1}{2x}$ and $g(x) = \frac{2}{x}$			

1) The graph of $f(x) = \sqrt{x}$ is horizontally compressed by a factor of $\frac{1}{9}$.
Determine an equivalent vertical transformation that produces the same function and graph.

2) The graph of $f(x) = x^2$ is vertically stretched by a factor of 4.
Determine an equivalent horizontal transformation that produces the same function and graph.

3) The graph of $f(x) = \frac{1}{x}$ is vertically stretched by a factor of 3.
Determine an equivalent horizontal transformation that produces the same function and graph.

Properties of Inverses of Functions

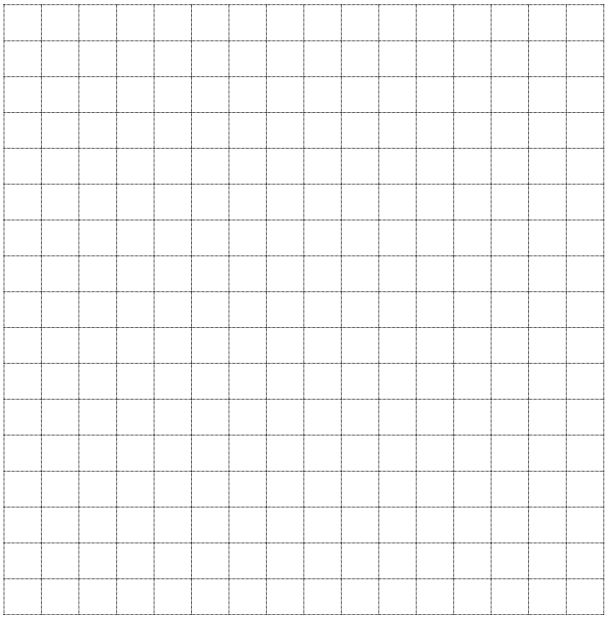
1. Given the function $g(x) = x^2 + 2$,

   a) List 7 ordered pairs that belong to $g(x)$ (use $x = -3, -2, -1, 0, 1, 2, 3$)

The ordered pairs of the <u>inverse</u> of $g(x)$ are found by <u>switching the coordinates</u> in each ordered pair.

   b) List 7 ordered pairs that belong to the inverse relation.

   c) Graph $g(x)$ and its inverse.



   d) Is the inverse relation a function? Explain.

   e) What is the relationship between the two graphs?
   Hint: find the mirror line…

   f) Compare the domain and range of $g(x)$ and its inverse.

Key Ideas:

- The inverse of a function is the relation that is created by _____
- The inverse of a function is not necessarily _____
- The inverse relation's graph is the reflection of the original function in the line _____
- The domain and range of the function become the _____ of the inverse (they switch).

Determining the algebraic model of inverse relations

4. The **function $f(x) = 5x + 8$ means**: take x → multiply by 5 → add 8.

The inverse relation means getting the reverse of the function.
So, we can apply opposite operations in the reverse order.

The **inverse relation of $f(x)$ is**: take x →

So the inverse relation is modeled by:

The **algebraic method** for determining the inverse relation is as follows:

1) Replace $f(x)$ with y.
2) Interchange x and y.
3) Isolate y.
4) Replace y with $f^{-1}(x)$ if the inverse is a function.

Example: Determine the algebraic model of the inverse relation for the following functions:

a) $f(x) = 4x - 8$

b) $f(x) = x^2 + 2$

c) $f(x) = \sqrt{3x - 6}$

d) $f(x) = \frac{1}{x}$

e) $f(x) = 2x^2 - 8x + 3$

f) $f(x) = \frac{3}{x-4} - 2$

Testing for Inverse Relations

5. From the previous example #4a), take $f(x) = 4x - 8$ and its inverse, _____.

a) Simplify $f(g)$.

b) Simplify $g(f)$.

c) What do you notice?

Key Idea: When $g(x)$ is the inverse relation of $f(x)$ we can conclude that

$f(g)$ and $g(f)$ will equal _____.

Example: Determine whether or not $f(x) = \frac{2}{x} + 1$ and $g(x) = \frac{1}{2x-2}$ are inverses of each other.

Unit 4:
Exponential Functions

Exponent Laws

$a^m \times a^n = a^{m+n}$	$(3^4)(3^3)$	$(-2)^3(-2)^5$	
$a^m \div a^n = a^{m-n}$	$\frac{6^5}{6^3}$	$\frac{5^7}{5}$	
$(a^m)^n = a^{m \cdot n}$	$(5^2)^4$	$\left(2^{\frac{1}{2001}}\right)^{2001}$	
$(ab)^m = a^m b^m$	$(5 \times 7)^2$	$(5xy^2)^3$	
$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{2}{5}\right)^4$	$\left(\frac{m}{4n^3}\right)^3$	$\left(\frac{6x^5y^3}{8y^4}\right)^2$
$a^0 = 1$, if $a \neq 0$	$(-4)^0$	-4^0	
$a^{-m} = \frac{1}{a^m}$, if $a \neq 0$	$\left(\frac{1}{3}\right)^{-2}$	-2^{-3}	$(-3)^{-2}$
$\left(\frac{a}{b}\right)^{-m} = \left(\frac{b}{a}\right)^m = \frac{b^m}{a^m}$	$\left(\frac{3}{2}\right)^{-3}$	$\left(\frac{7}{3}\right)^{-3}$	$\left(\frac{12x^2y}{15x^{-1}y^3}\right)^{-4}$

Note: $(-3)^2$ -3^2

Simplifying Expressions Involving Exponents

Simplify first and then evaluate.

a) $2^4 \times 2^6$

b) $\left(\frac{5^8}{5^2}\right)$

c) $(7^4)^3$

d) $(3^{-3})(3^5)(3^{-4})$

e) $(-2)^{-5} \div (-2)^{-2}$

f) $(4^{-1})^{-2}$

Evaluate.

a) $3^0 \div 3^3$

b) $4^2 - 2^{-1}$

Express your answer with positive exponents.

a) $(2a^{-3}b^{-2}c^4)^{-3}$

b) $\left(\frac{3x^{-4}y^3}{6x^2y^{-2}}\right)^{-2}$

Evaluate.

a) $\frac{2^{-4}+2^{-6}}{2^{-3}}$

b) $\frac{5^{-4}+5^{-6}}{5^{-3}+5^{-5}}$

We LOVE Exponential Equations

Solve the following. How? → By simplifying both sides into **equal bases** using exponent laws.

Example

$$2(3^{y-2}) = 18$$
$$3^{y-2} = 9$$
$$3^{y-2} = 3^2$$
$$y - 2 = 2$$
$$y = 4$$

Try it on your own!

a) $27(3^{3x+1}) = 3$

b) $\frac{27^x}{9^{2x-1}} = 3^{x+4}$

c) $2^{x+2} - 2^x = 48$ *common factor first!

d) Determine x and y if $\frac{16^{x+2y}}{8^{x-y}} = 32$ and $\frac{32^{x+3y}}{16^{x+2y}} = \frac{1}{8}$.

Lesson 4.1 Exponent Laws

Practice

Simplify each expression below, expressing your answer with positive exponents.

1. $y^{10}(y^4)^{-3}$

2. $\frac{(m^{-7})^2}{m(m^{-3})m^9}$

3. $\frac{(10x)^{-1}y^3}{15x^3y^{-3}}$

4. $\frac{p^{-5}(r^3)^2}{(p^2r)^2(p^{-1})^2}$

5. $\left(\frac{(x^3y)^{-1}(x^4y^3)}{(x^2y^{-3})^{-2}}\right)^{-1}$

Rational Exponents

We found last lesson that $(a^m)^n =$_____ according to power of a power rule.

If $m = \frac{1}{n}$, then $(a^m)^n$

$= \left(a^{\frac{1}{n}}\right)^n$

$=$

$=$

We get the _____ as the answer.

➔ Exponent n and exponent $\frac{1}{n}$ are _____ or _____ operations – they cancel each other out.

$a^{\frac{1}{n}} = \sqrt[n]{a}$, where $a \in \mathbb{R}$ and n is called the _____ of the radical.

$\sqrt[n]{a}$ is called "the nth root of a". The solution is the _____, when raised to the exponent of n, equals a.

e.g. $32^{\frac{1}{5}} = \sqrt[5]{32}$ → "what is the 5th root of 32," or "what is the base, when raised to the exponent of 5, equals 32?"

$=$

- $n = 2$ is the default index (when we need to _____) e.g. $4^{\frac{1}{2}} =$
- If n is an even number, then $a \geq 0$ for the nth root to be real.
  Ex. 1) $\sqrt[4]{16} =$ $\sqrt[4]{-16} =$
- If n is an odd number, then $a \epsilon \mathbb{R}$ for the nth root to be real.
  Ex. 2) $\sqrt[3]{8} =$ $\sqrt[3]{-8} =$

Ex. 3 $(-32)^{\frac{1}{5}}$

Ex. 4 $-16^{\frac{1}{4}}$

For $a^{\frac{m}{n}}$ where $m \in \mathbb{Z}$ (integers) and $n \in \mathbb{N}$ (natural numbers)

$a^{\frac{m}{n}}$

$= \sqrt[n]{a^m}$

$= \left(\sqrt[n]{a}\right)^m$

Ex 5: Rewrite the following as a power of x.

a) $\sqrt{x}$ b) $\sqrt{\sqrt{x}}$ c) $\sqrt[3]{x^2}$ d) $\sqrt{\sqrt[4]{x^3}}$

Example: Evaluate $4^{\frac{3}{2}}$.

Method 1	Method 2
Take the n^{th} root first, then apply the exponent	Apply the exponent first, then take the n^{th} root
$4^{\frac{3}{2}}$ $= \left(4^{\frac{1}{2}}\right)^3$ $= \left(\sqrt{4}\right)^3$ $= 2^3$ $= 8$	$4^{\frac{3}{2}}$ $= (4^3)^{\frac{1}{2}}$ $= (64)^{\frac{1}{2}}$ $= \sqrt{64}$ $= 8$

Evaluate the following.

1) $25^{\frac{3}{2}}$

2) $16^{\frac{3}{2}}$

3) $16^{\frac{3}{4}}$

4) $(-27)^{\frac{2}{3}}$

5) $27^{-\frac{4}{3}}$

6) $-81^{\frac{3}{2}}$

7) $(-81)^{\frac{3}{4}}$

8) $128^{-\frac{4}{7}}$

Exponential Equations...

Object: Solve for x → which is the base.

How? Raise both sides of the equation by the reciprocal of the exponent.

Ex. $x^{\frac{2}{3}} = 9$

$\left(x^{\frac{2}{3}}\right)^{\frac{3}{2}} = (9)^{\frac{3}{2}}$ → Note that the denominator is now even, so…

$x = \left(\pm\sqrt{9}\right)^3$

$x = \pm 27$

Now YOU try!

1) $x^{\frac{3}{4}} = 8$

2) $x^{\frac{1}{2}} = 12$

3) $x^{0.25} = 3$

4) $x^{\frac{5}{3}} = 32$

5) $x^{0.75} = 27$

6) $x^{-\frac{1}{3}} = 2$

7) $x^{\frac{4}{5}} - 3 = 13$

8) $x^{-\frac{2}{3}} + 4 = 8$

9) $x^{-\frac{3}{4}} = \frac{27}{64}$

Lesson 4.3 Intro to Exponential Functions

Warm-Up: Exponent Laws and Rational Exponents

Write out the exponent laws from lessons 4.1 and 4.2.

1. Evaluate the following without using a calculator. No decimals allowed!

a) $\left(\frac{3}{7}\right)^{-2}$ b) $\left(\frac{15}{25}\right)^{4}$ c) $\left(\frac{243}{81}\right)^{-3}$ d) $-\left(\frac{33}{24}\right)^{2}$

2. Evaluate $\frac{4^{-4}+4^{-6}}{4^{-3}+4^{-5}}$.

3. Express in radical form using positive exponents.

a) $y^{\frac{1}{4}}$ b) $(2x^3y^{-2})^{\frac{3}{2}}$ c) $(3x^{-5}y^{7})^{\frac{3}{2}}$

Lesson 4.3 Intro to Exponential Functions

4. Express as a single power.

a) $\sqrt[5]{m^7}$ b) $2\sqrt[3]{m}$ c) $\left(\frac{1}{\sqrt[12]{5}}\right)^4$

5. Simplify. Express final answers using positive exponents only.

a) $\left(\frac{12x^3y^9}{8y^4}\right)^2$ b) $\left(\frac{13a^{-3}b^3}{26a^5b^{-2}}\right)^{-3}$

6. Solve for x.

a) $x^{\frac{2}{5}} + 3 = 19$ b) $2x^{\frac{3}{4}} - 4 = 50$

Challenge: How would you solve this equation?

$2^{3x-2} = 4^{-2x+1}$

Investigation: Do You See the Difference?

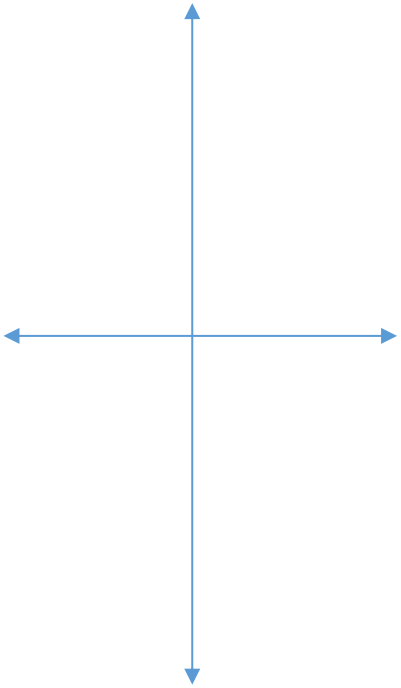
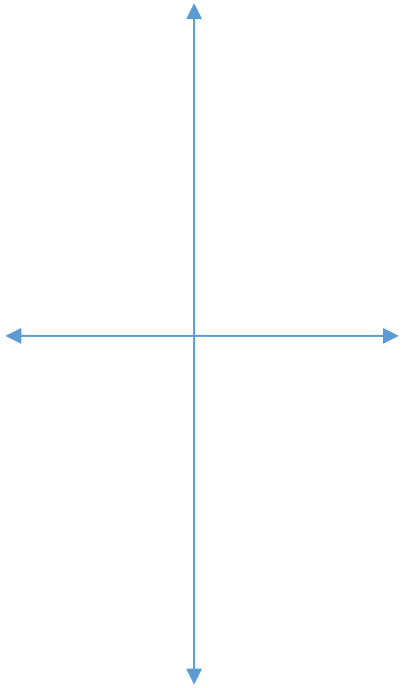
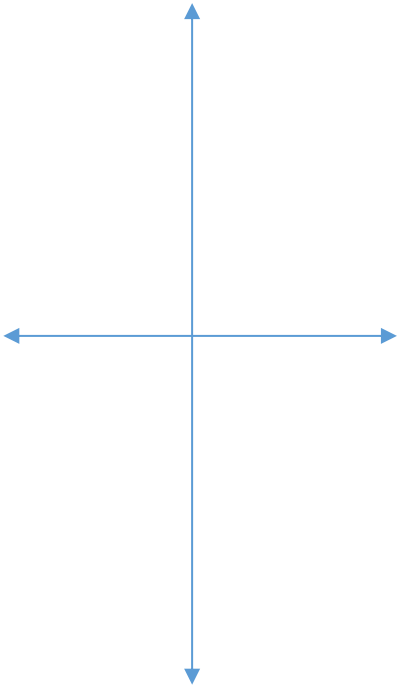
Part A: Compare and Contrast Algebraic Models

1) Explain the algebraic differences between linear, quadratic, and exponential functions.

	Coefficient	Base	Exponent
Linear $f(x) = 2x$			
Quadratic $g(x) = x^2$			
Exponential $h(x) = 2^x$			

Part B: Compare and Contrast Graphical Models

2) Use Desmos (www.desmos.com/calculator) to sketch the following functions.

Linear $f(x) = 2x$	**Quadratic** $g(x) = x^2$	**Exponential** $h(x) = 2^x$
		

Investigation: Do You See the Difference?

Part C: Compare and Contrast Numerical Models (tables of values)

x	$f(x) = 2x$	First Differences	Second Differences	Ratio between 1st diff.	Linear Functions have …
-3	-6				
-2	-4				
-1	-2				
0	0				
1	2				
2	4				
3	6				

x	$g(x) = x^2$	First Differences	Second Differences	Ratio between 1st diff.	Quadratic Functions have …
-3	9				
-2	4				
-1	1				
0	0				
1	1				
2	4				
3	9				

x	$h(x) = 2^x$	First Differences	Second Differences	Ratio between 1st diff.	Exponential Functions have …
-3	$\frac{1}{8}$				
-2	$\frac{1}{4}$				
-1	$\frac{1}{2}$				
0	1				
1	2				
2	4				
3	8				

Part D: Conclusions

Read over what you have written above and decide if your statements apply to all exponential, quadratic or linear functions. Explore further if needed. Then complete the following statements about all exponential functions.

A function is exponential when

- Algebraically it has…

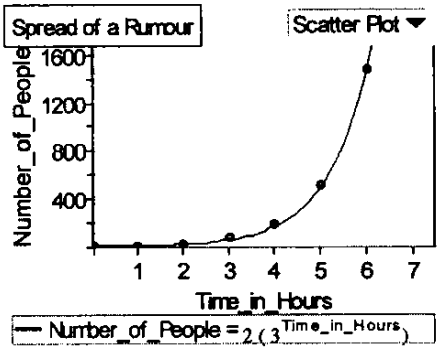
- In a numerical model (table of values) it has…

- Graphically it looks like…

What's Your Function?

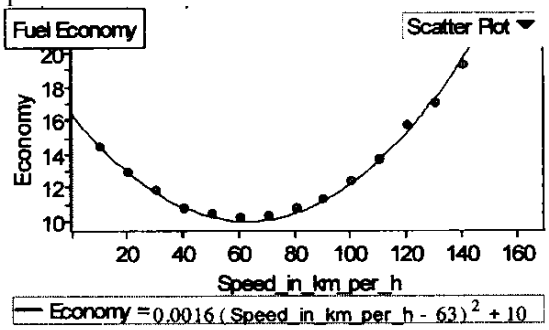
Given a function in its various forms (algebraic model, numerical model, or graph) determine whether the function is linear, quadratic, or exponential and explain how you know.

1. How fast does a rumour spread?



Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

2. The graph models the average fuel economy of a particular car, in litres per 100 km, at various speeds.



Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

3. The table describes the cooling of a cup of coffee.

Time (min)	0	4	8	12	16	20
Temperature (°C)	55	47	40	34	29	25

Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

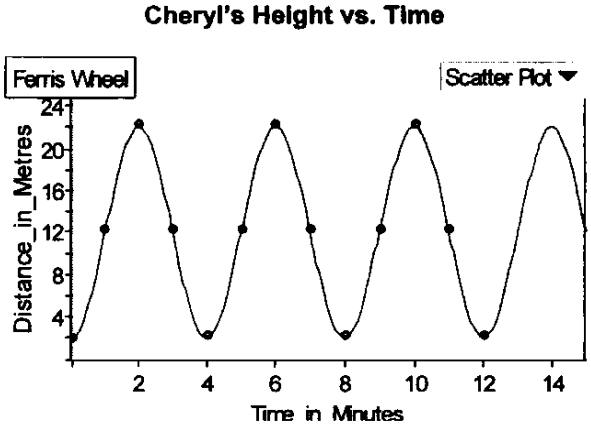
4. If you invest \$5000 in a stock that is increasing in value at a rate of 12% per year, then the value of your stock is given by $A = 5000(1.12)^n$, where A is the amount in dollars and n is the number of years.

Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

5. $(Pizza)^2$ charges \$10 for a large pizza plus \$2 per topping. The total cost of the pizza, C, can be modelled by $C = 2n + 10$, where n is the number of toppings.

Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

6. Cheryl is riding a Ferris wheel. The graph below shows Cheryl's height above the ground.



Function type	Reasons
☐ Linear ☐ Quadratic ☐ Exponential ☐ None of the above	

7. Identify the type of function represented by the table of values.

a)

x	y
-4	5
-3	8
-2	13
-1	20
0	29
1	40

b)

x	y
-5	32
-4	16
-3	8
-2	4
-1	2
0	1

c)

x	y
0.5	0.9
0.75	1.1
1	1.3
1.25	1.5
1.5	1.7
1.75	1.9

Warm-Up: Reviewing Properties of Exponential Functions

1. Write verbal descriptions of linear, quadratic, and exponential functions.

	Linear	**Quadratic**	**Exponential**
Algebraic model			
Numerical model			
Graphically			

Exploring Exponential Functions $f(x) = b^x$

Part A: Exploring $f(x) = b^x$ where $b > 1$

1) Use Desmos (www.desmos.com/calculator) to graph the following functions on the same axes:
   $f(x) = 2^x$
   $g(x) = 5^x$
   $h(x) = 10^x$

2) Discuss and answer the following:

   a) What do you notice about the values of $f(x), g(x)\ and\ h(x)$:

as x values increases?	as x values decreases?

   b) Does the graph cross the x-axis? If so, where?

   c) Does the graph have an asymptote? If so, where?

3) Fill in the table below.

Function	y-intercept	Domain	Range	x-intercept	Asymptote	Increasing or Decreasing Function
$y = 2^x$						
$y = 5^x$						
$y = 10^x$						

4) Where do you think $y = 7^x$ would lie relative to the curves graphed earlier? You can check your prediction by adding the graph to Desmos.

5) What happens to the graph as the b **value gets bigger**?

Exploring Exponential Functions $f(x) = b^x$

Part B: Exploring $f(x) = b^x$ where $0 < b < 1$

1) Use Desmos (www.desmos.com/calculator) to graph the following functions on the same axes:

$f(x) = \left(\frac{1}{2}\right)^x$

$g(x) = \left(\frac{1}{5}\right)^x$

$h(x) = \left(\frac{1}{10}\right)^x$

2) Discuss and answer the following:

a) What do you notice about the values of $f(x), g(x)$ *and* $h(x)$:

as x values increases?	as x values decreases?

b) Does the graph cross the x-axis? If so, where?

c) Does the graph have an asymptote? If so, where?

3) Fill in the table below.

Function	y-intercept	Domain	Range	x-intercept	Asymptote	Increasing or Decreasing Function
$y = \left(\frac{1}{2}\right)^x$						
$y = \left(\frac{1}{5}\right)^x$						
$y = \left(\frac{1}{10}\right)^x$						

4) Where do you think $y = \left(\frac{1}{7}\right)^x$ would lie relative to the above curves? You can check your prediction by adding the graph to Desmos.

5) What happens to the graph as the b **value gets smaller**?

Exploring Exponential Functions $f(x) = b^x$

Part C: Exploring $f(x) = b^x$ where $b = 1$ or $b = 0$

1) Use Desmos (www.desmos.com/calculator) to graph the following functions. Then, fill in the table below.

$f(x) = 1^x$	$g(x) = 0^x$	$h(x) = (-2)^x$
x | y -2 | -1 | 0 | 1 | 2 |	x | y -2 | -1 | 0 | 1 | 2 |	x | y -2 | -1 | 0 | 1 | 2 |
Function type: Domain:	Function type: Domain:	Function type: Domain:
Explain why you get the graph that you do.	Explain why you get the graph that you do.	Explain why you get the graph that you do.

4) Discuss as a group:
What **restrictions on** b are necessary to create a continuous, increasing/decreasing exponential function?

Part D: Things that make you go hmmmmmm!!

Graph $y = 2^x$ and $y = \left(\frac{1}{2}\right)^x$. What do you notice?

Discuss as a group: What transformation of $y = 4^x$ will give $y = \left(\frac{1}{4}\right)^x$ as its image?

SUMMARY

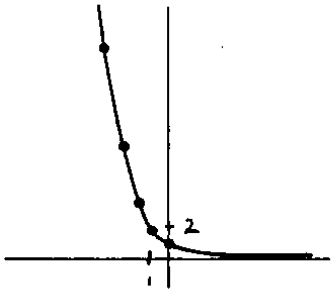
Properties	$y = b^x, b > 1$	$y = b^x, 0 < b < 1$
Y-intercept		
Domain		
Range		
Asymptote type and equation		
As x-values increase, does the function increase or decrease?		
Base points and how to identify the equation from the graph		

Note: The graph of $y = \left(\frac{1}{b}\right)^x$ or $y = b^{-x}$ is a _____ of $y = b^x$ in the ____ - axis.

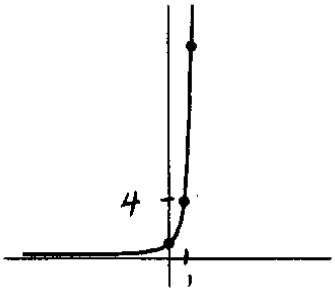
Part A

1. The graphs below have equations of the form $y = b^x$. Answer the following questions about each of the functions:

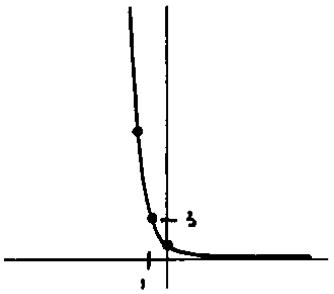
a.



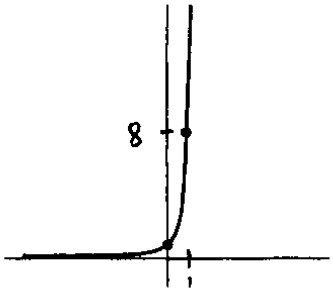
b.



c.



d.



i) What is the value of the y-intercept?

ii) Is the function always increasing or always decreasing?

iii) Is $b > 1$ or is $0 < b < 1$?

iv) What is the value of y when $x = 1$ or $x = -1$?

v) Use the information you gained in parts **i** to **iv** write the equation of the function.

Part B

Application

2. If f is a function defined by $f(x) = b^x$, where $b > 0$, what can be stated about
   a. the sign of $f(x)$?
   b. the growth behaviour of f?
   c. $f(0)$?

Thinking/Inquiry/ Problem Solving

3. If f is a function defined by $f(x) = b^x$, where $0 < b < 1$, what can be stated about
   a. the sign of $f(x)$?
   b. the growth behaviour of f?
   c. $f(0)$?

Communication

4. Describe how you determine the equation of a function of the form $y = b^x$, $b > 0$, if you are given its graph. You can use only your knowledge of exponential functions and not your calculator. Include sketches to help you describe the procedure you use.

Lesson 4.5 Transformations of the form $f(x) = ab^x + c$

Exploring Exponential Functions $f(x) = ab^x + c$

Part A: $f(x) = b^x + c$

1) Use Desmos to graph the following functions and state the transformations for $g(x)$ and $h(x)$.

$f(x) = 2^x$
$g(x) = 2^x + 4$ →
$h(x) = 2^x - 3$ →

2) Fill in the table.

	$f(x) = 2^x$	$g(x) = 2^x + 4$	$h(x) = 2^x - 3$
Horizontal asymptote			
Y-intercept			
Range			

Part B: $f(x) = ab^x$

3) Use Desmos to graph the following functions and state the transformations $g(x)$ and $h(x)$.

$f(x) = 2^x$
$g(x) = 3(2^x)$ →
$h(x) = \frac{1}{2}(2^x)$ →

4) Fill in the table.

	$f(x) = 2^x$	$g(x) = 3(2^x)$	$h(x) = \frac{1}{2}(2^x)$
Horizontal asymptote			
Y-intercept			

Part C: *Reflection*

5) Use Desmos to graph the following functions and state the transformations for $g(x)$.

$f(x) = 2^x$

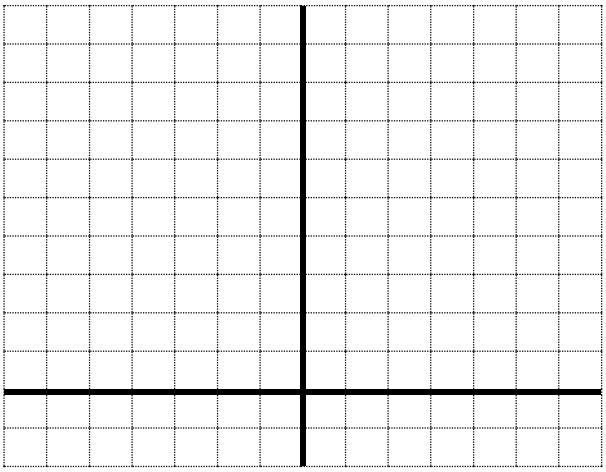
$g(x) = -2^x$ →

6) Fill in the table below.

	$f(x) = 2^x$	$g(x) = -2^x$
Range		
Y-intercept		
Increasing/Decreasing		

Lesson 4.5 Transformations of the form $f(x) = ab^x + c$

Part D: Practice Ex. 1) Given $f(x) = 2(2^x) + 1$, fill in the table below.

	Steps for Graphing
Base	1) Identify the base value, b, and determine the base points: $\left(-1, \frac{1}{b}\right), (0, 1), (1, b), (2, b^2)$
Transformations (in order)	2) Use mapping notation: $(x, y) \rightarrow \left(\frac{1}{k}x + d, ay + c\right)$
Equation of asymptote	
Y-intercept	
Increasing or decreasing function	
Domain	
Range	

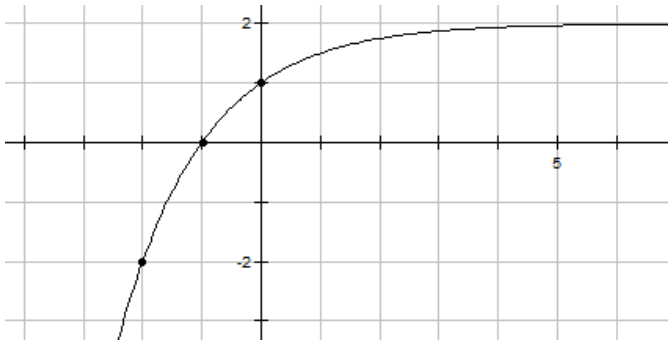
Lesson 4.5 Transformations of the form $f(x) = ab^x + c$

Ex 2) Determine the algebraic model of the exponential function that has:

- a horizontal asymptote $y = -2$,
- a y-intercept of -2.5, and
- passes through the point (3, -6).

What do we **know**?	What do we **want** to know?
H.A. of $y = -2$ → this means that y-intercept of -2.5 → this means that $(x, y) =$ passes through (3, -6) → $(x, y) = (3, -6)$	$f(x) = ab^x + c$ → we need to determine the values of
How can we get the answer? 1) Substitute $c =$ into the algebraic model 2) Create two equations using $(x, y) =$ (the y-intercept) and $(x, y) = (3, -6)$ 3) Solve for a and b using _____ or _____	

You try: Given the graph of $f(x) = ab^x + c$ below, determine its algebraic model.



Lesson 4.5 Transformations of the form $f(x) = ab^x + c$

Part A

Knowledge/ Understanding

1. For each of the following, state

i) the equation of the horizontal asymptote,

ii) whether the function is increasing or decreasing,

iii) the y-intercept.

a. $y = 3^x - 5$ b. $y = 2^x + 4$ c. $y = 4\left(\frac{1}{3}\right)^x$

d. $y = \left(\frac{1}{2}\right)^x + 2$ e. $y = 2(5^x) - 1$ f. $y = 5\left(\frac{2}{3}\right)^x + 1$

Knowledge/ Understanding

2. For the curve $y = 3(4^x) + 5$,

a. determine

i) the horizontal asymptote,

ii) the y-intercept,

iii) whether the curve is increasing or decreasing,

iv) the domain and range.

Application

b. Sketch the curve.

Part B

Knowledge/ Understanding

3. For the curve $y = 2\left(\frac{2}{3}\right)^x - 4$,

a. determine

i) the horizontal asymptote,

ii) the y-intercept,

iii) whether the curve is increasing or decreasing,

iv) the domain and range.

Application

b. Sketch the curve.

Communication and Thinking/Inquiry/ Problem Solving

4. Describe how you can draw a quick mental picture of the graph for a function of the form $y = ab^x + c$, $b > 0$, by using your knowledge of the effect of changing the parameters a, b, and c. Use specific equations and sketches to help you describe the process.

Answers: 1. **a)** i) y=-5 ii) increasing iii) (0, -4) **b)** i) y=4 ii) increasing iii) (0, 5) **c)** i) y=0 ii) decreasing iii) (0, 4) **d)** i) y=2 ii) decreasing iii) (0, 3) **e)** i) y=-1 ii) increasing iii) (0, 1) **f)** i) y=1 ii) decreasing iii) (0, 6) **2. a)** i) y=5 ii) (0, 8) iii) increasing iv) D:$\{x \in \mathbb{R}\}$ R: $\{y > 5 | y \in \mathbb{R}\}$ **3. a)** i) y=-4 ii) (0, -2) iii) decreasing iv) D:$\{x \in \mathbb{R}\}$ R: $\{y > -4 | y \in \mathbb{R}\}$

Lesson 4.6 – Transformations of the form $f(x) = ab^{(x-d)} + c$

Exploring Exponential Functions $f(x) = ab^{k(x-d)}$

Part A: $f(x) = ab^{x-d}$

1) Use Desmos to graph the following functions and state the transformations for $g(x)$ and $h(x)$.

$f(x) = 2^x$
$g(x) = 2^{x+4}$ →
$h(x) = 2^{x-3}$ →

2) Fill in the table.

	$f(x) = 2^x$	$g(x) = 2^{x+4}$	$h(x) = 2^{x-3}$
Horizontal asymptote			
Y-intercept			

Part B: $f(x) = ab^{kx}$

3) Use Desmos to graph the following functions and state the transformations $g(x)$ and $h(x)$.

$f(x) = 2^x$
$g(x) = 2^{2x}$ →
$h(x) = 2^{\frac{1}{2}x}$ →

4) Fill in the table.

	$f(x) = 2^x$	$g(x) = 2^{2x}$	$h(x) = 2^{\frac{1}{2}x}$
Horizontal asymptote			
Y-intercept			

Part C: *Reflection*

5) Use Desmos to graph the following functions and state the transformations for $g(x)$.

$f(x) = 2^x$
$j(x) = \left(\frac{1}{2}\right)^x$ →

$g(x) = 2^{-x}$ →
$k(x) = \left(\frac{1}{2}\right)^{-x}$ →

6) Fill in the table below.

	$f(x) = 2^x$	$j(x) = \left(\frac{1}{2}\right)^x$	$g(x) = 2^{-x}$	$k(x) = \left(\frac{1}{2}\right)^{-x}$
Range				
Y-intercept				
Increasing/ Decreasing				

Part D: Equivalent Transformations

1) Graph each pair of functions on the same set of axes, using Desmos:

a) $f(x) = 2^{-x}$ $g(x) = \left(\frac{1}{2}\right)^{x}$

b) $f(x) = 2^{x}$ $g(x) = \left(\frac{1}{2}\right)^{-x}$

c) $h(x) = 9^{x}$ $k(x) = 3^{2x}$

d) $p(x) = 8^{x}$ $q(x) = 2^{3x}$

2) What do you notice about the pairs of graphs? Why does this make sense according to exponent laws?

3) For the following functions, give two equivalent functions that will produce the same graph.

a) $f(x) = 8(4)^{-x+2} - 3$

b) $g(x) = \frac{1}{27}(3)^{2x+6} + 4$

Summary of Transformations of the Form $f(x) = ab^{k(x-d)} + c$

Base		**Base points**	
Transformations			
Vertical stretch/compress, factor of	Horizontal stretch/compress, factor of	Horizontal translation	Vertical translation
If $a < 0$ If $\|a\| > 1$ If $\|a\| < 1$ Default: $a = 1$	If $k < 0$ If $\|k\| > 1$ If $\|k\| < 1$ Default: $k = 1$	If $d < 0$ If $d > 0$ Default: $d = 0$	If $c < 0$ If $c > 0$ Default: $c = 0$
Affects _____ by multiplication	Affects _____ by multiplication	Affects _____ by addition	Affects _____ by addition
Graphical Features			
Equation of horizontal asymptote	**Y-intercept**	**Domain** **Range** If $a > 0$ If $a < 0$	
Increasing or decreasing			
Increasing if b Then with every reflection,		Decreasing if b Then with every reflection,	

Graphing $f(x) = ab^{k(x-d)} + c$

1) Determine base value b to identify base points $\left(-1, \frac{1}{b}\right), (0, 1), (1, b), (2, b^2)$
2) Apply mapping notation formula $(x, y) \rightarrow \left(\frac{1}{k}x + d, ay + c\right)$
3) Draw and label the horizontal asymptote $y = c$

Lesson 4.6 – Transformations of the form $f(x) = ab^{(x-d)} + c$

Example: Fill in the table, then graph $f(x) = -3^{\frac{1}{2}x+1}$.

<table>
<tr><td colspan="2">Values of $a, b, k, d,$ and c</td><td rowspan="3">$f(x) = ab^{k(x-d)} + c$

Base → Transformed
$(x, y) \rightarrow \left(\frac{1}{k}x + d, ay + c\right)$

$\left(-1, \frac{1}{b}\right)$
$(0,1)$
$(1, b)$
$(2, b^2)$</td></tr>
<tr><td colspan="2">Base value</td></tr>
<tr><td colspan="2">Transformations (in order)</td></tr>
<tr><td>Equation of horizontal asymptote</td><td>Y-intercept</td><td rowspan="3">$f(x) = -3^{\frac{1}{2}x+1}$</td></tr>
<tr><td>Domain</td><td>Range</td></tr>
<tr><td colspan="2">Increasing or decreasing function</td></tr>
</table>

Modeling Exponential Growth and Decay

Data that appears to have a rapidly increasing or decreasing, nonlinear pattern can be modeled by the general exponential function $A(t) = A_0 r^{\frac{t}{n}}$, where

$A(t)$ represents	A_0 represents	r represents	t represents	n represents

- **Exponential growth** occurs when situations show increases in the value of a function, where the increases grow larger with time
  - This occurs when _____ (i.e. when the growth ratio is greater than 1)

- **Exponential decay** occurs when situations show decreases in the value of a function, where the decreases grow smaller with time
  - This occurs when _____ (i.e. when the decay ratio is positive but less than 1)

- The value of r represents the constant ratio between the values of the function
  - The constant ratio is found by dividing y-values (e.g. $\frac{y_2}{y_1}$)

Solving for the Initial or Final Amounts

Ex. 1) A colony of bacteria doubles in population every hour. There are currently 1000 bacteria.

a) Complete the following table to show the growth of bacteria.

Time (hours)	Number of bacteria
-3	
-2	
-1	
0	1000
1	
2	
3	

What is the rate of growth r?

$r =$

b) Write the function that models the scenario.

➔ Substitute your known values.
➔ Write let statements to introduce your unknown variables.

c) How many bacteria will there be after 24 hours?

d) How many bacteria were there 5 hours ago?

Ex.2) You invest $5000 at 6% interest per annum, compounded annually.

a) Complete the following chart to see how much money you have after 4 years. Determine a pattern that arises with each year's calculation – how can you determine the amount with one calculation?

Time (years)	Calculation	Amount ($)
0		
1		
2		
3		
4		

b) Model the scenario with an exponential function. Remember to write let statements!

c) Determine the worth of your investment after 10 years.

Ex. 3) The population of RGSS is increasing at a rate of 10% every 5 years. The population now is 1200 students. Write the function that models this situation and predict how many students there will be in 30 years when your children attend RGSS.

Ex. 4) The population of a town decreases at a rate of 8% every 4 years. The population now is 25 000. What was the town's population 20 years ago?

Solving for the Rate

If we do not know the rate of increase or decrease, we need to use reciprocal exponents or the t^{th} root to solve for the unknown base. First, choose the appropriate form of the exponential function!

*Note: When we increase by $i\%$, the rate is written as $r = 1 + i$, i as a decimal.
When we decrease by $i\%$, the rate is written as $r = 1 - i$, i as a decimal.

Ex. 5) The population of a town in 1980 was 24 000. In 1990 the population was 29 000. What is the yearly percentage increase in the population of the town?

Ex. 6) The value of a car today is $3800. If the car was purchased seven years ago for $19 500, determine the depreciation rate of the car's value.

Solving for Time

When we solve for an unknown exponent, we have to use either equal bases or the log button on our calculator.

Ex. 7) The population of a certain strain of e-coli bacteria triples every 2 hours. There are 240 bacteria now.

a) Write the function that models this situation. Remember to write let statements!

b) How long will it take for our bacteria population to grow to 19440 bacteria? (Use **equal bases** to solve this one…)

Ex. 8) The value of a car depreciates (decreases) about 20% per year. You buy a car today worth $24 000.

a) Write the function that models this situation. Remember to write let statements!

b) How many years would it take for the car to be worth half of its original value? (We need to guess and check, or use the **log button on our calculator** for this one…)

Half-Life

A special type of exponential decay is called **half-life**. The expression modeling half-life is

$$A(t) = A_0\left(\frac{1}{2}\right)^{\frac{t}{h}}$$

Ex. 1) Polonium^{210} has a half-life of 140 days. Assume that we start with 20g of polonium^{210}.

a) What is the significance of 140 days?

b) Write a function that models this situation. Remember to write let statements!

c) How much polonium^{210} will be left after 280 days?

Ex. 2) An isotope of radium is used by the hospital for cancer radiation. The half-life of radium is 1620 years.

a) If the hospital initially had 10mg of radium, write the function that models this situation.

b) How long will it be until there is only 6.25% of original radium sample remaining? (Hint: first convert 6.25% to a fraction…)

Example (homework part B #8): If a bacteria population doubles in 5 days,

a) When will it be 16 times as large as its present population?

b) When was it 1/4 of its present population?

Summary for Exponential Growth and Decay Application Problems:

The general form of the exponential growth and decay function is $A(t) = A_0 r^{\frac{t}{n}}$.

We use each of the following variations:

$A(t) = A_0(1+i)^{\frac{t}{n}}$	$A(t) = A_0(1-i)^{\frac{t}{n}}$	$A(t) = A_0\left(\frac{1}{2}\right)^{\frac{t}{h}}$

When we want to solve for an exponent (time or growth/decay period):	When we want to solve for a growth/decay rate:

Exponential Growth and Decay – Practice

1) A bacteria population of 1000 increases in population by 4.5% every three weeks. What will be the bacteria population after six months? (Note: there are 52 weeks in a year…)

2) Jessica purchased a diamond ring in 1997 for $3250. If the ring appreciates (increases) in value by 1.25% every year, what is the doubling period?

3) A car depreciates in value by i% per year. If Karim bought his car eight years ago for $30 000 and today it is worth $9800, determine the depreciation rate, i.

4) Carbon^{14} has a half-life of 5730 years. How much would remain today of a 50g sample of Carbon^{14}, if the sample began to decay 57 300 years ago?

5) A radioactive substance has a half-life of 20 minutes. How long will it take for the substance to decrease to 1/64th of its original amount?

Exponential Growth and Decay

1. The population of a city is 810 000. If it is increasing by 4% per year, estimate the population in four years.

2. A painting, purchased for $10 000 in 1990, increased in value by 8% per year. Find the value of the painting in the year 2000.

3. A river is stocked with 5000 salmon. The population of salmon increases by 7% per year.
   a. Write an expression for the population t years after the salmon were put into the river.

   b. What will the population be in
      i) 3 years?
      ii) 15 years?
   c. How many years does it take for the salmon population to double?

Knowledge/ Understanding

4. A house was bought six years ago for $175 000. If real-estate values have been increasing at the rate of 4% per year, what is the value of the house now?

5. A used-car dealer sells a five-year-old car for $4200. What was the original value of the car if the depreciation is 15% a year?

Knowledge/ Understanding

6. In the early 1990s, the Canadian dollar was declining in value due to inflation at the rate of 8.3% per year. If the situation continued, what would the dollar be worth five years later?

Part B

Application

7. To determine whether a pancreas is functioning normally, a tracer dye is injected. A normally functioning pancreas secretes 4% of the dye each minute. A doctor injects 0.50 g of the dye. Twenty minutes later, 0.35 g remain. If the pancreas were functioning normally, how much dye should remain?

8. If a bacteria population doubles in 5 d,
   a. when will it be 16 times as large?
   b. when was it $\frac{1}{2}$ of its present population?
   c. when was it $\frac{1}{4}$ of its present population?
   d. when was it $\frac{1}{32}$ of its present population?

9. Inflation is causing things to cost roughly 2% more per year.
   a. A bag of milk costs $3.75 now. Estimate its cost in five years.
   b. i) A movie ticket costs $8.50 now. If inflation continues at 2% per year, when will the ticket cost $10.00?
      ii) How long ago did the movie ticket cost $4.25?

10. An element is decaying at the rate of 12%/h. Initially we have 100 g.
    a. How much remains after 10 h?
    b. How much remains after 30 h?
    c. When will there be 40 g left?

Application

11. A research assistant made 160 mg of radioactive sodium (Na^{24}) and found that there was only 20 mg left 45 h later.
    a. What is the half-life of Na^{24}?
    b. Find a function that models the amount A left after t hours.
    c. If the laboratory requires 100 mg of Na^{24} 12 h from now, how much Na^{24} should the research assistant make now? (Ignore the 20 mg she currently has.)
    d. How much of the original 20 mg would be left in 12 h?

12. A bacteria colony grows at the rate of 15%/h.
    a. In how many hours will the colony double in size?
    b. In 10 h the bacteria population grows to 1.3×10^3. How many bacteria were there initially?

Application

14. The population of a city was estimated to be 125 000 in 1930 and 500 000 in 1998.
    a. Estimate the population of the city in 2020.
    b. If the population continues to grow at the same rate, when will the population reach 1 million?

Part C

Thinking/Inquiry/ Problem Solving

15. On the day his son is born, an excited father wants to give his new son a season's ticket to watch the father's favourite sports team. A season's ticket costs $900. The father realizes there is no point in buying tickets for a baby only a few hours old, so he decides to put the money aside until the boy is six years old. If inflation is assumed to be 3% per year, how much money should the father put aside so that he can purchase the season's ticket in six years?

Thinking/Inquiry/ Problem Solving

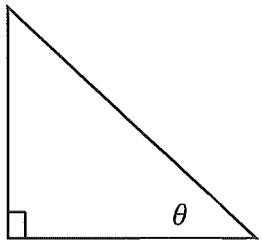
16. Two different strains of cold virus were isolated and put in cultures to grow. Virus A triples every 8 h while virus B doubles every 4.8 h. If each culture has 1000 viruses to start, which has more after 24 h?

1. 948 000
2. $21 600
3. a. $P = 5000(1.07)^5$ **b.** i) 6125 ii) 13 800 **c.** $10\frac{1}{4}$ years
4. $221 000
5. $9500
6. $0.65
7. 0.22 g
8. a. 20 days **b.** 5 days ago **c.** 10 days ago **d.** 25 days ago
9. a. $4.14 **b.** i) 8 years ii) 35 years ago
10. a. 28 g **b.** 2 g **c.** 7 h
11. a. 15 h **b.** $A = 160\left(\frac{1}{2}\right)^{\frac{t}{15}}$ **c.** 174 mg **d.** 11.5 mg
12. 5 h **b.** 320
14. a. 783 000 **b.** 2032
15. $1075
16. B

Unit 5:
Trigonometric Functions

Trigonometry Review - The Primary Trigonometric Ratios

The three primary trig ratios are the ratios of the lengths of two sides of a right triangle.



Label the sides *adjacent*, *opposite* and *hypotenuse* to the angle θ in the triangle. Write the general equations for $\sin\theta$, $\cos\theta$, and $\tan\theta$.

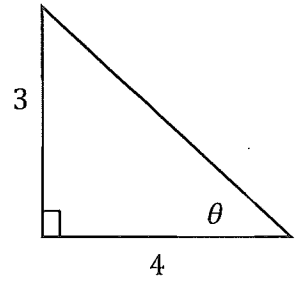
$\sin\theta =$ $\cos\theta =$ $\tan\theta =$

These ratios really mean the following:

$\sin\theta =$ how long the opposite side is compared to the hypotenuse $\cos\theta =$ $\tan\theta =$

Ex 1 Determine the primary trig ratios for the given triangle.

(*First, find the hypotenuse by using Pythagorean theorem, $c^2 = a^2 + b^2$, where c is the hypotenuse...)



$\sin\theta =$ $\cos\theta =$ $\tan\theta =$

The opposite side of angle θ is as long as the hypotenuse.

To find the measure of an angle θ

The trig ratios are $\sin\theta$, $\cos\theta$ and $\tan\theta$. The **inverse of the ratios** are called arcsin ($\sin^{-1}$), arcos ($\cos^{-1}$) and arctan ($\tan^{-1}$); the inverse is what gives us the measure of angle θ. The inverse operation can be accessed on your calculator by using the SHIFT or 2nd button.

Ex 2 Determine the measure of θ from example 1.

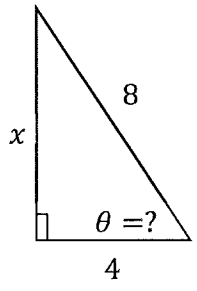
$\theta = \sin^{-1}$ or $\theta = \cos^{-1}$ or $\theta = \tan^{-1}$

The Reciprocal Trigonometric Ratios

The **RECIPROCAL** trig ratios are the reciprocal of the primary trig ratios.

Cosecant is the reciprocal of sine.	Secant is the reciprocal of cosine.	Cotangent is the reciprocal of tangent.

Ex 3 Determine the reciprocal ratios of angle θ. Then determine the measure of θ.



$\csc\theta =$ $\sec\theta =$ $\cot\theta =$

$\theta =$

Ex 4 Solve the triangle with $\angle D = 90°$, $e = 20\text{cm}$ and $f = 14\text{cm}$.

*Note: to *solve* a triangle means to determine

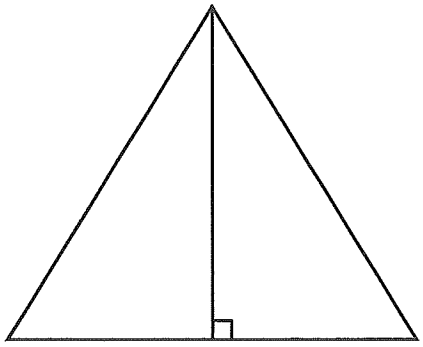
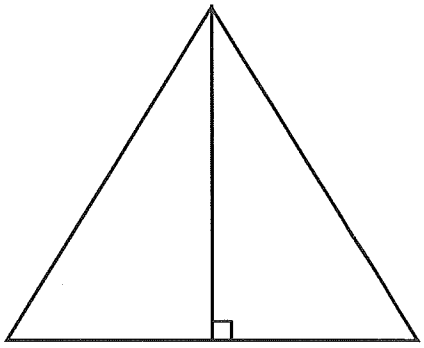
Investigation Part 1: The Primary Trig Ratios of 30° and 60°

Definitions

- **Equilateral triangles:**

- Primary trigonometric ratios
  Sine ratio: **Cosine** ratio: **Tangent** ratio:

Given the two equilateral triangles of different side lengths below, compare their primary trigonometric ratios.

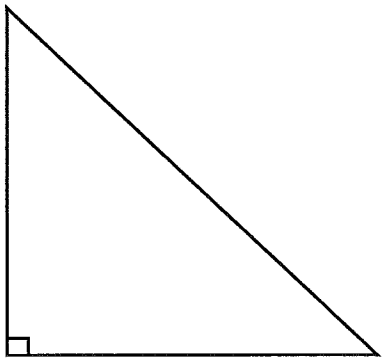
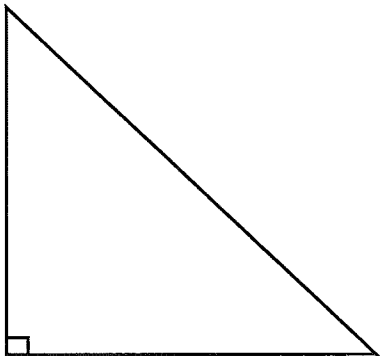
Instructions:	Example #1: Equilateral triangle of side length 2 units	Example #2: Equilateral triangle of side length 8 units
Cut the equilateral triangle in **half** to create two right triangles. Label the two triangles with their new angles and side lengths.		
Using the triangles, determine the primary trig ratios of 30° and 60°. Keep your answers in reduced fraction form.	$\sin 30° =$ $\cos 30° =$ $\tan 30° =$	$\sin 30° =$ $\cos 30° =$ $\tan 30° =$
	$\sin 60° =$ $\cos 60° =$ $\tan 60° =$	$\sin 60° =$ $\cos 60° =$ $\tan 60° =$

Investigation Part 2: The Primary Trig Ratios of 45°

Definition

- **Right isosceles triangles:**

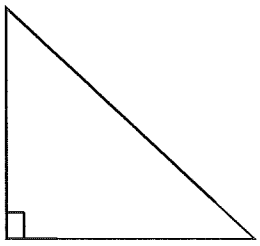
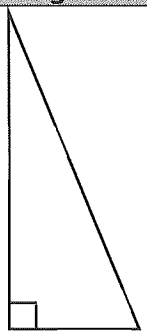
Complete the investigation by filling in the table below.

Instructions:	Example #1: Right isosceles triangle of equal side lengths 1 unit	Example #2: Right isosceles triangle of equal side lengths 6 units
Label the angles and the equal side lengths given. Calculate the length of the hypotenuse. Leave your answer as an exact value.		
Determine the primary trig ratios; keep your answers in reduced fraction form.	$\sin 45° =$ $\cos 45° =$ $\tan 45° =$	$\sin 45° =$ $\cos 45° =$ $\tan 45° =$

Conclusion: Regardless of the size (side length) of the triangle, the trig ratios for the special angles $30°$, $45°$ and $60°$ are...

Summary – The Exact Trigonometric Ratios of Special Angles

These special right triangles will help you remember the exact values of trig ratios for the special angles 30°, 45°, and 60°.

The 45°-45°-90° or 1-1-$\sqrt{2}$ triangle	The 30°-60°-90° or 1-2-$\sqrt{3}$ triangle
	

If you prefer to remember the table below, there is a useful pattern that will help you remember the ratios.

	$\theta = 30°$	$\theta = 45°$	$\theta = 60°$
$\sin\theta = \frac{opp}{hyp}$			
$\cos\theta = \frac{adj}{hyp}$			
$\tan\theta = \frac{opp}{adj}$			
$\csc\theta = \frac{hyp}{opp}$			
$\sec\theta = \frac{hyp}{adj}$			
$\cot\theta = \frac{adj}{opp}$			

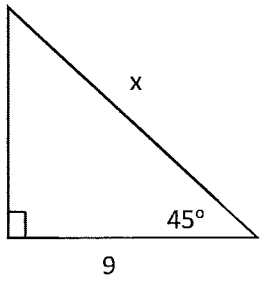
Solving for unknown sides in a right triangle

You can use what you've just learned about the special trigonometric ratios to solve for an unknown side length in a right triangle.

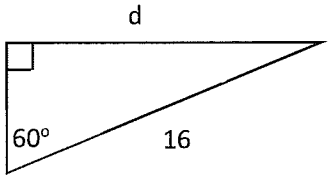
Given	Two sides	One side and an angle (not 90°)	All angles
Use			
Steps	1. Determine which side you need – the hypotenuse or one of the 'legs' 2. To determine the hypotenuse: To determine one of the 'legs'	1. With respect to the angle that you have, label the side you are given and the side you want to find (adj/opp/hyp) → E.g. Given adj side, need the opp side 2. Use the appropriate trig ratio that has both sides → Use $\tan\theta = \frac{opp}{adj}$ 3. Solve for the unknown side length using the appropriate operation → $opp = adj \cdot \tan\theta$	

Ex 5 Solve for the indicated side length without using a calculator.

a)



b)



Lesson 5.1 Recap

Primary Trig Ratios

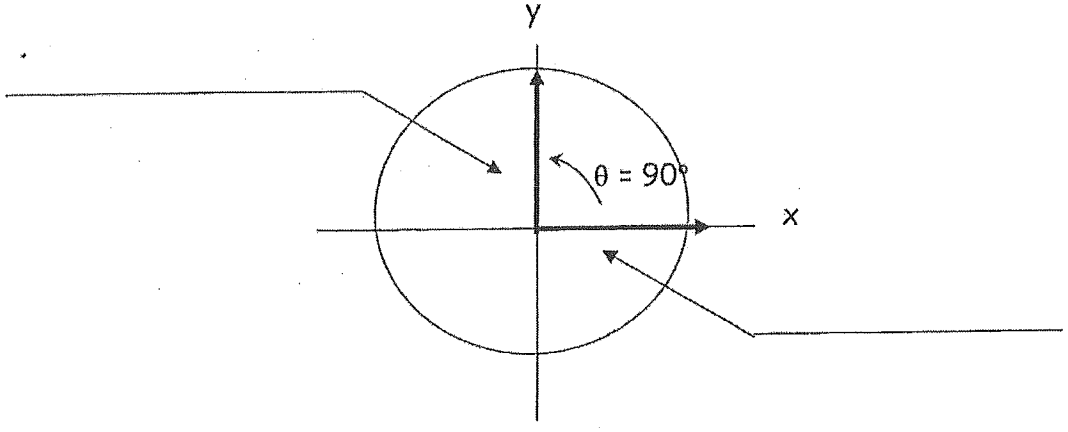
Reciprocal Trig Ratios

Special Triangles / Table – Trig Ratios of Special Angles

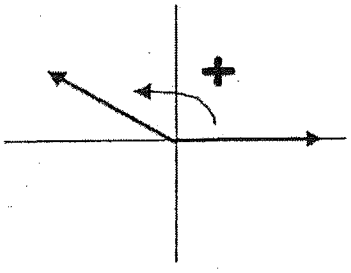
The Standard Position of an Angle and Coterminal Angles

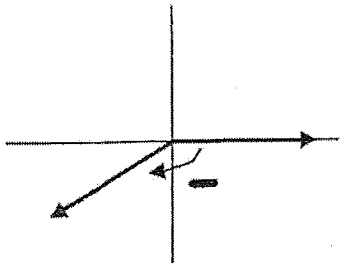
STANDARD POSITION is a common convention used to describe the size and orientation of angles. Every angle has an 'initial arm' and a 'terminal arm.' When using STANDARD POSITION, we represent the initial and terminal arms of an angle on an x-y axes system with the initial arm of the angle ALWAYS placed on the x-axis pointing right.

Label the initial arm and terminal arm for the angle 90° below.



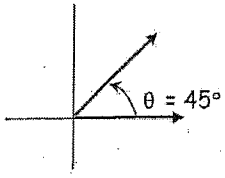
An angle can be either positive or negative. If the terminal arm of an angle opens in a counter-clockwise direction, the angle is positive. If the terminal arm opens in a clockwise direction, the angle is negative.





Ex 1: Draw the following angles in standard position. The first one has been done for you.

a) $45°$ b) $120°$ c) $-45°$ d) $420°$



COTERMINAL ANGLES are angles whose terminal arms have the same standard position.

Ex 2: For each angle below, find one positive and one negative coterminal angle.

	a) $30°$	b) $310°$	c) $-90°$
Positive coterminal angle			
Negative coterminal angle			

a) How many angles are coterminal with $30°$? Explain how you know.

b) Write an expression representing all angles coterminal with $30°$.

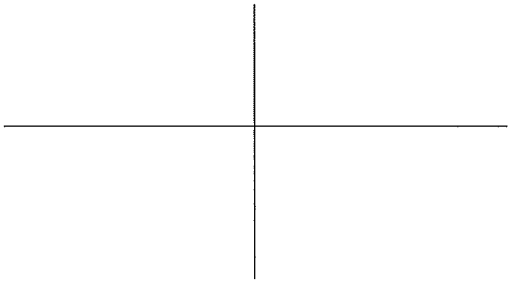
Determining the Reference Angle

The reference angle α is the acute angle located between the x-axis and the terminal arm of the angle in standard position.

Ex 1 Draw the following angles in standard position. Then determine the measure of the reference angle α.

a) $60°$ b) $150°$ c) $240°$ d) $315°$

Given an angle θ, calculate reference angle α using the following formulas:

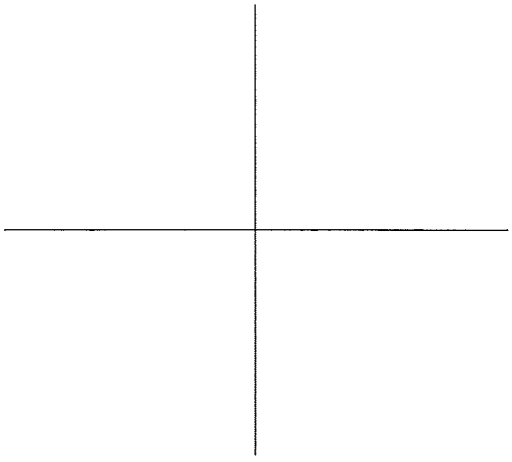


Defining Trig Ratios Given Coordinates

Standard position and reference angles will allow us to determine trigonometric ratios for angles that are greater than 90°. First, let's determine another way to express ratios other than using the terms 'adjacent', 'opposite' and 'hypotenuse'.

Take, for example, the acute angle θ (between 0° and 90°).

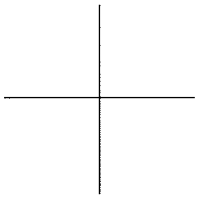
1. Pick a point $P(x, y)$ on the terminal arm.
2. Drop a <u>vertical</u> line to the x-axis to construct a right triangle.
3. Label the acute angle inside the triangle (between the terminal arm and the x-axis) as α.
4. Use the new right triangle constructed to determine the trig ratios.



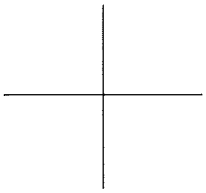
$\sin\theta =$	$\cos\theta =$	$\tan\theta =$
$\csc\theta = \frac{1}{\sin\theta} =$	$\sec\theta =$	$\cot\theta =$

Ex 1: Determine i) the primary trig ratios of the angle whose terminal arm passes through the given point and ii) the measure of the angle.

a) P (-3, 4)

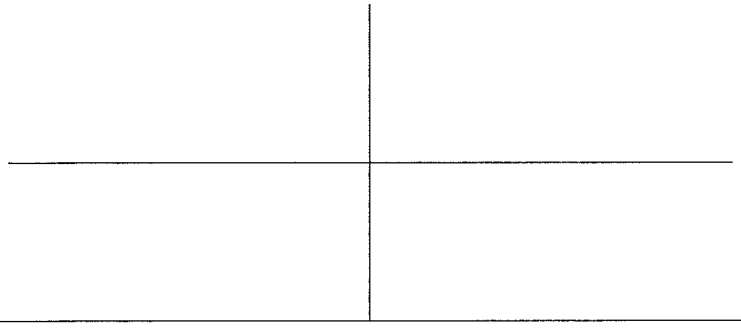


b) P (3, -4)



Steps to determining the exact values of the trig ratios and the angle passing through a point

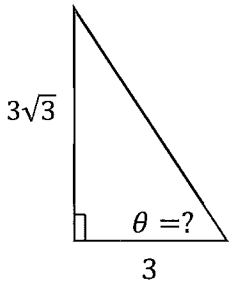
1. Given two sides of a right triangle, calculate the third side using _____.
2. State the ratios. Ensure that the ratio has the appropriate positive/negative sign.
3. Select the <u>positive</u> ratio in order to calculate the reference angle, α.
4. Calculate θ by using one of these formulas, depending on in which quadrant the terminal arm of θ lands:



Warm-Up: Recap of Lesson 5.2

1. Draw the special triangles/table that help you remember the trigonometric ratios of special angles.

2. Determine the indicated angle in the triangle below, without using a calculator.



3. Determine i) the primary trig ratios of the angle whose terminal arm passes through the point (-4, -9) and ii) the measure of the angle. Include a diagram.

Quadrant 2

Angles between 90° and 180° (and their coterminal angles):

x		$\sin\theta$	
y		$\cos\theta$	
r		$\tan\theta$	

Quadrant 1

Angles between 0° and 90° (and their coterminal angles):

x	+	$\sin\theta$	
y	+	$\cos\theta$	
r	+	$\tan\theta$	

Quadrant 3

Angles between 180° and 270° (and their coterminal angles):

x		$\sin\theta$	
y		$\cos\theta$	
r		$\tan\theta$	

Quadrant 4

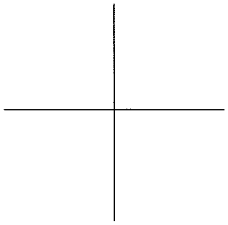
Angles between 270° and 360° (and their coterminal angles):

x		$\sin\theta$	
y		$\cos\theta$	
r		$\tan\theta$	

***** The trigonometric ratio of all angles that share the same reference angle will be the _____, but have different _____ because the triangles are all congruent.**

THE CAST RULE:

- We find the trig ratios based on which quadrant the terminal arm of the angle lands
  - All three trig ratios are positive in the first quadrant
  - Only _____ is positive in the second quadrant
  - Only _____ is positive in the third quadrant
  - Only _____ is positive in the fourth quadrant



Determining the exact value of the trig ratio using special triangles, α, and CAST rule

Steps to determining the trig ratio of an angle larger than 90°	Example
If the angle is not between 0° and 360°, then add/subtract increments of 360° to determine the coterminal angle. 1. Determine in which quadrant the terminal arm of your angle θ is in 2. Determine whether the ratio will be positive or negative (CAST rule) 3. Determine the measure of the reference angle α 4. Use the special triangles and the angle α to determine the exact trig ratio value; make sure to put the appropriate positive or negative sign.	Question: Determine the exact value of $\sin 240°$. Steps: 1. Terminal arm of 240° is in quadrant 2. Sine is in quadrant 3. In quadrant , $\alpha =$ 4. Refer to the special triangles/table $\sin 240°$

Ex 1: Determine the exact value of the following ratios:

a) $\sin 585°$ b) $\tan 330°$ c) $\sec 300°$

Determining the angle(s) given the trig ratio

Steps to determining the angle(s) given the trig ratio	Example
If the given ratio is a reciprocal ratio, change it to its corresponding primary ratio. 1. Determine in which quadrant(s) your angle will lie (CAST rule) 2. Select the positive ratio in order to calculate the reference angle, α. This ensures your calculator (if applicable) will give you the acute angle (since all ratios are positive in Quadrant 1). Use the appropriate special triangle (or calculator if permitted) 3. Calculate the measure of θ (depends on the quadrants)	Determine θ for $\cos\theta = \frac{1}{2}$, $0° \le \theta \le 360°$. 1. Cosine ratio is ; angle must lie in quadrants and 2. $\cos\theta = \frac{1}{2}$ means 1 is and 2 is ; referring to the special triangles/table, $\alpha =$ 3. In quadrant , $\theta =$ so $\theta_1 =$ In quadrant , $\theta =$ so $\theta_2 =$ $\therefore$ If $\cos\theta = \frac{1}{2}$, then $\theta =$ and

Ex 2: Determine θ for the given ratios if $0° \le \theta \le 360°$.

a) $\csc\theta = -2$ b) $\tan\theta = 1$ c) $\sec\theta = \frac{2\sqrt{3}}{3}$

Ex 3: (Use a calculator) Determine θ for the given ratios if $0° \le \theta \le 360°$.

a) $\sin\theta = 0.8432$ b) $\cot\theta = 0.2438$ c) $\cos\theta = 0.5921$

Warm-Up

1. Draw the special triangles/table that help you remember the trigonometric ratios of special angles.

2. Show CAST rule and explain what it tells us about trigonometric ratios.

3. Determine the exact value of the ratios given, without using a calculator:
   a) $\cos 240°$ b) $\sin 135°$ c) $\cot 330°$

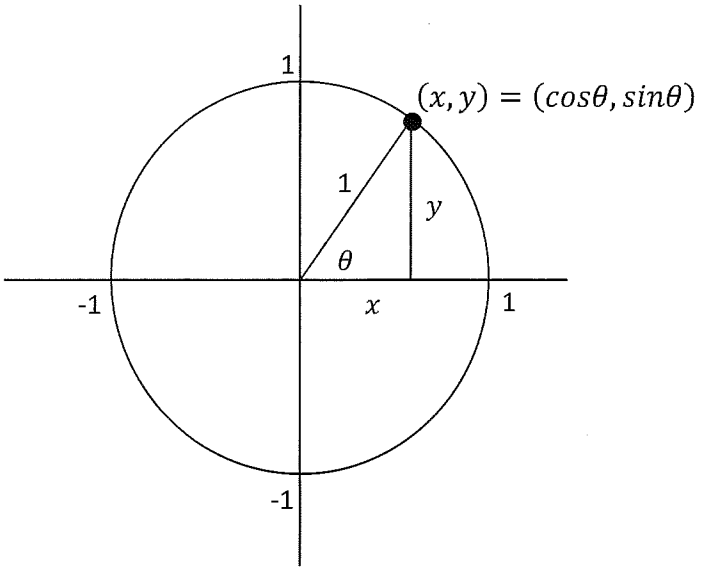
4. Determine θ for the given ratios if $0° \leq \theta \leq 360°$.
   a) $\sin\theta = -\frac{1}{2}$ b) $\cot\theta = \frac{\sqrt{3}}{3}$ c) $\cos\theta = -0.7660$

The Unit Circle

The **unit circle** is a <u>circle that has radius of 1 unit</u>. In a unit circle, the ratios can be defined as:

$\sin\theta = \frac{y}{r} =$ $\quad\quad$ $\cos\theta = \frac{x}{r} =$ $\quad\quad$ $\tan\theta =$

So for any point P on the unit circle, the coordinates of P are $P(x, y) = (\cos\theta, \sin\theta)$



The values of coordinates ARE the trig ratios themselves...NEAT!!

**Note: This works ONLY for the circle with radius of 1 unit!*

Ex 1: Use the unit circle to determine the following trig ratios:

a) $\cos 180°$ $\quad\quad$ b) $\csc 360°$ $\quad\quad$ c) $\tan 270°$

Ex 2: Determine θ for the given ratios, $0° \leq \theta \leq 360°$, using the unit circle.

a) $\sin\theta = 1$ $\quad\quad$ b) $\sec\theta = -1$ $\quad\quad$ c) $\cot\theta =$ undefined

Ex 3: Determine the coordinates (x, y) at which $\theta = 210°$ intersects the unit circle.

Summary of Lessons 5.1-5.4

Special triangles, unit circle:

	$\theta = 0°$	$\theta = 30°$	$\theta = 45°$	$\theta = 60°$	$\theta = 90°$
$\sin\theta = \frac{opp}{hyp}$					
$\cos\theta = \frac{adj}{hyp}$					
$\tan\theta = \frac{opp}{adj}$					

Standard Position, Reference Angles	Coterminal Angles	New Definitions of Trig Ratios	CAST rule

Grade 11 Trig Review

1. Predict whether each value will be positive or negative. Sketch each angle on a coordinate grid.
   a) tan 167° b) sin 99° c) cos 132°

2. Given that $0° \le \theta \le 180°$, determine the value(s) of θ to 1 decimal place.
   a) $\cos\theta = 0.4772$ b) $\tan\theta = -0.2272$ c) $\sin\theta = 0.5476$
   d) $\tan\theta = 1.6191$ e) $\sin\theta = 0.3486$ f) $\cos\theta = -0.5577$

3. Angle θ is obtuse.
   a) $\tan\theta = -0.4452$; calculate $\sin\theta$ to 4 decimal places.
   b) $\sin\theta = 0.9707$; calculate $\cos\theta$ to 4 decimal places.

8. Sketch each angle θ in standard position, then write a coterminal angle.
   a) $\theta = 170°$ b) $\theta = 293°$ c) $\theta = -30°$ d) $\theta = -320°$
   e) $\theta = 450°$ f) $\theta = 600°$ g) $\theta = -370°$ h) $\theta = 200°$

9. Determine two angles between 0° and 360° that have each trigonometric function value. Write the angle to the nearest degree.
   a) $\sin\theta = 0.42$ b) $\cos\theta = -0.31$ c) $\tan\theta = 3.46$

10. The point P(4, –15) lies on the terminal arm of an angle θ in standard position. Determine each trigonometric function value to 3 decimal places.
    a) $\sin\theta$ b) $\cos\theta$ c) $\tan\theta$

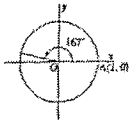
11. The terminal arm of an angle θ lies in Quadrant II on the line with equation $4x + 3y = 0$. Determine each trigonometric function value.
    a) $\sin\theta$ b) $\cos\theta$ c) $\tan\theta$

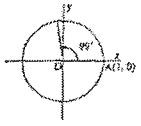
12. State each exact value. Do not use a calculator.
    a) cos 135° b) tan 225° c) sin 210° d) $\frac{1}{\tan 60°}$

13. Simplify each expression. Do not use a calculator.
    a) sin 30° + cos 60° b) tan 45° + tan 225° c) sin 240° + cos 300°

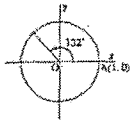
Answers:

1. a) Negative b) Positive





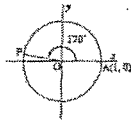
c) Negative



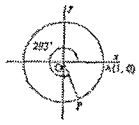
2. a) 61.5° b) 167.2° c) 33.2°, 146.8°
   d) 58.3° e) 20.4°, 159.6° f) 123.9°

8. Coterminal angles may vary.

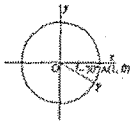
a) 530°



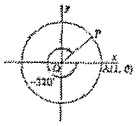
b) 653°



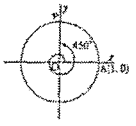
c) 330°



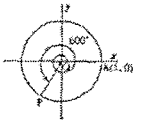
d) 40°



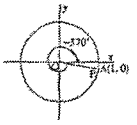
e) 90°



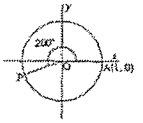
f) 240°



g) 350°



h) 560°



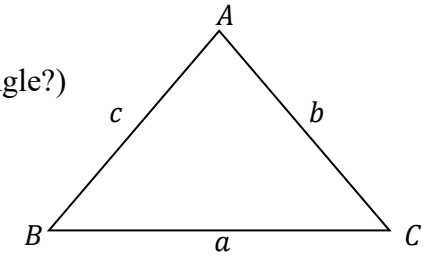
9. a) 25°, 155° b) 108°, 252° c) 74°, 2
10. a) –0.966 b) 0.258 c) –3.750
11. a) $\frac{4}{5}$ b) $-\frac{3}{5}$ c) $-\frac{4}{3}$
12. a) $-\frac{1}{\sqrt{2}}$ b) 1 c) $-\frac{1}{2}$ d) $\frac{1}{\sqrt{3}}$
13. a) 1 b) 2 c) $\frac{1-\sqrt{3}}{2}$

Cosine Law and 3-D Triangle Problems

Remember from last year that in order to solve non-right angle triangles, we used either the Sine Law or Cosine Law:

Cosine Law → $c^2 = a^2 + b^2 - 2ab\cos C$	**Sine Law** → $\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$

(Remember how to label a triangle?)



When do we use cosine law?

When do we use sine law?

Ex.1) In a triangle, $\angle A = 75°$, $b = 24cm$ and $c = 18cm$. Solve the triangle.

Ex. 2) In a triangle, $a = 15cm$, $b = 18cm$ and $c = 23cm$. Solve the triangle.

Application Problems – remember to draw a diagram first!

3-D Triangle Problems

Ex. 3) Mike is standing at the top of a building that is 5m tall. In one direction, he can see the bus stop at an angle of depression of 25º. In the other direction, he can see Richmond Green at an angle of depression of 10º. If the angle between his two sightlines is 125º, find the distance between the bus stop and Richmond Green.

*Draw a 3-D diagram!

- Mike – top of a 5m-tall building
- In one direction: bus stop at an angle of depression of 25º (same as the **angle of elevation** from the bus stop up to Mike)
- In another direction: RGSS at an angle of depression of 10º
- Angle between sightlines is 125º (triangle on the ground)

Ex. 4) Samantha leaves her house walks 850m to the local convenience store. After buying an ice cream cone, she turns right through an angle of 120^o and walks up a hill with an angle of elevation of 20^o to reach the library. At this point, she has walked a total distance of 1500m. If there is a direct path from the library back to Samantha's house, determine how far it is and what the angle of depression would be.

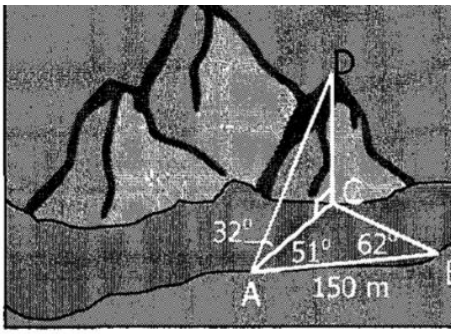
*Draw a 3-D diagram! You may want to separate it into several different views of the triangles.

- Starts at her house
- Walks 850m to the convenience store
- Turns right through an angle of 120^o
- Walks up a hill with an angle of elevation of 20^o to the library
- Total distance is 1500m (house → convenience store → library)

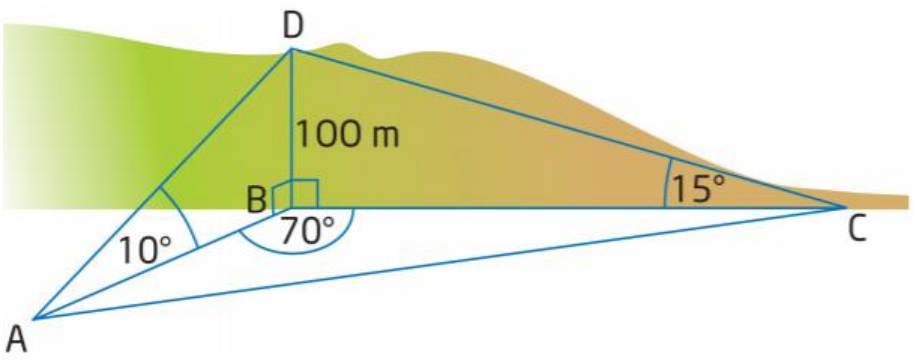
Reviewing Lesson 5.5 – 3D Trig Problems

Ex.1)

A surveyor is on one side of a river. On the other side is a cliff of unknown height that she wants to measure. The surveyor lays out a baseline AB of length 150 m. From point A, she selects point C at the base of the cliff and measures $\angle CAB$ to be 51°. She selects point D on top of the cliff directly above C and measures am angle of elevation of 32°. She moves to point B and measure $\angle CBA$ as 62°. Determine the height of the cliff.



Ex.2) Determine the length of AC.



Warm-Up: Review of Solving Triangles

	Given a right triangle…	Given a SSS or SAS triangle...	Given a SSA or ASA or AAS triangle…
Diagram			
What trigonometric concept(s) do we use to solve the triangle?			
Formulae			

The Ambiguous Case of the Sine Law (SSA)

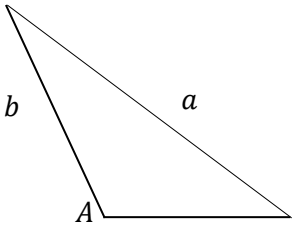
The ambiguous case in SSA triangles occurs when we are given two sides and a non-included angle, and it is unclear how many triangles can be formed.
This situation happens when:

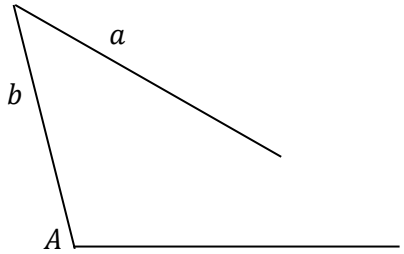
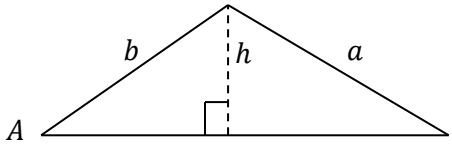
- The angle given is acute
- The side across from the given angle is shorter than the other side
- The height of the triangle is shorter than the two given sides

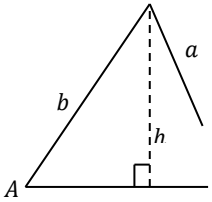
e.g. Draw two different triangles that satisfy the given information: $\angle A = 53°,\ a = 23cm,\ b = 26cm.$

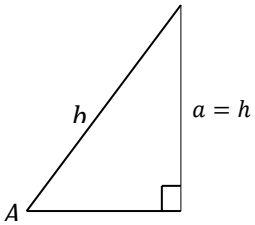
Determining the Number of Solutions in SSA Triangles

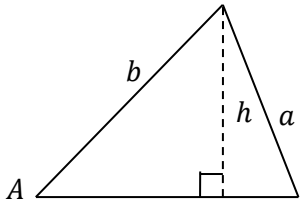
To determine how many possible triangles exist when given **angle A, side a, and side b**, we consider whether angle A is acute or obtuse.

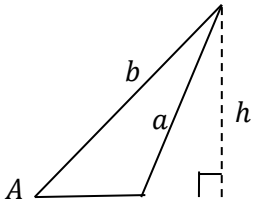
<table>
<tr><th>If $\angle A$ is obtuse…</th><th>If $\angle A$ is acute…</th></tr>
<tr><td>If $\boldsymbol{a > b}$, one triangle exists.
 

If $\boldsymbol{a \leq b}$, no triangle exists.
 
</td><td>If $\boldsymbol{a \geq b}$, one triangle exists.
 

If $\boldsymbol{a < b}$:
a) Draw the perpendicular line h from the point where the two sides a and b meet to the opposite side.
b) Calculate h by using the sine ratio.
1) If $\boldsymbol{a < h}$, no triangle exists.
 

2) If $\boldsymbol{a = h}$, one triangle exists.
 

3) If $\boldsymbol{h < a < b}$, two triangles exist.
 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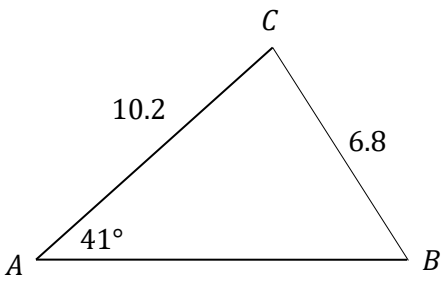
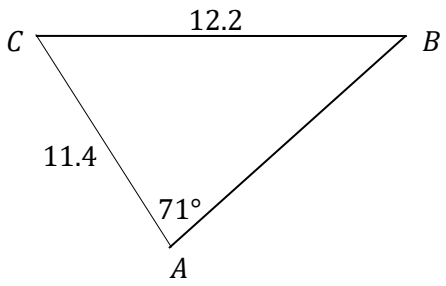
Ex. 1) Determine whether each triangle has no solution, one solution or two solutions.

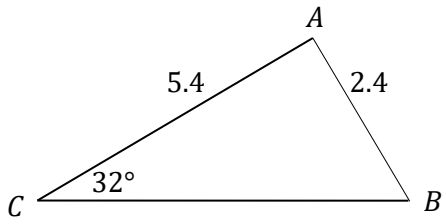
a)



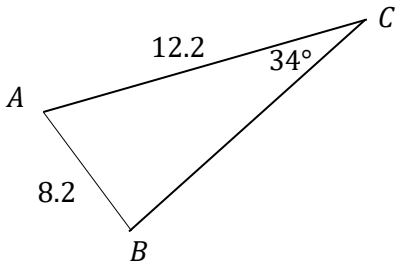
b)



c)

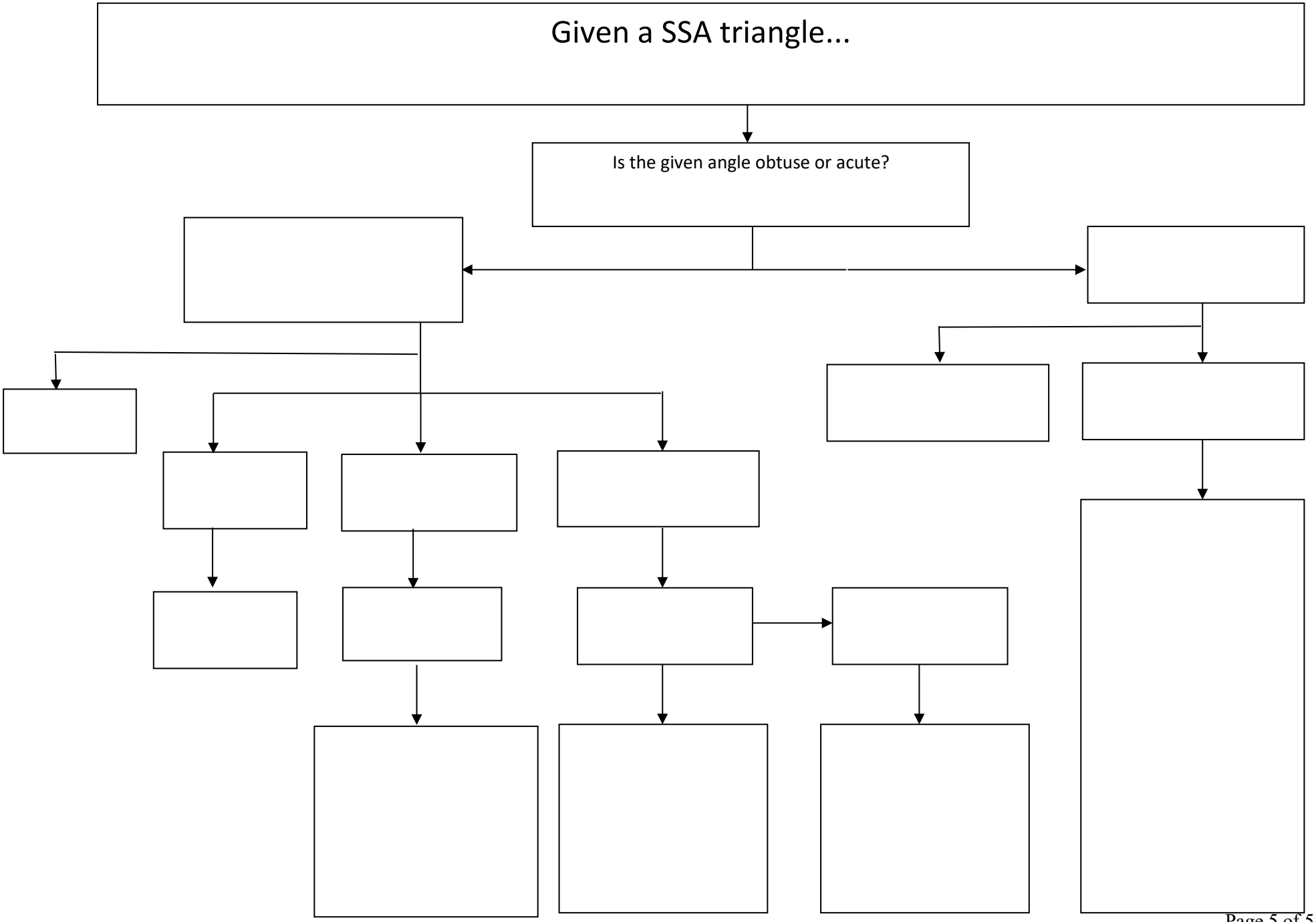


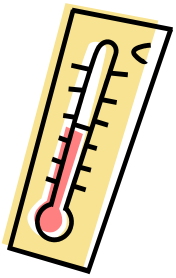
d)



Ex.2) ΔABC is given with $\angle A = 41°$, $a = 23$ and $b = 28$. Solve ΔABC to the nearest degree and lengths to one decimal place.

Ex.4) Nicole and Jessica are part of a scientific team studying thunderclouds. The team is about to launch a weather balloon into an active part of a cloud. Nicole's rope is 7.8m long and makes an angle of 36^{o} with the ground. Jessica's rope is 5.9m long. How far, to the nearest tenth of a metre, is Nicole from Jessica?


Given a SSA triangle...
Is the given angle obtuse or acute?



Warm-Up!!

In triangle ABC, side a = 14 cm, side b = 17 cm, and A = 54°. Solve the triangle.

Lesson 5.7 Trigonometric Identities Part I

Proving Trigonometric Identities

An **identity** is a mathematical statement which is true for all values of the variable for which the statement is defined.

e.g. $(x+1)^2 = x^2 + 2x + 1$ is true for all $x, x \in \mathbb{R}$.
$sin\theta = \sin(\theta + 360°)$ is true for all $\theta \in \mathbb{R}$.

An identity is different from an equation because an equation is only true for certain values of the variable

e.g. $\sin\theta = \frac{1}{2}$ is only true for $\theta = 30°, 150°$ and their coterminal angles (but not true for all values of $\theta \in \mathbb{R}$.)

Recall: $sin\theta =$ $cos\theta =$ $tan\theta =$

A famous Pythagorean Trig Identity: $\sin^2\theta + \cos^2\theta = 1 \rightarrow$ same as $(\sin\theta)^2 + (\cos\theta)^2 = 1)$

Proof of the Pythagorean Trig Identity:

Other identities you need to know:

Reciprocal Identities			Quotient Identities	
$\sec\theta = \frac{1}{\cos\theta}$	$\csc\theta = \frac{1}{\sin\theta}$	$\cot\theta = \frac{1}{\tan\theta}$	$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$

The Importance of Separating Left Side from Right Side

When proving a trigonometric identity, you MUST separate the L.S. (left side) from the R.S. (right side)! It is unacceptable (mathematically incorrect) to move terms from one side of the equal sign to the other; remember, you are not trying to solve an equation, you are trying to PROVE that the left side equals right side. As soon as you move a term from one side to the other, your entire proof is incorrect.

Also, you must **work with one side only**! This is the proper mathematical way to prove identities.

Lesson 5.7 Trigonometric Identities Part I

Reciprocal Identities			Quotient Identities		Pythagorean Identity
$\sec\theta = \frac{1}{\cos\theta}$	$\csc\theta = \frac{1}{\sin\theta}$	$\cot\theta = \frac{1}{\tan\theta}$	$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$	$\sin^2\theta + \cos^2\theta = 1$

Tips to proving a trigonometric identity:	For full marks, remember to
• Work with the more complex side first; visualize how you can make one side look like the other • Change reciprocal identities to primary ones • If you see $\sin^2\theta$ or $\cos^2\theta$, use Pyathagorean identity • Factor where appropriate • If there are fractions, simplify using the LCD (lowest common denominator) • Use the conjugate binomial if necessary (see example 3)	• Separate into L.S. and R.S. • Your last step should state L.S. = R.S. if you have proved it to be so • Work on one side only • NEVER move terms from one side to the other side

Let's try…

1. $\sin\theta \cdot \sec\theta = \tan\theta$

2. $\sin^2\theta \cot^2\theta + \sin^2\theta = 1$

3. $\frac{\cos^2\theta}{1-\sin\theta} = 1 + \sin\theta$

4. $\frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} = \csc\theta \sec\theta$

Lesson 5.7 Trigonometric Identities Part I

Reciprocal Identities			Quotient Identities		Pythagorean Identity
$\sec\theta = \frac{1}{\cos\theta}$	$\csc\theta = \frac{1}{\sin\theta}$	$\cot\theta = \frac{1}{\tan\theta}$	$\tan\theta = \frac{\sin\theta}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$	$\sin^2\theta + \cos^2\theta = 1$

5. $(\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2 = 2$

6. $\frac{\sin\theta - 1}{\cos\theta} = \tan\theta - \sec\theta$

HOMEWORK: Prove the following identities.

1. $\dfrac{\sin^2 A + \cos^2 A}{\cos^2 A} = \sec^2 A$

2. $\sin^2 A + \tan^2 A + \cos^2 A - \sec^2 A = 0$

3. $\sec A = \dfrac{\cot^2 A + 1}{\csc A \cdot \cot A}$

4. $\dfrac{\sin A}{\csc A} + \dfrac{\cos A}{\sec A} = 1$

5. $\sec^2 A + \csc^2 A = \sec^2 A \cdot \csc^2 A$

6. $\tan B \cdot \cot B = \sin^2 B + \cos^2 B$

7. $\tan B + \cot B = \sec B \cdot \csc B$

8. $\tan B \cdot \sin B = \sec B - \cos B$

9. $\dfrac{1}{\sin A \cdot [\tan A + \cot A]} = \cos A$

10. $\dfrac{\cot B - \cos B}{\cot B \cdot \cos B} = \dfrac{\cot B \cdot \cos B}{\cot B + \cos B}$

Trigonometric Identities Review

Without looking at your notes, state as many identities as you remember:
(Hint: reciprocal, quotient, Pythagorean…)

Prove the following identities. **Make sure to work on one side only!**

1. $\frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}$

2. $1 - \cos^2\theta = \sin\theta\cos\theta\tan\theta$

3. $\frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos^2\theta - \sin^2\theta$

More Trigonometric Identities Practice

1. $\dfrac{1-\cos^2 x}{\sin x} = \sin x$

2. $\sin^2\theta + \dfrac{\sin\theta\cos\theta}{\tan\theta} = 1$

3. $1 + \dfrac{1}{\tan^2\beta} = \dfrac{1}{\sin^2\beta}$

4. $2\sin^2\theta - 1 = \sin^2\theta - \cos^2\theta$

5. $\dfrac{1}{\cos\Phi} - \cos\Phi = \sin\Phi\tan\Phi$

6. $\sin\alpha + \tan\alpha = \tan\alpha(1+\cos\alpha)$

7. $\dfrac{1}{1-\sin^2 x} = 1 + \tan^2 x$

8. $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$

9. $\sin^2\theta + \cos^2\theta + \tan^2\theta = \dfrac{1}{\cos^2\theta}$

10. $\dfrac{\sin\phi}{\sin\phi + \cos\phi} = \dfrac{\tan\phi}{1+\tan\phi}$

11. $\dfrac{1+\tan^2\theta}{1-\tan^2\theta} = \dfrac{1}{\cos^2\theta - \sin^2\theta}$

12. $\dfrac{1}{\sin^2\theta} + \dfrac{1}{\cos^2\theta} = \dfrac{1}{\sin^2\theta\cos^2\theta}$

13. $\dfrac{1}{1-\cos x} + \dfrac{1}{1+\cos x} = \dfrac{2}{\sin^2 x}$

14. $(\sin x + \cos x)^2 = 1 + 2\sin x\cos x$

15. $(1-\cos^2\theta)\left(1 + \dfrac{1}{\tan^2\theta}\right) = 1$

16. $\left(1+\tan^2\beta\right)\left(1-\cos^2\beta\right) = \tan^2\beta$

17. $\dfrac{\sin x - 1}{\sin x + 1} = \dfrac{-\cos^2 x}{(\sin x + 1)^2}$

18. $\sin^4\theta + 2\sin^2\theta\cos^2\theta + \cos^4\theta = 1$

19. $\dfrac{4}{\cos^2\phi} - 5 = 4\tan^2\phi - 1$

20. $\dfrac{\cos\alpha - \sin\alpha - \cos^3\alpha}{\cos\alpha} = \sin^2\alpha - \tan\alpha$

21. $\dfrac{\sin^2\beta - 6\sin\beta + 9}{\sin^2\beta - 9} = \dfrac{\sin\beta - 3}{\sin\beta + 3}$

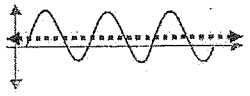
Unit 6:
Sinusoidal Functions

Exploring Periodic Motion

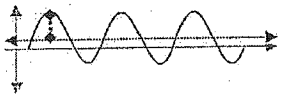
Important Definitions for Periodic Functions

Word Bank: Central axis, maximum/peak, minimum/trough, cycle, period, amplitude, domain, range, phase shift, vertical displacement

1. _____ The set of x-values upon which a function is defined.
2. _____ The horizontal line that lies exactly half-way between the maximum and minimum values of the graph of a periodic function.



3. _____ The amount by which a periodic function is horizontally translated.
4. _____ The length of one cycle; *graphically*, the amount along the horizontal axis it takes a periodic function to complete one cycle.
5. _____ The height of the central axis of a periodic function or the amount by which a periodic function is vertically translated.
6. _____ Half the distance between the maximum and the minimum values of a periodic function; the distance from the central axis to the minimum or maximum value of a periodic function.



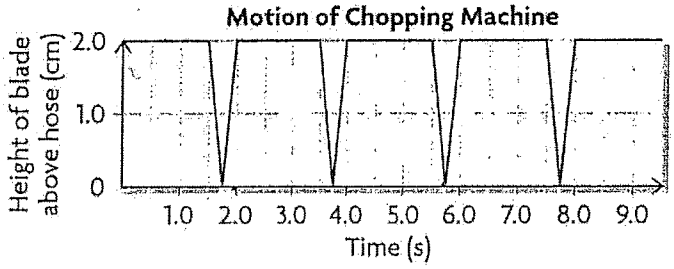
7. _____ The set of y-values for which a function is defined.
8. _____ A single complete execution of a periodically repeated phenomenon.
9. _____ The largest value of a function; the highest y-value on the graph of a function.
10. _____ The smallest values of a function; the lowest y-value on the graph of a function.

Key Idea: When you repeat a motion in exactly the same way, the graph of that motion is periodic. By looking at its graph, you can determine details of the motion and make predictions.

Practice:

1) Tanya's mother works at a factory that makes rubber hose. A chopping machine cuts each hose to 5.0 m lengths. How can Tanya interpret the graph that shows the process?

Study the graph below and try to identify certain patterns.



a) The cutting blade cuts a new section of hose every _____ since the graph has a pattern that repeats during this time (the period).

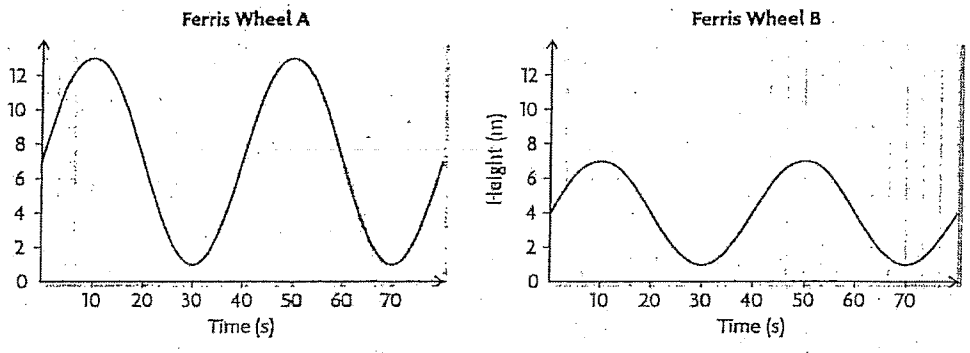
b) The height (peak) is always _____ or less, so the blade cannot be higher than this length.

c) When the height is _____, the blade is hitting the cutting surface (trough).

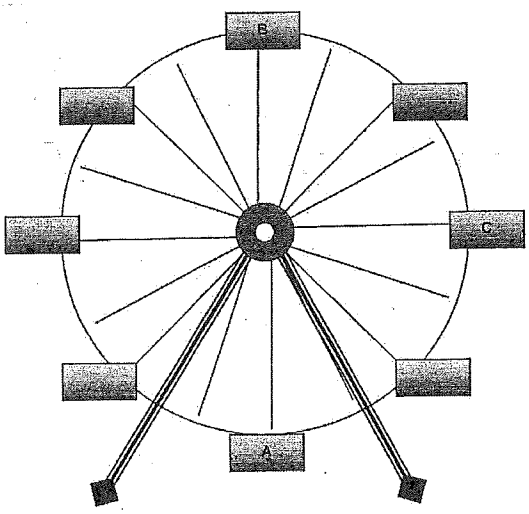
d) Flat sections like the one from 2.0 to 3.5 means _____

e) From the graph, it takes the blade _____ to go down and _____ to go up.

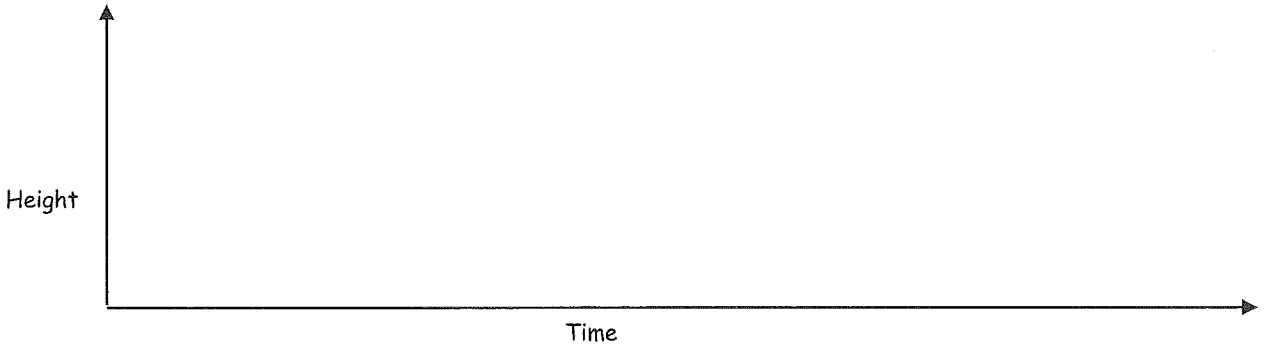
2) Maria, Shannon and Rachel decide to go to an amusement park. Maria rides on Ferris Wheel A while Shannon rides on Ferris Wheel B. Rachel collects the data while enjoying an ice cream waffle. Compare the periods, central axis and the amplitude of the functions.



3) Imagine you are taking a Ferris wheel ride. You climb on at the bottom and are slowly raised up, rotating in a counter-clockwise direction.



a) Sketch a graph to show how the height of car A varies with time. You may assume the Ferris wheel makes at least three revolutions.

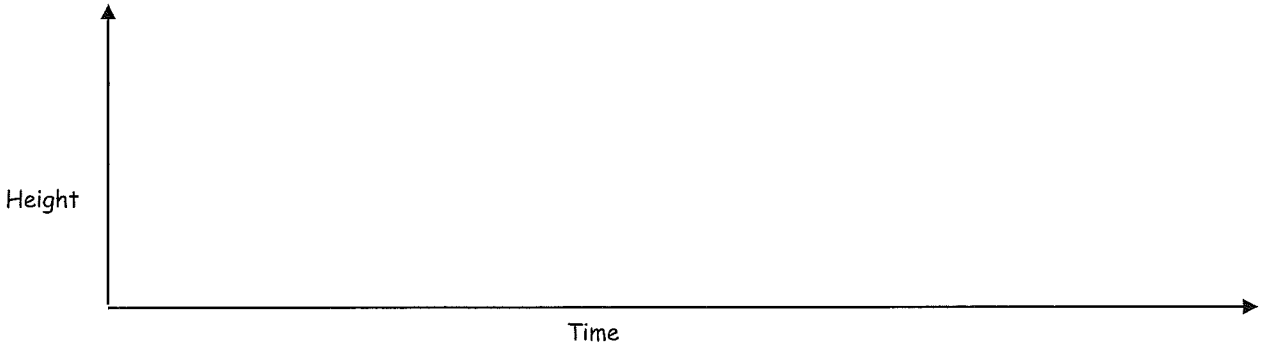


b) How will your graph change if the wheel turns faster? Sketch on the same axes.

c) How will your graph change if the radius of the wheel is larger? Sketch.

d) How will your graph change if the axle of the wheel is higher up off the ground? Sketch.

e) Using different coloured pens/pencils, sketch two more graphs to show how the height of car B and the height of car C vary with time.



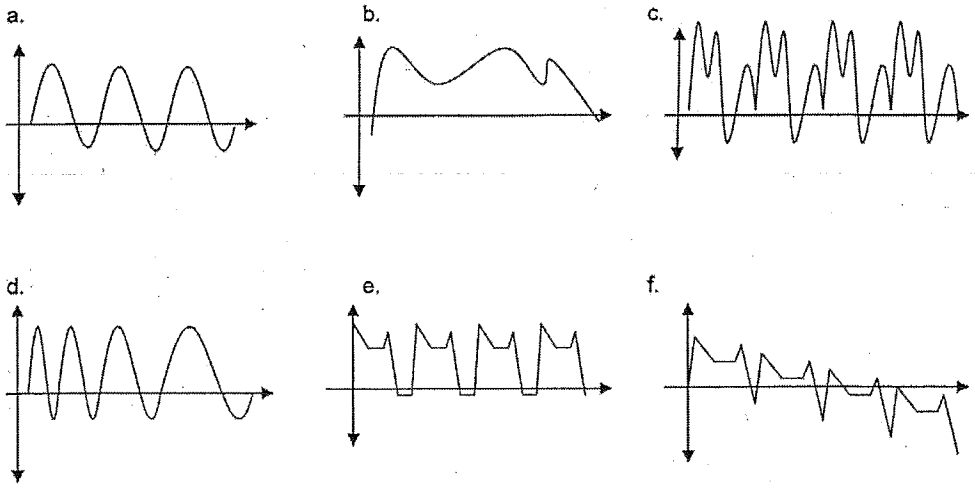
f) How are these graphs different from the graph in a)? How are they the same?

Sinusoidal Functions

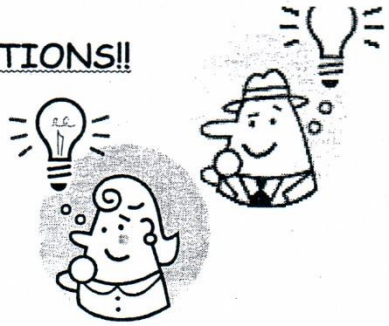
A periodic function is a function where the y-values are repeated over equal intervals of x-values.

A sinusoidal function is a type of periodic function that has a characteristic wave shape.

Practice: Cross out the functions below which are NOT periodic.



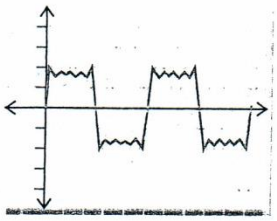
... CHECK YOUR UNDERSTANDING OF PERIODIC FUNCTIONS!!



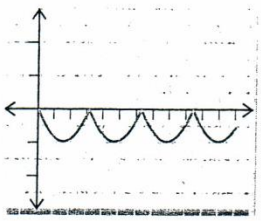
Class/home work!

1. Which graphs represent periodic functions? Justify your decision.

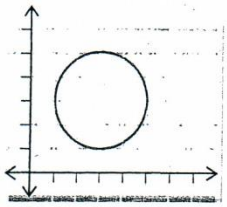
a)



b)



c)

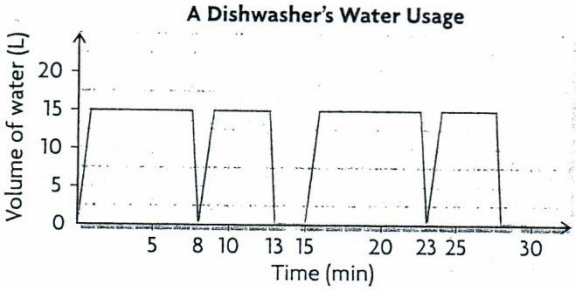


4. A coach measures the velocity of air as a gymnast inhales and exhales after working out.

a) Plot the data from the table and draw a curve.
b) State the period of this function.
c) Explain why some velocities are positive and others are negative.
d) Extend the graph for the period between 9 s and 19 s.
e) Predict when the velocity of air will be 0 m/s for the period between 9 s and 19 s.

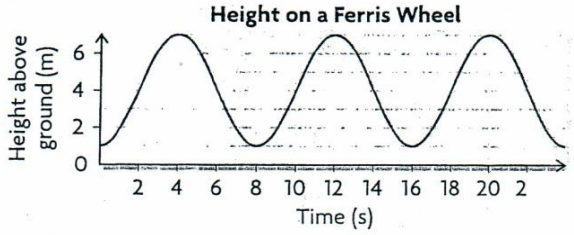
Time (s)	Velocity of Air (L/s)
1.0	1.75
1.5	1.24
2.0	0
2.5	−1.24
3.0	−1.75
3.5	−1.24
4.0	0
4.5	1.24
5.0	1.75
5.5	1.24
6.0	0
6.5	−1.24
7.0	−1.75
7.5	−1.24
8.0	0
8.5	1.24
9.0	1.75

5. The graph shows the amount of water used by an automatic dishwasher as a function of time.



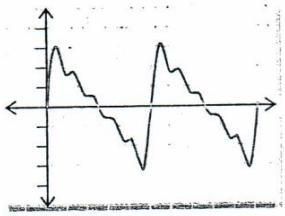
a) Why does the operation of the dishwasher model a periodic function?
b) What is the period? What does one complete cycle mean?
c) Extend the graph for one more complete cycle.
d) How much water is used if the dishwasher runs through eight complete cycles?
e) For part (d), state the domain and range of the function.

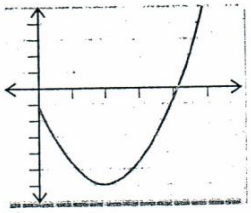
6. This is a graph of Nali's height above the ground in terms of time while riding a Ferris wheel.

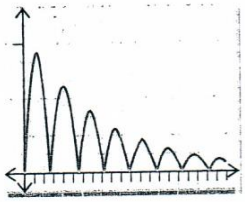


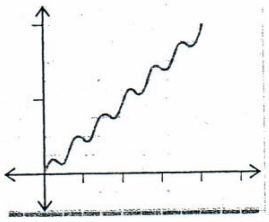
a) What is the period of this function?
b) What does the period represent?
c) What is the diameter of the Ferris wheel? How do you know?
d) Approximately how high above the ground is Nali at 10 s?
e) At what times is Nali at the top of the wheel?
f) When is Nali 4 m above the ground?

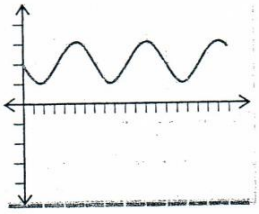
7. Which graphs represent periodic functions? Explain how you know.

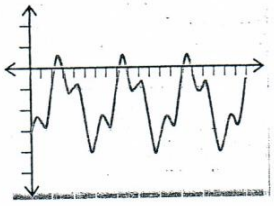
a)


c)


e)


b)


d)


f)


8. Which graphs in #7 represent sinusoidal functions?

Investigating the Sine and Cosine Function

Let's examine the base **sinusoidal functions** $y = \sin\theta$ and $y = \cos\theta$.

Complete the table of values for $y = \sin\theta$. Keep your answers to 2 decimal places.

θ°	0	30	45	60	90	120	135	150	180	210	225	240	270	300	315	330	360
$\sin\theta$																	

Complete the table of values for $y = \cos\theta$. Keep your answers to 2 decimal places.

θ°	0	30	45	60	90	120	135	150	180	210	225	240	270	300	315	330	360
$\cos\theta$																	

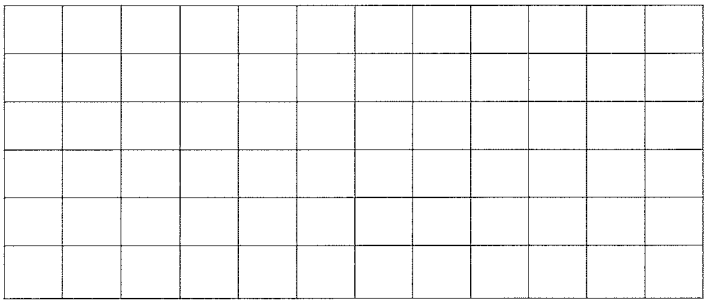
Graph both sinusoidal functions below.

	$y = \sin\theta$	$y = \cos\theta$
Domain		
Range		
X-intercepts		
Y-intercepts		
Amplitude		
Period		

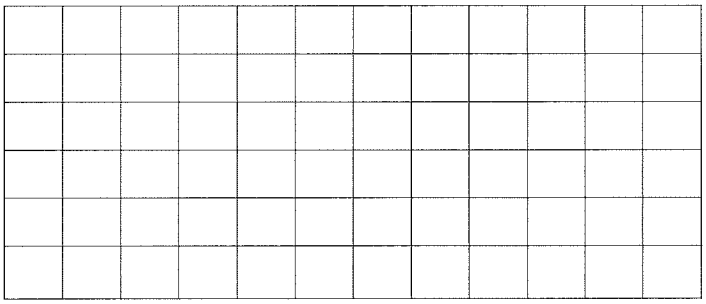
The five-point method for graphing:

Graph $y = \sin\theta$ and $y = \cos\theta$ using five major points per cycle (peak, trough, central axis points).

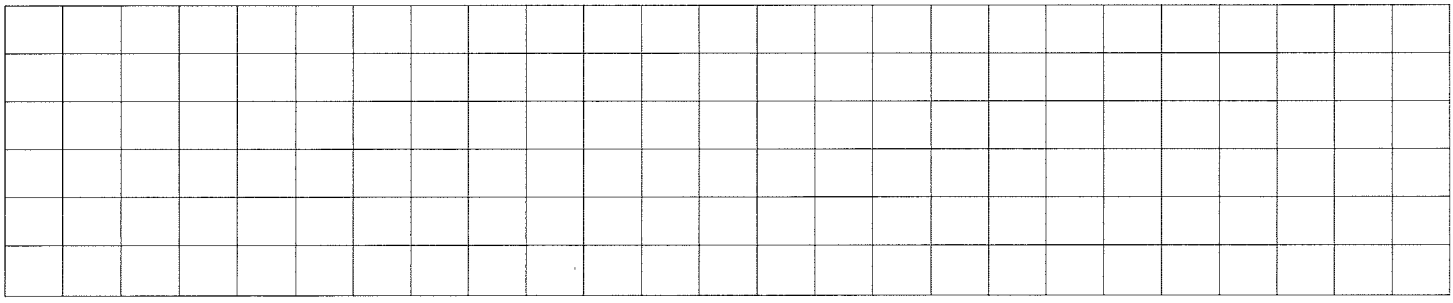
$y = \sin\theta$



$y = \cos\theta$



Practice! Cover the above and graph $y = \sin\theta$ and $y = \cos\theta$ using five major points per cycle. Show two cycles.

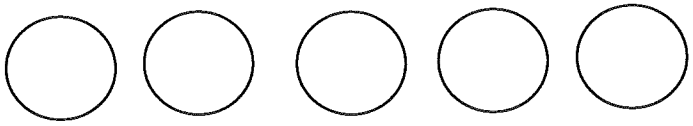


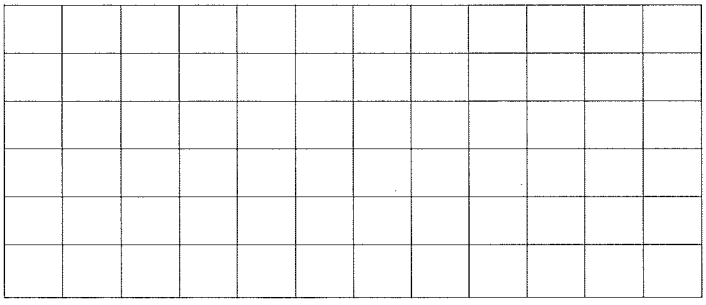
Key Ideas:

• A sine wave, or sinusoid, is the graph of the sine functions in trigonometry • Sinusoidal functions repeat at regular intervals ➔ Graphs that are periodic and have the same shape and characteristics of the sine function (form symmetrical waves) • A cosine wave is the graph of the sine function translated $\frac{1}{4}$ of the period to the left	Situations modelled by sinusoidal functions include: • Motion of objects in a circular orbit • Motion of a pendulum • Motion of vibrating objects • # of hours of sunlight for a particular latitude • Current for an AC circuit • Ocean, sound and light waves

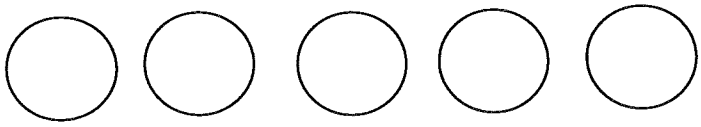
Example:
Beckham is riding a Ferris wheel. The wheel has a radius of 10m and turns counter-clockwise. Let the origin be at the centre of the wheel and assume that Beckham begins his ride at the point where the radius from the centre of the wheel to Beckham's car is along the positive x-axis.

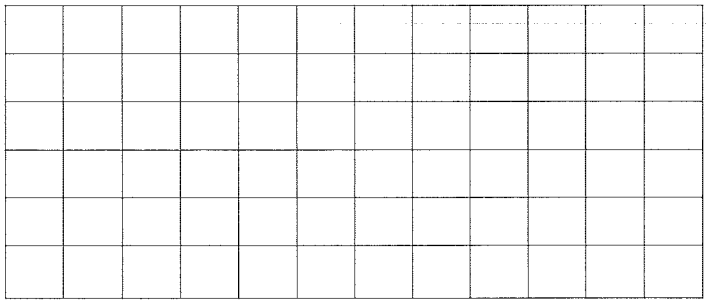
a) Sketch the graph of Beckham's horizontal displacement versus the angle through which he turns for one rotation of the Ferris wheel. Is this graph best represented by a sine or a cosine function?

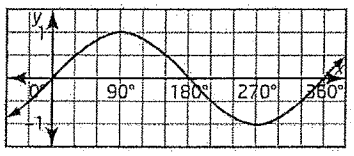
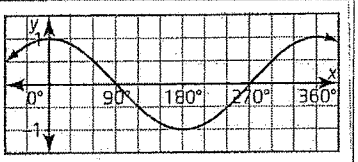




b) Sketch the graph of Beckham's vertical displacement versus the angle through which he turns for one rotation of the Ferris wheel. Is this graph best represented by a sine or a cosine function?



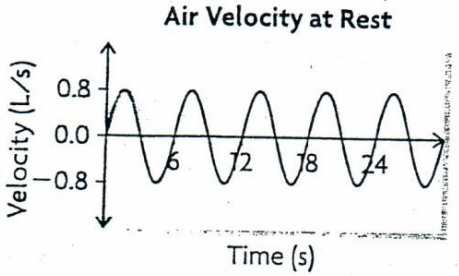


Properties	Sine Function ($y = sinx$)	Cosine Function ($y = cosx$)
Sketch graph		
Max. value and Min. value	1 and -1	1 and -1
Amplitude (vertical distance from axis to min. or max. value)	1	1
Domain	$\{x \in \mathbb{R}\}$	$\{x \in \mathbb{R}\}$
Range	$\{y \in \mathbb{R} \mid -1 \leq y \leq 1\}$	$\{y \in \mathbb{R} \mid -1 \leq y \leq 1\}$
Period (change in x's between 2 max. values)	360°	360°
Axis defined by: $(y = \frac{max.value + min.value}{2})$	y = 0	y = 0
Key Points	(0°, 0) (90°, 1) (180°, 0) (270°, -1) (360°, 0)	(0°, 1) (90°, 0) (180°, -1) (270°, 0) (360°, 1)
$x - intercepts$	0°, 180°, and 360° over one cycle	90° and 270° over one cycle
$y - intercepts$	0	1

Practice:

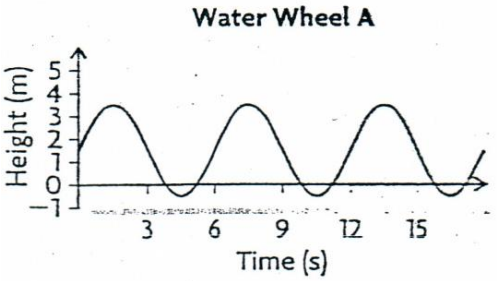
3. Sketch a height-versus-time graph of the sinusoidal function that models each situation. Assume that the first point plotted on each graph is at the lowest possible height.
   a) A Ferris wheel with a radius of 9 m, whose axle is 10 m above the ground, and that rotates once every 60 s
   b) A water wheel with a radius of 2 m, whose centre is at water level, and that rotates once every 20 s
   c) A bicycle tire with a radius of 35 cm and that rotates once every 30 s
   d) A girl lying on an air mattress in a wave pool that is 3 m deep, with waves 1 m in height that occur at 10 s intervals

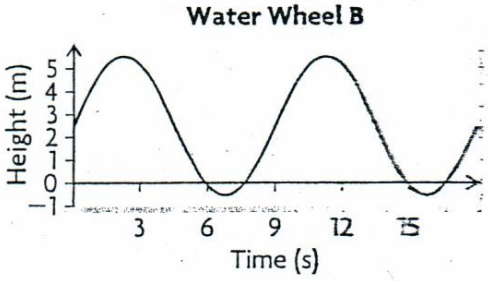
4. When you breathe, the velocity of the air entering and exiting your lungs changes in terms of time. The air entering your lungs has a positive velocity, and the air exiting your lungs has a negative velocity. If a person is at rest, this relationship can be modelled using the graph shown.



a) What is the equation of the axis, and what does it represent in this situation?
b) What is the amplitude of this function?
c) What is the period of the function, and what does it represent in this situation?
d) State the domain and range of the function.

6. Maire is visiting a historical site that has two working water wheels. K She notices that there is a large spike on the circumference of each wheel. She studies the motion of each spike, collects data, and constructs the following graphs:





Analyze the graphs and compare the water wheels. Refer to

- the radius of each wheel
- the height of the axle relative to the water
- the time taken to complete one revolution
- the speed of each wheel

- I congratulate you for making it this far

7. Marcus has two different-sized wheels, on which he conducted four similar experiments.

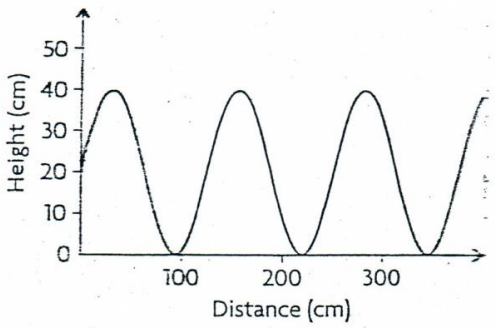
Experiment 1: Marcus took the larger wheel and stuck a piece of tape on the wheel's side so that the tape touched the ground. He rolled the wheel and recorded the height of the tape above the ground relative to the distance the wheel travelled. He graphed his data.

Experiment 2: Marcus took the same wheel and stuck the tape on the side of the wheel 10 cm in from the edge so that the tape would never touch the ground. He rolled the wheel, collected his data, and graphed the data.

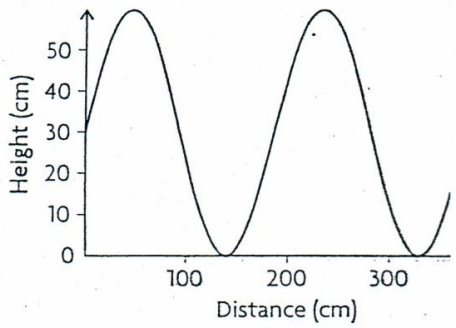
Experiment 3: Marcus took the smaller wheel and stuck the tape on the side of the wheel so that the tape touched the ground. He rolled the wheel, collected his data, and graphed the data.

Experiment 4: Using the smaller wheel, Marcus stuck the tape on the side of the wheel 10 cm in from the edge so that the tape would never touch the ground. He rolled the wheel, collected his data, and graphed the data.

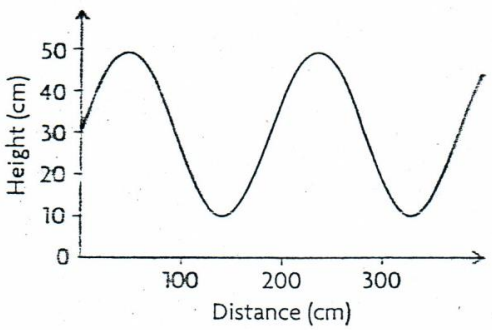
a)



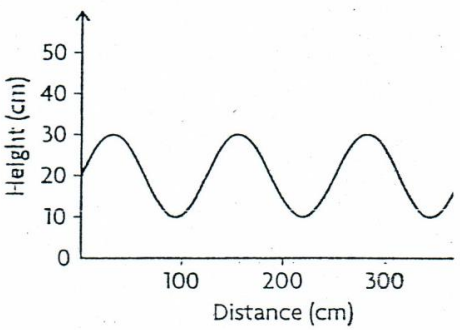
c)



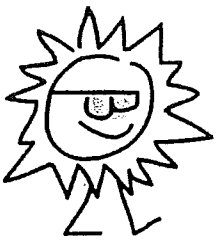
b)



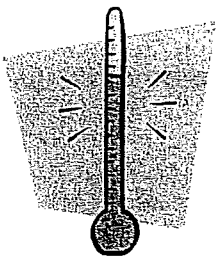
d)



a) Match each graph to the appropriate experiment.
b) What is the equation of the axis for each graph? What does the equation of the axis represent for each graph?
c) What is the amplitude for each graph?
d) What is the period for each graph? What does the period represent in each graph?
e) What is the radius of the larger wheel?

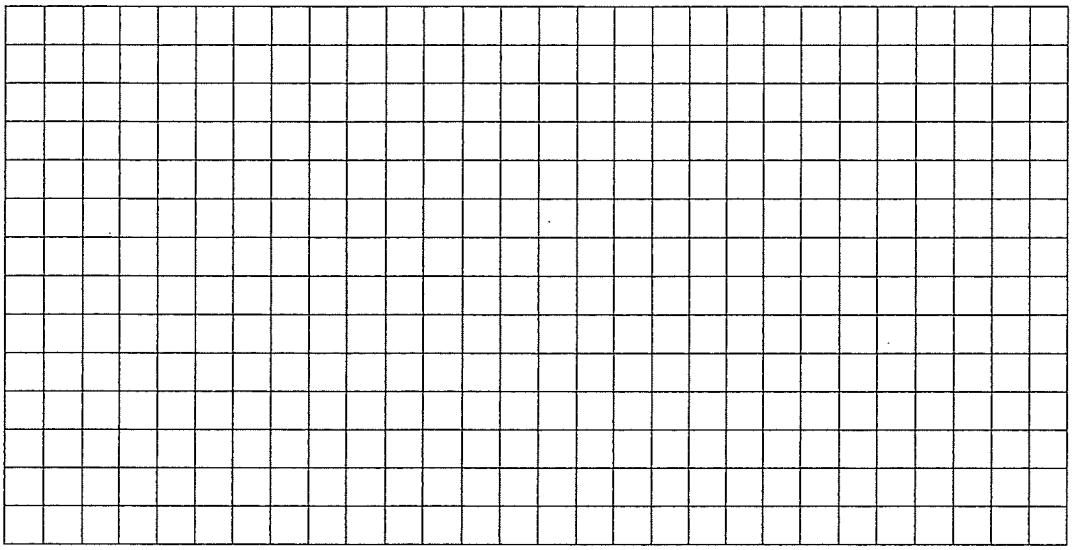


Warm-Up!!

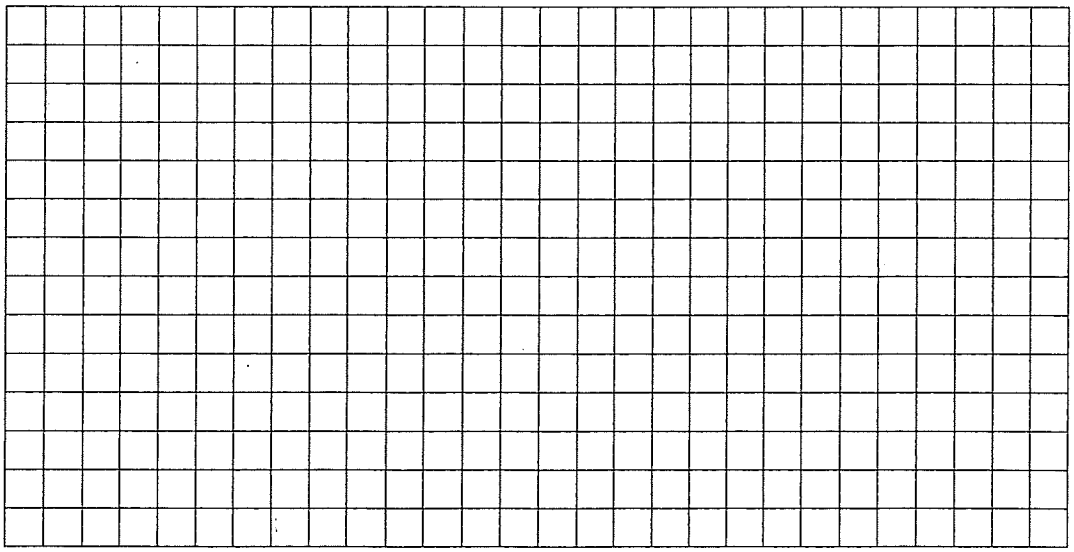


1) On the grids below, graph one cycle of the functions f(x) = sinx and f(x) = cosx, for the interval $0° \leq x \leq 360°$.

a) f(x) = sinx



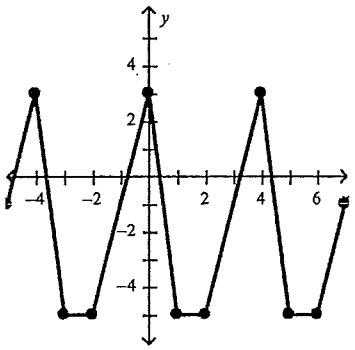
b) f(x) = cosx



2) Summarize the key features of both the sine and cosine graphs.

	y = sinx	y = cosx
Maximum and minimum values		
Amplitude		
Domain		
Range		
Period		
Equation of the Central Axis		
x-intercepts		
y-intercepts		

3) Consider the graph of the periodic function shown.



a) Determine the period of the function.

b) Determine the amplitude of the function.

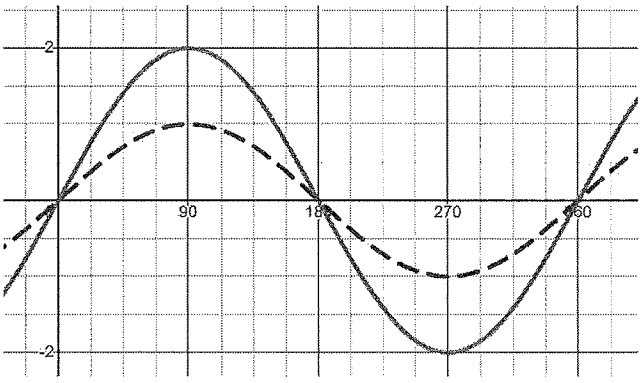
c) The equation of the central axis.

d) $f(-3) =$

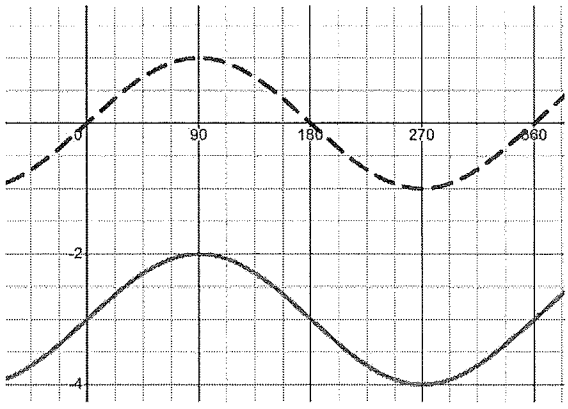
e) $f(60) =$

Concept Attainment

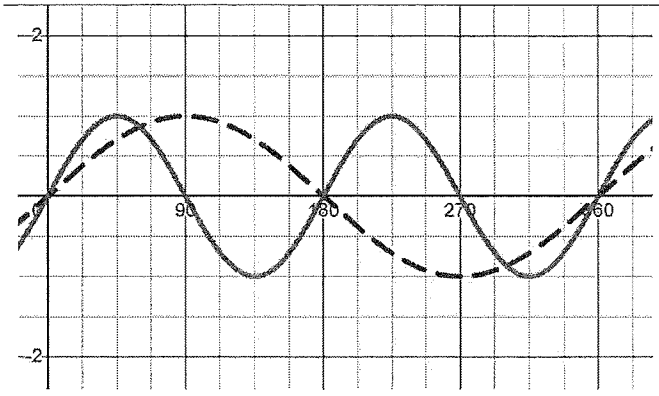
Analyze each of the graphs given. Determine where you can look at the graph/which features indicate the values of a, k, d, and c in $f(\theta) = a \sin k(\theta - d) + c$ and $f(\theta) = a \cos k(\theta - d) + c$.



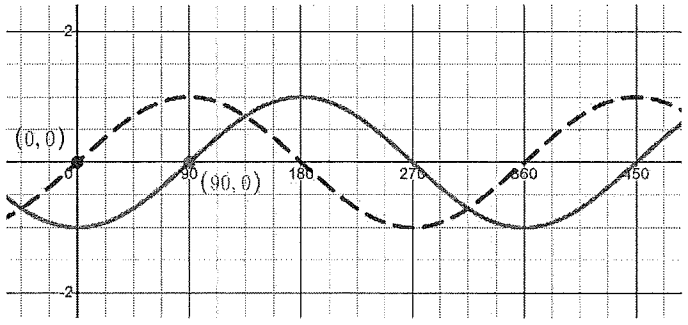
$y = \sin\theta$ (dashed) vs $y = 2\sin\theta$



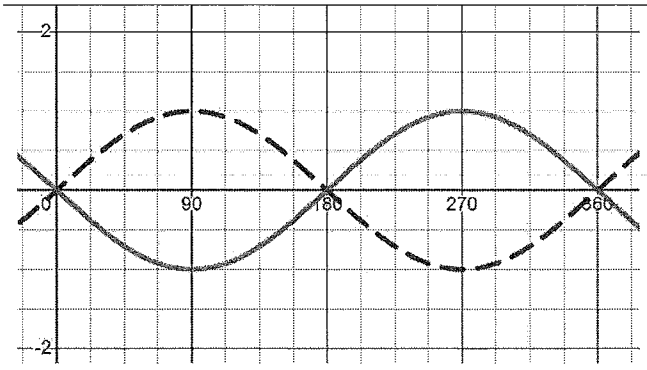
$y = \sin\theta$ (dashed) vs $y = \sin\theta - 3$



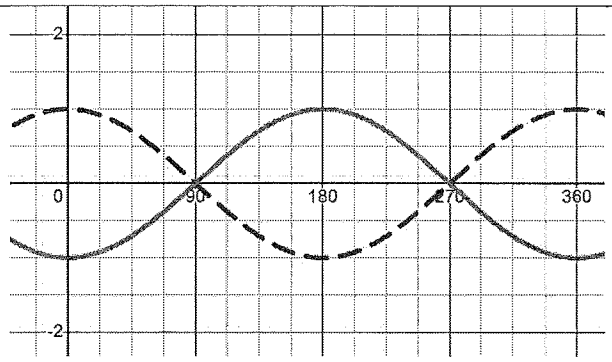
$y = \sin\theta$ (dashed) vs $y = \sin 2\theta$



$y = \sin\theta$ (dashed) vs $y = \sin(\theta - 90°)$



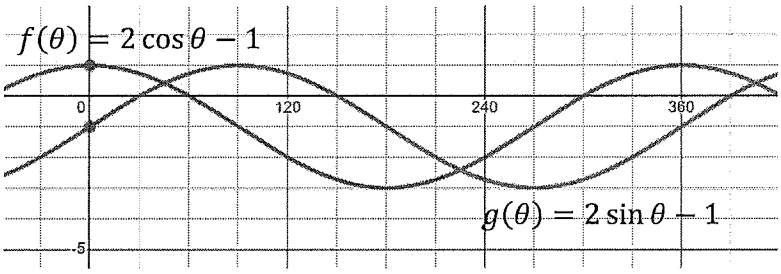
$y = \sin\theta$ (dashed) vs $y = -\sin\theta$



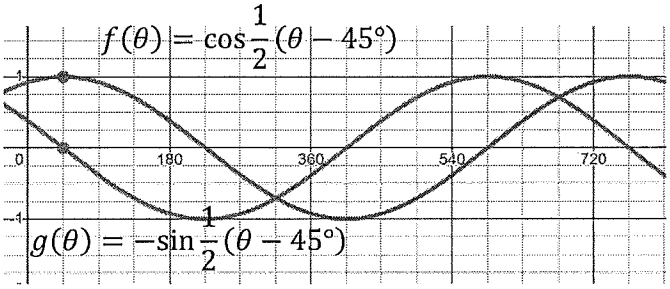
$y = \cos\theta$ vs $y = -\cos\theta$

Lesson 6.3 Transformations of Trigonometric Functions

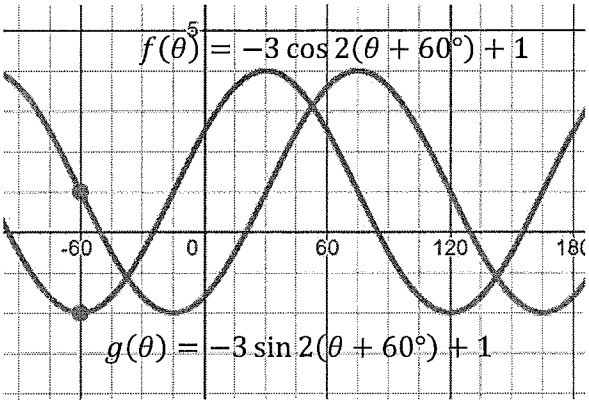
Apply your understanding – identify the key features and difference between each pair of graphs.



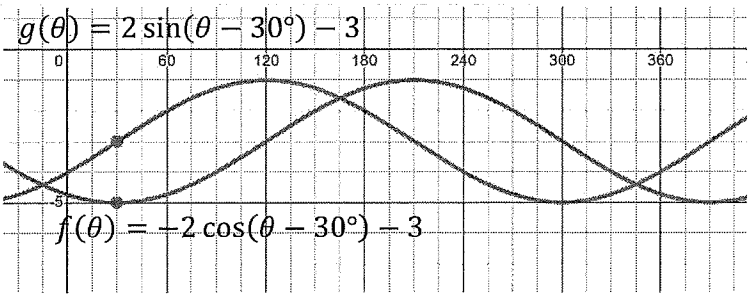
Amplitude:	Period:	Phase shift:	Central axis:
$f(\theta)$ (+cosine) starts at		$g(\theta)$ (+sine) starts at	



Amplitude:	Period:	Phase shift:	Central axis:
$f(\theta)$ (+cosine) starts at		$g(\theta)$ (-sine) starts at	



Amplitude:	Period:	Phase shift:	Central axis:
$f(\theta)$ (-cosine) starts at		$g(\theta)$ (-sine) starts at	



Amplitude:	Period:	Phase shift:	Central axis:
$f(\theta)$ (-cosine) starts at		$g(\theta)$ (+sine) starts at	

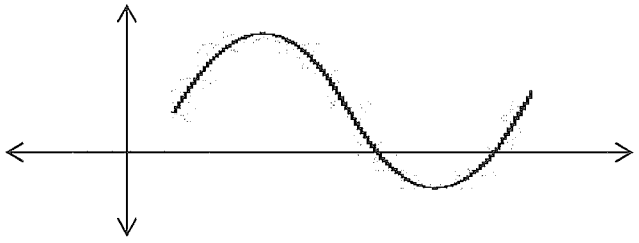
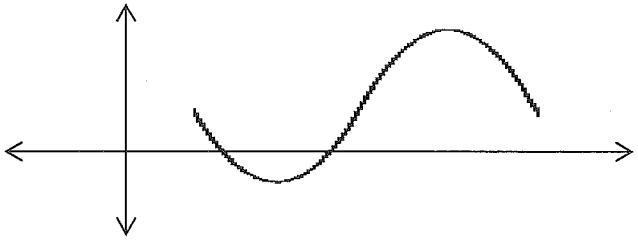
Summary: Graphing sinusoidal functions of the form

$f(\theta) = a\sin k(\theta - d) + c$ **or** $g(\theta) = a\cos k(\theta - d) + c$

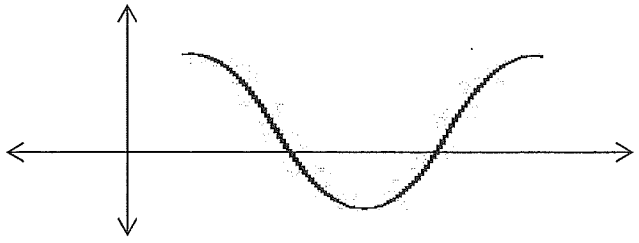
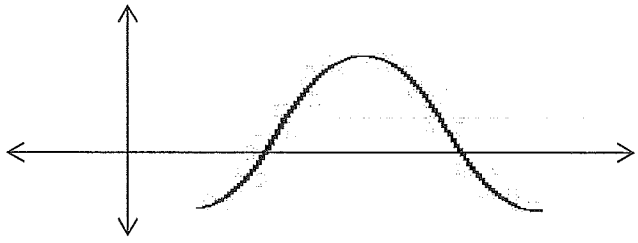
Amplitude:	Period:	Phase shift:	Central axis:
Maximum/peak:	Minimum/trough:	Range:	

Using Step Pattern to Graph Transformations of Sinusoidal Functions

$f(\theta) = a\sin k(\theta - d) + c$ → Starts at (d, c)

If $a > 0$: starts at (d, c)				If $a < 0$: starts at (d, c)			
¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ down
							

$g(\theta) = a\cos k(\theta - d) + c$ → Starts at maximum (if $a > 0$) or minimum (if $a < 0$)

If $a > 0$: starts at $(d, c + \lvert a\rvert)$				If $a < 0$: starts at $(d, c - \lvert a\rvert)$			
¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ up	¼ period → $\lvert a\rvert$ down	¼ period → $\lvert a\rvert$ down
							

Reminder: instead of this step pattern method, you may use **mapping notation** from previous units:

$$(x, y) \to \left(\frac{1}{k}x + d, ay + c\right)$$

Determining the best x-axis scale for sinusoidal functions: To guarantee that points always land on the intersection of gridlines…

1) Determine the phase shift	2) Determine the new period	3) Determine ¼ of the period	4) Determine greatest common factor between phase shift and ¼ period

e.g. Determine the best x-axis scale for $f(\theta) = -2\cos\frac{1}{3}(\theta + 45°) + 1$:

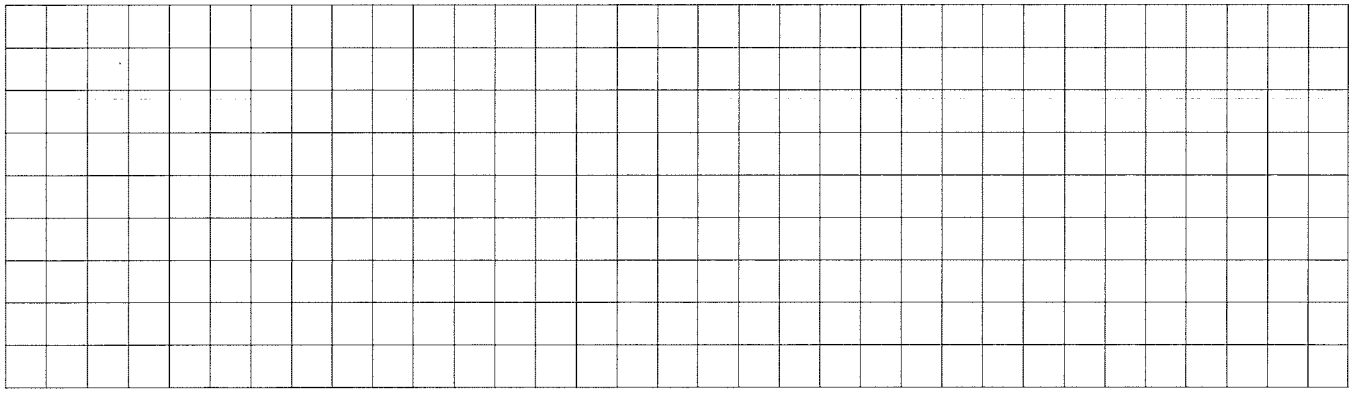
1) Phase shift	2) New period	3) ¼ of the period	4) GCF between P.S. and ¼ period

Practice

1. For $y = 2\sin(\theta - 60°) + 1$, state the characteristics of the function. Then graph one complete cycle.

Amplitude:	Central axis:	Max:	Min:	Period:	¼ Period:	Phase shift:

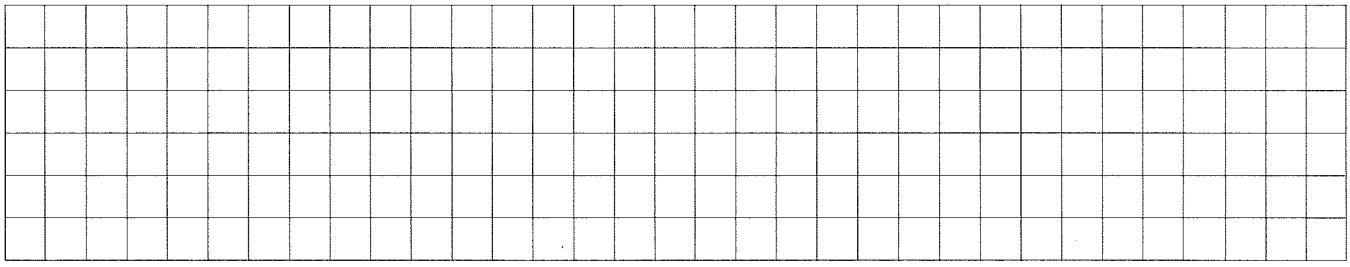
Range:	Starting point, direction:	Best scale:



2. For $y = \frac{1}{2}\sin(\theta + 240°) - 1$, state the characteristics of the function. Then graph one complete cycle.

Amplitude:	Central axis:	Max:	Min:	Period:	¼ Period:	Phase shift:

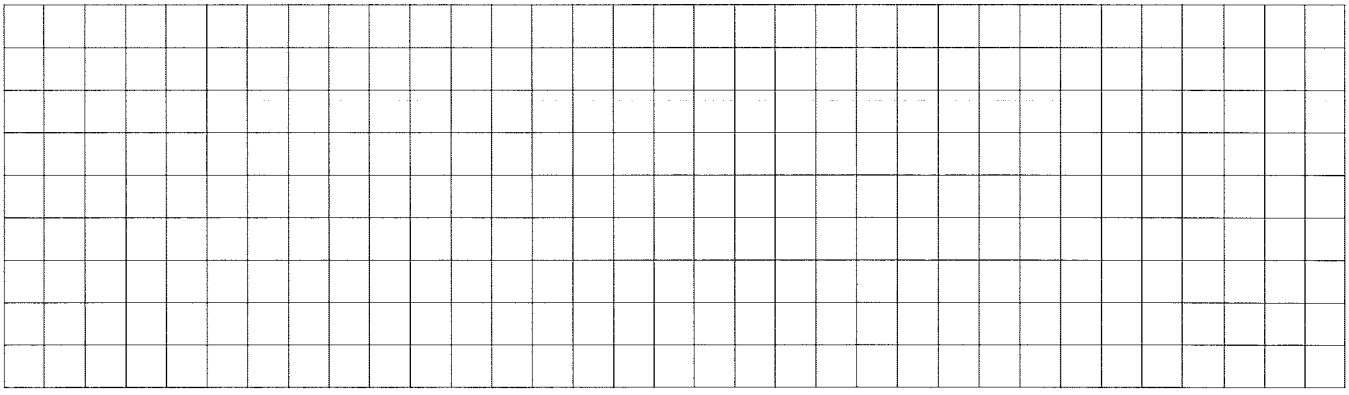
Range:	Starting point, direction:	Best scale:



3. For $y = -3\cos(\theta - 45°) + 1$, state the characteristics of the function. Then graph one complete cycle.

Amplitude:	Central axis:	Max:	Min:	Period:	¼ Period:	Phase shift:

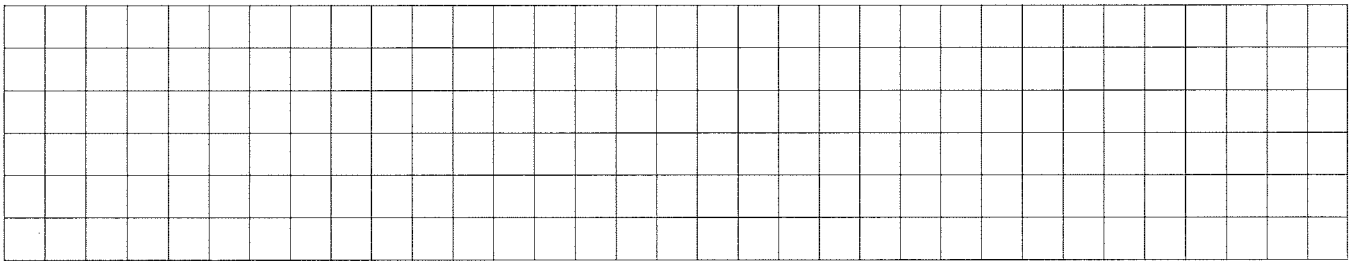
Range:	Starting point, direction:	Best scale:



4. For $y = -\sin(2\theta - 180°)$, state the characteristics of the function. Then graph one complete cycle.

Amplitude:	Central axis:	Max:	Min:	Period:	¼ Period:	Phase shift:

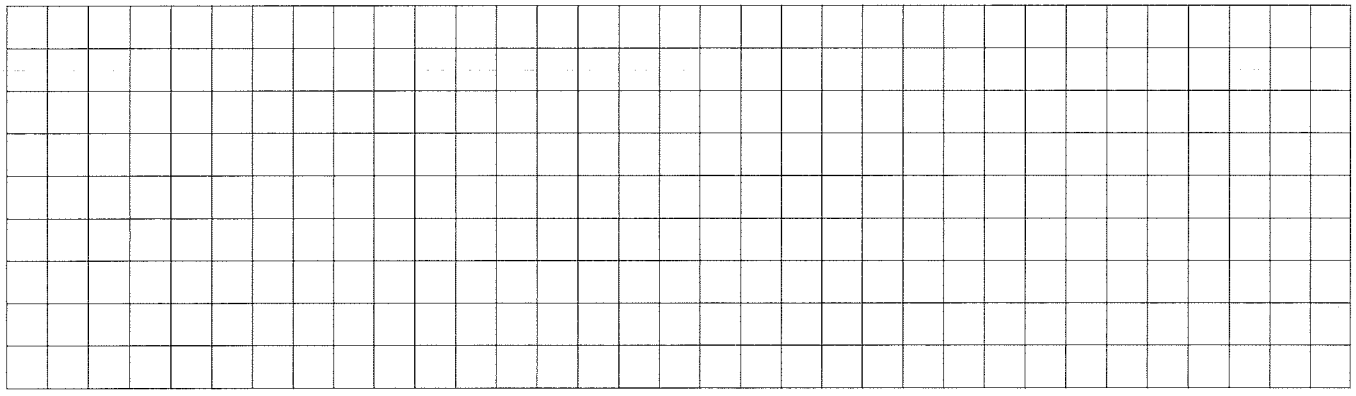
Range:	Starting point, direction:	Best scale:



5. For $y = 2\cos(4\theta + 180°) + 2$, state the characteristics of the function. Then graph one complete cycle.

Amplitude:	Central axis:	Max:	Min:	Period:	¼ Period:	Phase shift:

Range:	Starting point, direction:	Best scale:



Transformations of Trig Functions - Homework

1) Graph **ONE** complete cycle of the following function. Choose an appropriate scale (do not add your own gridlines)! Show all appropriate work.

$y = 4sin(\theta - 45°) + 2$

Amp _____
Period _____
P.S. _____
C.A. _____

Scale: _____

2) Graph **ONE** complete cycle of the following function. Choose an appropriate scale (do not add your own gridlines)! Show all appropriate work.

$$y = cos\left(\frac{1}{2}\theta + 60°\right) + 1$$

Amp _____
Period _____
P.S. _____
C.A. _____

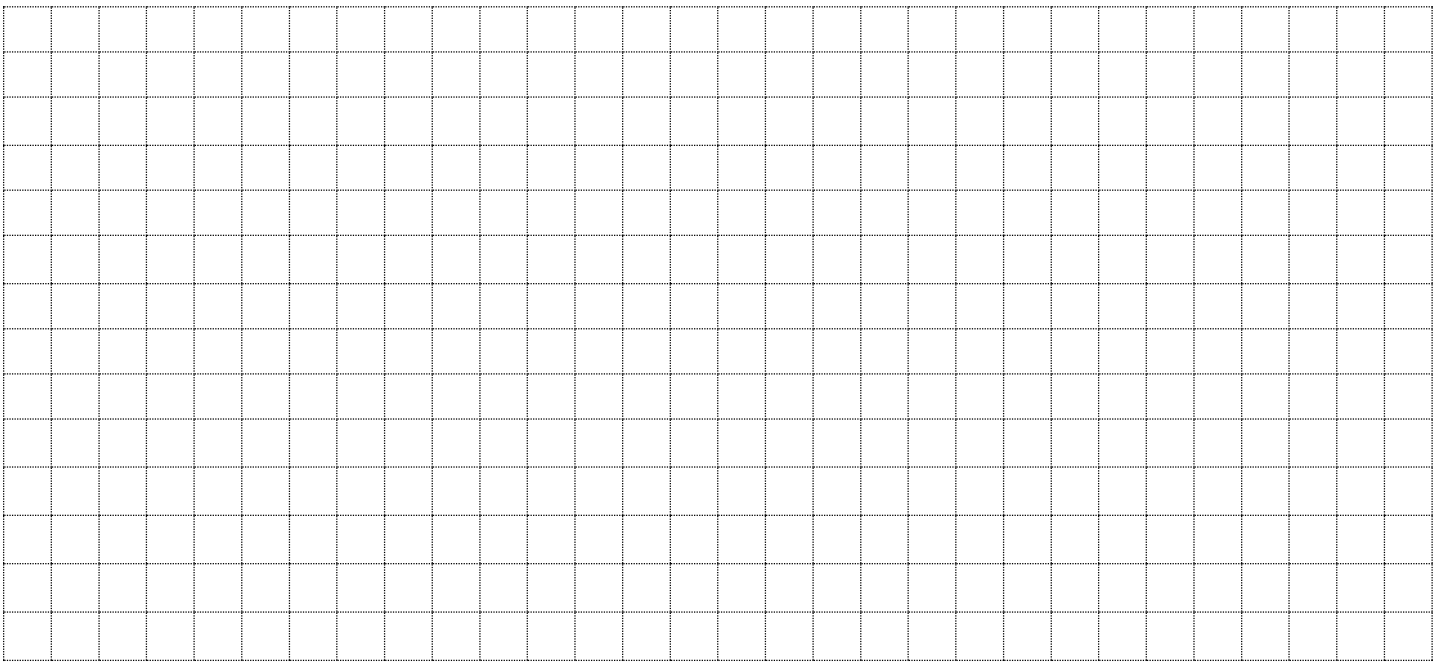
Scale: _____

3) Graph **ONE** complete cycle of the following function. Choose an appropriate scale (do not add your own gridlines)! Show all appropriate work.

$y = -2sin(2\theta - 60°) - 3$

Amp _____
Period _____
P.S. _____
C.A. _____

Scale: _____

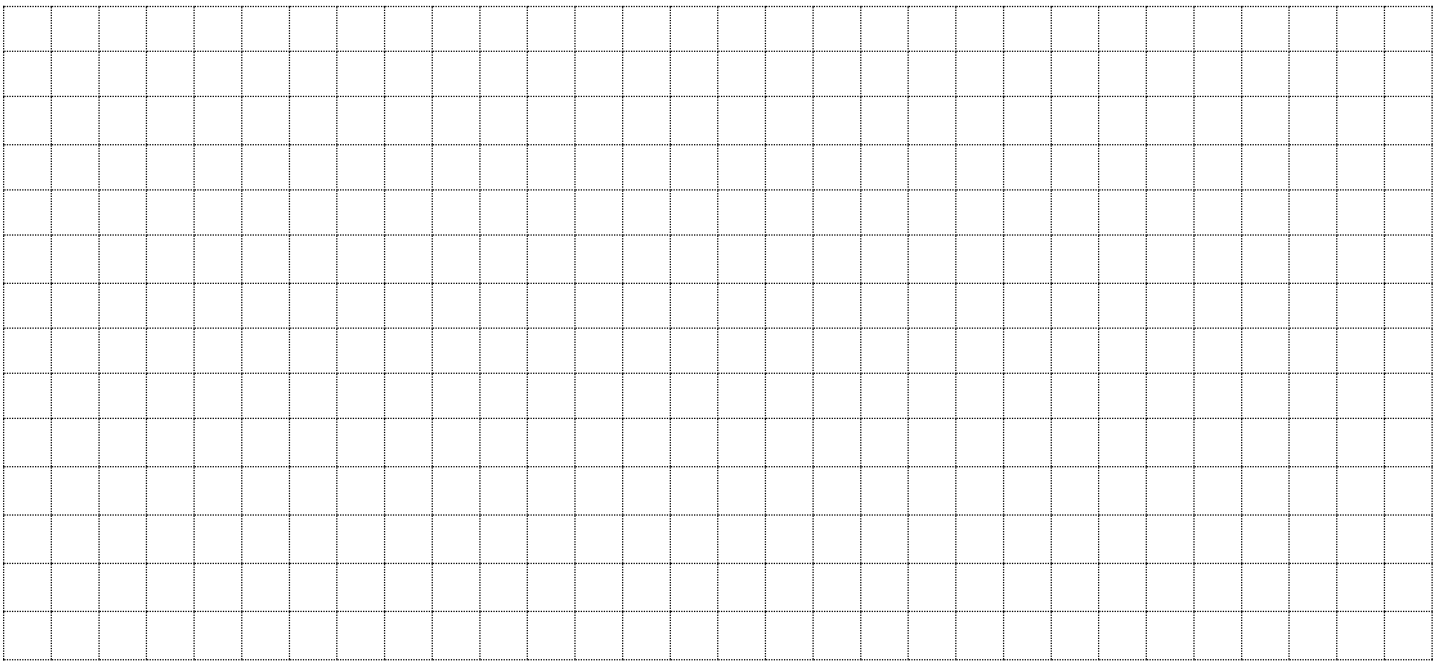


4) Graph **ONE** complete cycle of the following function. Choose an appropriate scale (do not add your own gridlines)! Show all appropriate work.

$$y = -\frac{1}{2}cos(3\theta + 60°) - 1$$

Amp _____
Period _____
P.S. _____
C.A. _____

Scale: _____

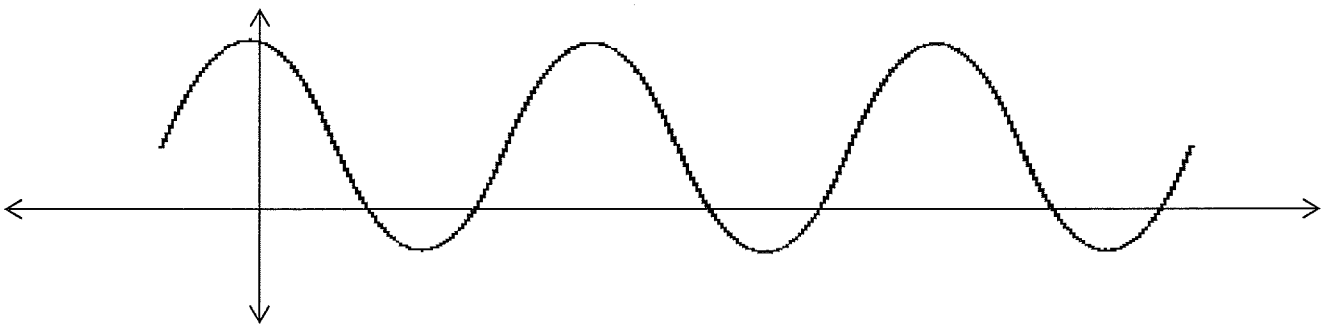


Determining the Trigonometric Relation $f(\theta) = a\sin[k(\theta - d)] + c$

Complete the following chart. or $g(\theta) = a\cos[k(\theta - d)] + c$

Function	Amplitude	Period	Phase Shift	Vertical Displacement	Maximum	Minimum
1) $f(\theta) = 3\cos\theta$						
2) $f(\theta) = \cos 4\theta$						
3) $f(\theta) = \sin\theta - 2$						
4) $f(\theta) = -\sin(\theta - 60°)$						
5) $f(\theta) = 2\cos \frac{1}{2}\theta$						
6) $f(\theta) = \sin(2\theta - 180°)$						
7) $f(\theta) = 2\cos(3\theta - 270°) + 4$						
8) $f(\theta) = \frac{1}{2}\sin\frac{1}{3}(\theta + 90°) - 1$						
9) $f(\theta) =$ sin	2	360°	left 180°	none		
10) $f(\theta) =$ cos	½	90°	right 60°	down 3		
11) $f(\theta) =$ sin	3	720°	left 45°	up 1		
12) $f(\theta) =$ cos	1	540°	none	down 2		

Steps to finding the Relation from the Graph of a Sinusoidal Function



1) Locate the **central axis** → same as

2) Determine the **amplitude** → same as

3) Determine the **period** → same as

Then calculate the value of k by using:

4) Choose a "starting point"

If the "starting point" is on the central axis (and the function immediately increases), use the sine function	If the "starting point" is on the central axis (and the function immediately decreases), use the negative sine function	If the "starting point" is on a peak, use the cosine function	If the "starting point" is on a trough, use the negative cosine function

5) Determine the **phase shift** → same as

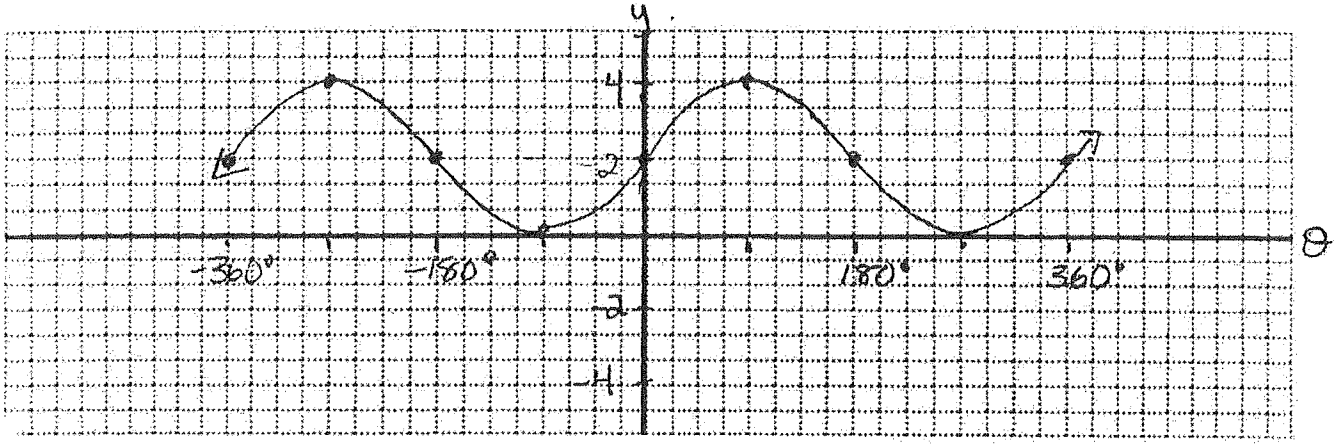
6) Substitute the values of $a, k, d,$ and c into the appropriate algebraic model

Finding the Relation from the Graph

1) State the central axis, amplitude, period, and phase shift of the following graphed sinusoidal functions. Use this information to state the equation of the sinusoidal function.

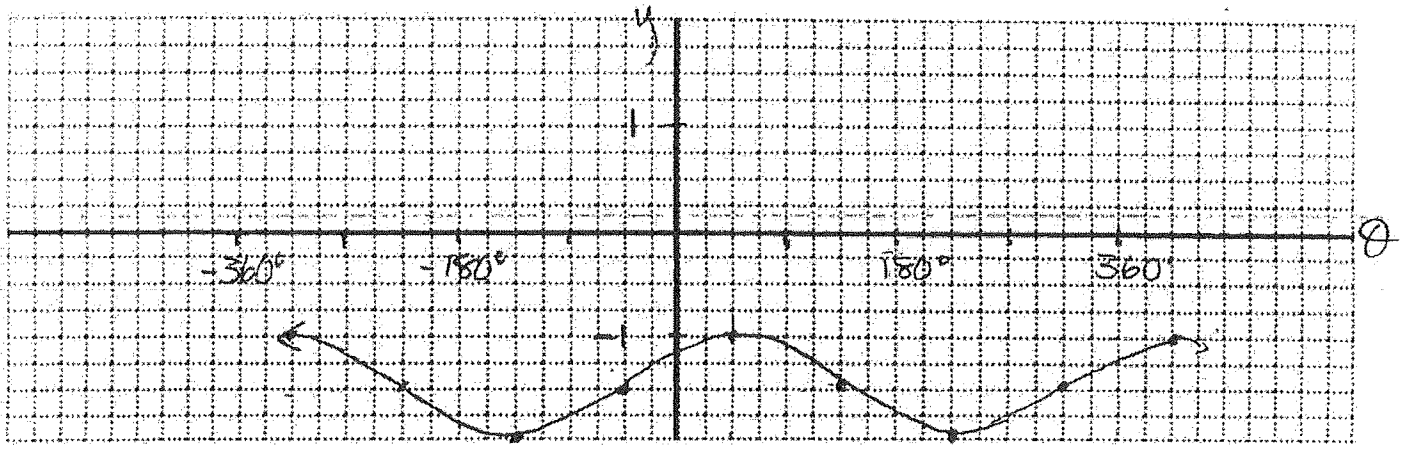
a) Central axis: _____ Amplitude: _____ Period: _____

Starting point: _____ Phase shift: _____ Relation: _____



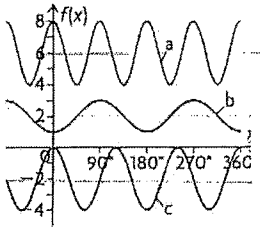
b) Central axis: _____ Amplitude: _____ Period: _____

Starting point: _____ Phase shift: _____ Relation: _____



FINDING THE EQUATION OF A TRIG FUNCTION!

CHECK Your Understanding



1. Determine an equation for each sinusoidal function at the right.
2. Determine the function that models the data in the table and does not involve a horizontal translation.

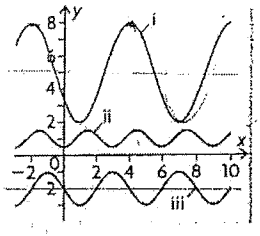
	0°	45°	90°	135°	180°	225°	270°
	9	7	5	7	9	7	5

3. A sinusoidal function has an amplitude of 4 units, a period of 120°, and a maximum at (0, 9). Determine the equation of the function.

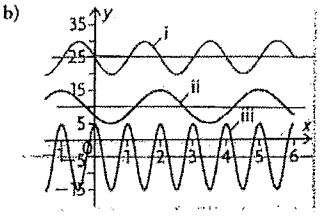
PRACTISING

4. Determine the equation for each sinusoidal function.

a)



b)



5. For each table of data, determine the equation of the function that is the simplest model.

a)

x	0°	30°	60°	90°	120°	150°	180°
y	3	2	1	2	3	2	1

b)

x	−180°	0°	180°	360°	540°	720°	900°
y	17	13	17	21	17	13	17

c)

x	−120°	−60°	0°	60°	120°	180°	240°
y	−4	−7	−4	−1	−4	−7	−4

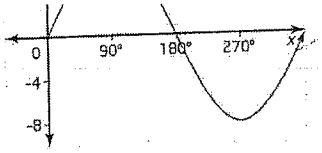
d)

x	−20°	10°	40°	70°	100°	130°	160°
y	2	5	2	−1	2	5	2

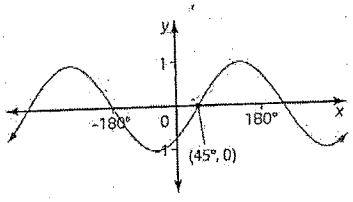
Write an equation for each sine function.

a)

b)



c)

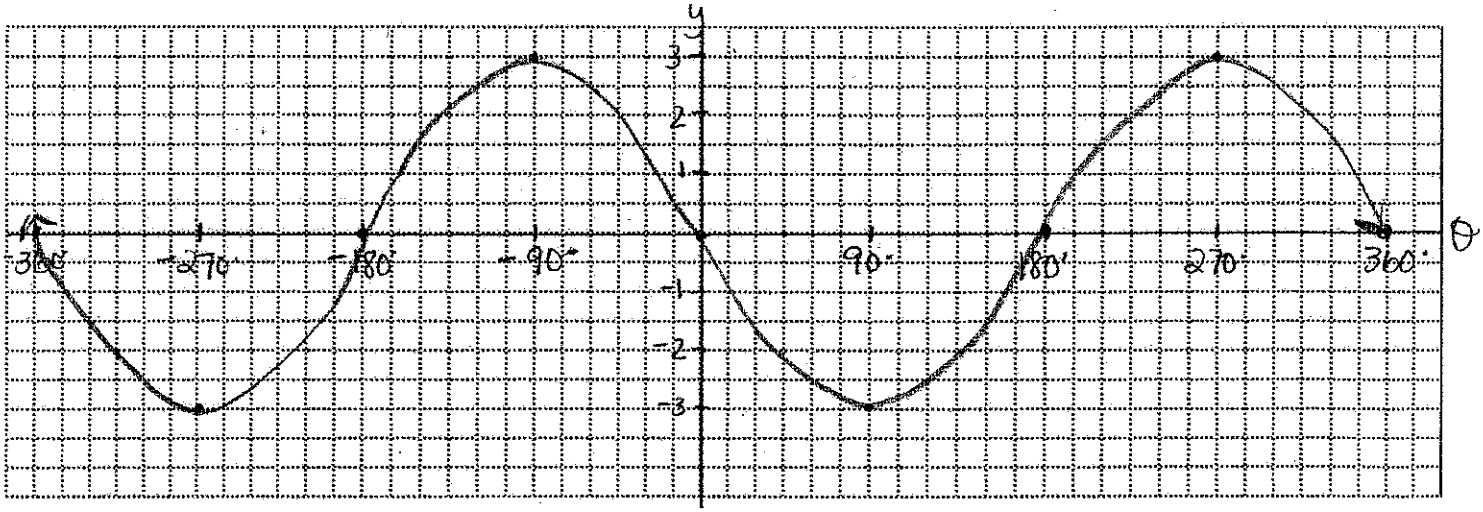


Finding the Equation from the Graph... some more!

1) State the amplitude, period, phase shift and vertical displacement of the following graphed trig functions. Use this information to state the equation of the trig function.

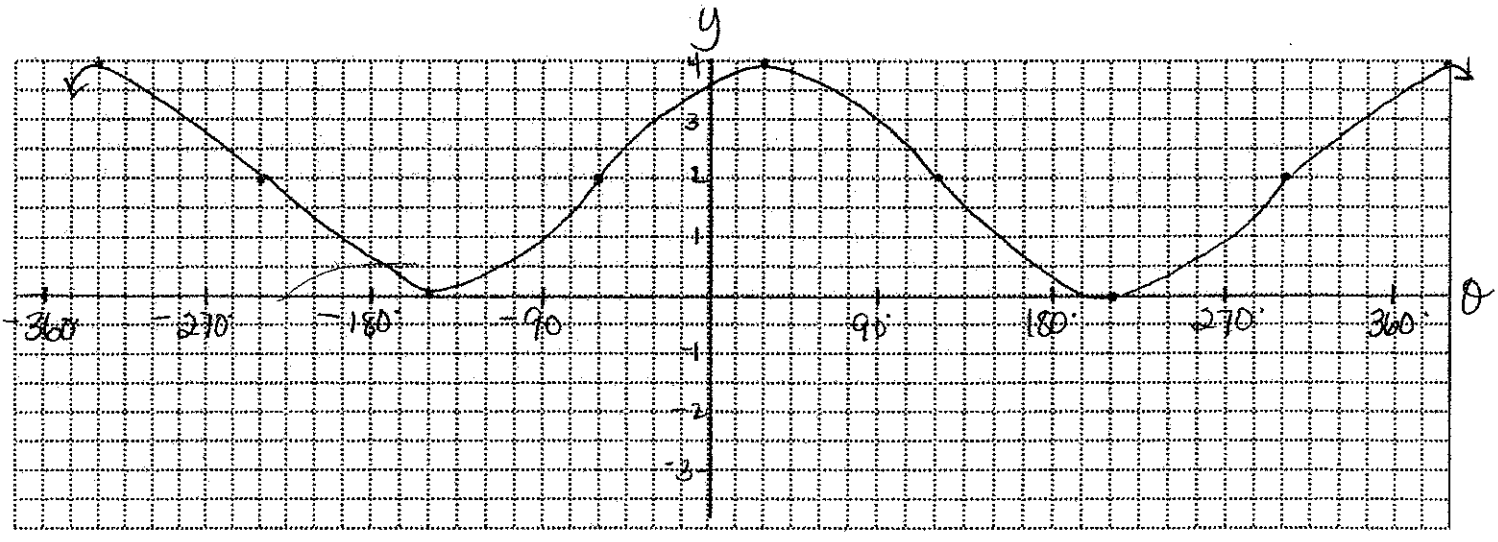
a) amplitude: _____ phase shift: _____ period: _____

vertical displacement: _____ equation: _____



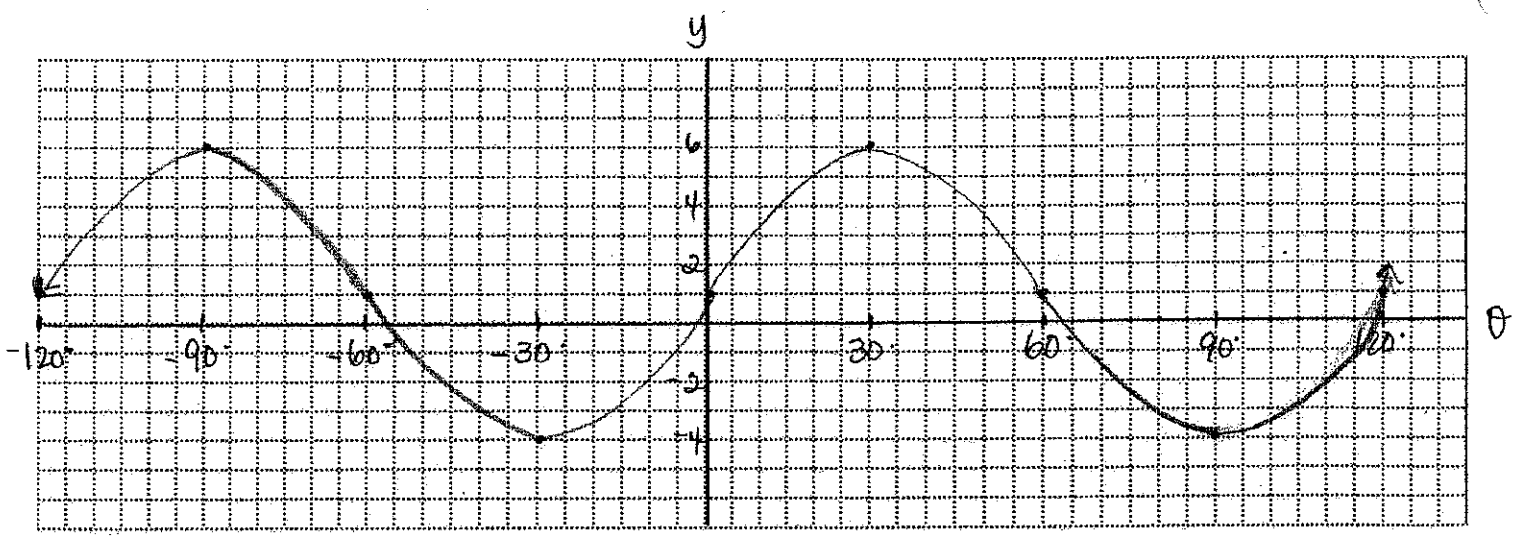
amplitude: _____ phase shift: _____ period: _____

vertical displacement: _____ equation: _____



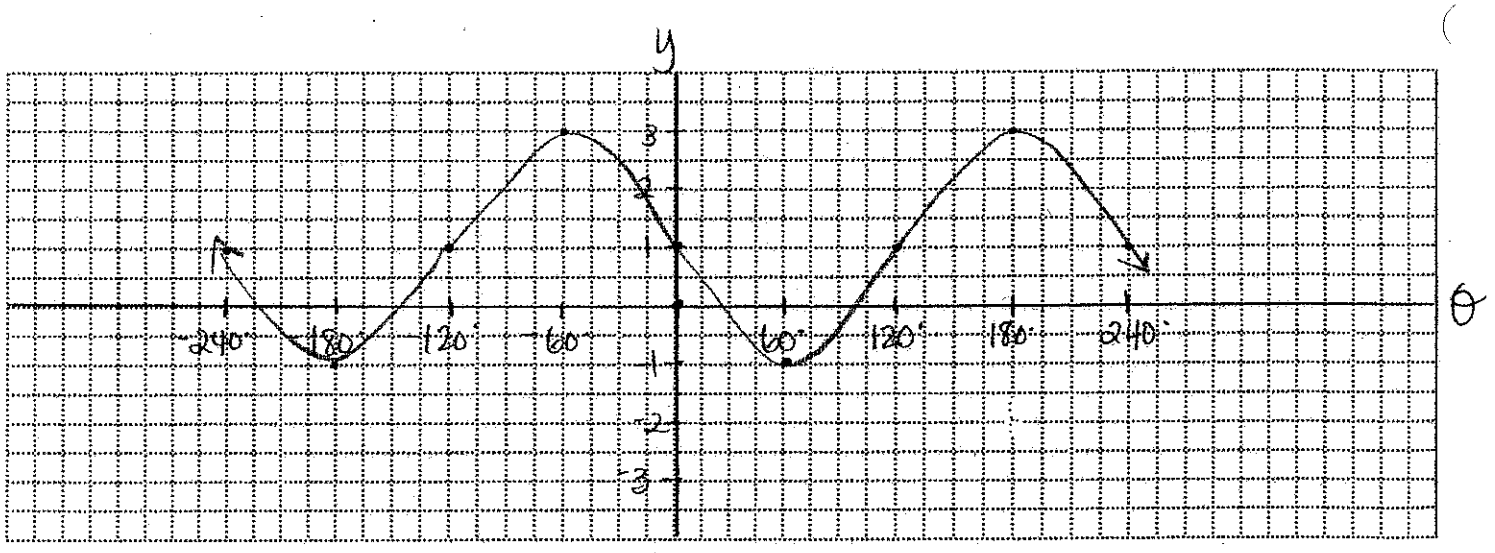
c) amplitude: _____ phase shift: _____ period: _____

vertical displacement: _____ equation: _____



d) amplitude: _____ phase shift: _____ period: _____

vertical displacement: _____ equation: _____



Modeling Periodic Phenomena

The following periodic phenomena can be modeled graphically and algebraically. When creating a graphical model, be sure to

- use meaningful variables for the two quantities that vary in each case
- label key values (min, central axis, max) on the y-axis and number your x-axis scale
- draw and label the central axis

When creating an algebraic model, use meaningful variables that match the ones you used in your graphical model.

- Determine the period → use $\frac{360}{period}$ to calculate the value of k
- Determine the amplitude → $|a| = \frac{max-min}{2}$
- Determine the central axis → $c = \frac{max+min}{2}$ (or $min + |a|$ or $max - |a|$)
- Determine the starting point (based on if this is a sine or cosine function) → x-coordinate is the same as d

Example 1: A Ferris wheel has a diameter of 36 m, and its centre is 19 m above the ground. The wheel completes one revolution every 30 s. The rider begins at the lowest position on the wheel.

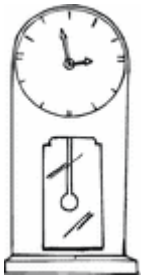
a) Sketch the graph of a person's height in metres, to time, in seconds, for two revolutions of the wheel.

b) Determine the trigonometric function that models this relationship.

c) Determine the height of the rider at 100s. Round your answer to 2 decimal places, if needed.

Example 2: A pendulum clock is designed so that it takes 4 seconds for the pendulum to swing from its centre position, to the right, back to centre, to the left and then back to centre again to complete one cycle. The pendulum swings a maximum horizontal distance of 3 cm from its centre position.

a) Graph the horizontal displacement, D(t), of the pendulum from its centre position, to time, t, in seconds, for a 10-second interval.



b) Determine the trigonometric function that models this relationship.

c) Determine the times at which the pendulum is 2cm to the right of its centre position during its first 10-second interval.

Example 3:

The Bay of Fundy, which is between New Brunswick and Nova Scotia, has the highest tides in the world. During 'low tide' there is no water on the shore at all, while at 'high tide' the water covers the shore with depth of 16 m. High tide occurs roughly 12 hours after low tide. On a particular day, the first low tide occurs at 6am.

a) Model the depth of water, D(t), in meters, to time, t, in hours for two days.

b) Determine the time(s) at which the depth of tide is 10m during the first two days.

Practice

1. A Ferris wheel has a diameter of 24 m and it completes one revolution every 40 s. People climb onto the Ferris wheel from a platform that is 2 m above the ground. Model the relationship, graphically and algebraically, between a person's height in metres, to time, in seconds, during a 2-minute ride for a rider who begins
    a) at the lowest position on the wheel.
    b) at the '3-o'clock' position, with the wheel rotating counter-clockwise.

2. Bob jumps up and down on a trampoline. Mary determines that he jumps about 12 times each minute, consistently reaching a maximum height of about 9.6 m. The trampoline ensures he is never closer than 0.8 m to the ground.

    a) Model Bob's height, in metres, to time, in seconds, for a time interval of 30 seconds (both graphically and algebraically).

    b) Determine at what times Bob is closest to the ground over the 30-second interval.

3. A weight is hung on the end of a spring and rests 50 cm above a table top. The weight is pulled down 25 cm and released at time t = 0. This creates a periodic up and down motion. It takes the weight 1.6 s to return to the lowest position. Model the height of the bobbing weight over the table, in centimetres, to time, t, in seconds, for at least three cycles (both graphically and algebraically).

4. The volume of air in the lungs during normal breathing is a periodic function of time. Suppose a person's lungs contain from 2200 mL to 2800 mL of air during normal breathing. Suppose a normal breath takes 4 seconds and that time t = 0 s corresponds to when a person has finished exhaling (i.e. minimum lung volume). Model the relation (both graphically and algebraically) between V, the volume in mL, and t, time in seconds, for a person breathing for 20 seconds.

5. Towers and skyscrapers appear to be rigid by they are designed to sway with the wind.

    a) Suppose you are standing on the glass floor of the CN Tower on a particularly windy day. You determine that this part of the CN Tower is swaying back and forth across a total distance of 80 m and that it takes 12 seconds to sway back and forth once. Let D be your horizontal distance from vertical (i.e. when the tower is perfectly straight up) in metres and let t be the amount of time you are swaying. Model how your distance from vertical changes over time, assuming t = 0 corresponds to a perfectly upright CN Tower.

    b) You climb up to the Sky Pod, the highest point a person can be (legally!). At the Sky Pod you determine you are now swaying back and forth through a greater distance. How would the graph of D vs. t change in (a) now that you are in the Sky Pod.

Unit 7:
Sequences and Series

Arithmetic and Geometric Sequences

A sequence is a set of numbers, shapes, letters, etc. that are in a distinct or recognizable pattern.

E.g. 1, 3, 5, 7, … 12, 24, 48, …. $\frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \ldots$

Sequences are **discrete functions**, unlike the functions we've studied to date which are **continuous functions**.

- Discrete functions –
- Continuous functions –

The **domain** of sequences is the set/subset of **natural numbers**. The **range** is the **terms** of the sequence.

The first term is referred to as t_1, the second term t_2, the third term as t_3, and so on. The n^{th} term is referred to as t_n. Notice the punctuation in a sequence $t_1, t_2, t_3, \ldots$

- The terms are separated by commas
- … (called an **ellipsis)** indicates the pattern continues indefinitely.

A sequence may be arithmetic, geometric or neither. An **arithmetic sequence** has terms that are separated by a **constant difference** (add a constant to get the next term). For example,

5, 7, 9, 11 → constant difference = ; D: R:

20, 16, 12, 8, … → constant difference = ; D: R:

A **geometric sequence** has terms that are separated by a **constant ratio** (multiply by a constant to get the next term). For example,

2, 6, 18, 54 → constant ratio = ; D: R:

80, 40, 20, 10, … → constant ratio = ; D: R:

Sometimes a pattern can lead to a general rule or formula for finding the terms of a sequence.
The rule is called the **general term** for the n^{th} term, t_n.

n	1	2	3	4	…	n
t_n	a	$a+d$			…	

The **general term** of an arithmetic sequence is _____

Arithmetic sequences are represented by _____ functions.

n	1	2	3	4	…	n
t_n	a	ar			…	

The **general term** of a geometric sequence is_____

Geometric sequences are represented by an _____ function.

Question Type 1: "What is/find the n$^{\text{th}}$ term" – i.e. determine the value of t_n

Ex. 1: Find the first 3 terms in the sequence described by $t_n = 3^n$.

(Example of _____ sequence)

Ex. 2: Find the 15$^{\text{th}}$ term in the sequence $f(n) = 2n^2 - 1$. **(Example of _____)**

Question Type 2: "What term is/does _____ belong to the sequence" – i.e. determine the value of n

➔ if n is NOT a natural number, the term does not belong to the sequence.

Ex. 7: Given the sequence $t_n = 3n - 2$, determine

a) What term is 775

b) Whether or not 1023 belongs to the sequence.

Question Type 3: To determine the general term of a sequence given two terms …

"Find the general term of" means determine the formula $t_n = a + (n-1)d$ or $t_n = ar^{n-1}$ or another formula that relates n and t_n (remember, not all sequences are arithmetic or geometric!)

Ex.4: If $t_3 = 17$ and $t_8 = 37$ in an arithmetic sequence, determine the general term of the sequence.

Ex. 5: If $t_3 = 24$ and $t_7 = \frac{3}{2}$ in a geometric sequence, determine the general term of the sequence.

• If you are given two terms in an **arithmetic sequence**: 1) Determine the constant difference, d, between the terms 2) Use the d and one of the given terms to determine the initial value, a 3) Substitute d and a into $t_n = a + (n-1)d$ and simplify	• If you are given two terms in a **geometric sequence**: 1) Determine the constant ratio, r, between the terms 2) Use the r and one of the given terms to determine the initial value, a 3) Substitute r and a into $t_n = ar^{n-1}$

- If you are given two terms in general **sequence** (not arithmetic or geometric), try to determine the pattern that gives you the term, t_n, based on its term number, n.

Ex. 6: Determine the general term of the sequence $\frac{3}{4}, \frac{5}{9}, \frac{7}{16}, \ldots$

A few more practice questions…

1. Find the general term of the sequence 3, -12, 48, -192, … **(Example of _____ sequence)**

2. The year 2000 was a leap year. Leap years occur every four years. Determine if the year 6388 will be a leap year or not. **(Example of _____ sequence)**

3. Given the arithmetic sequence with $t_2 = -5$ and $t_9 = -33$, determine
   a) The general term of the sequence

   b) Whether or not -200 belongs in the sequence.

Arithmetic Series

An **arithmetic series** is the indicated **sum of the terms of an arithmetic sequence**.

e.g. 4, 9, 14, 19, … is an arithmetic sequence

$4 + 9 + 14 + 19 + \ldots$ is an arithmetic series.

How the sum formula is derived:

Write the sum formula:	$S_n = t_1 + t_2 + t_3 + \cdots + t_n$ $S_n = a + (a + d) + (a + 2d) + \cdots + (t_n - 2d) + (t_n - d) + t_n$
Write in backwards order:	$S_n = t_n + (t_n - d) + (t_n - 2d) + \cdots + (a + 2d) + (a + d) + a$
Add the two sum equations together:	$S_n = a + (a + d) + (a + 2d) + \cdots + (t_n - 2d) + (t_n - d) + t_n$ $S_n = t_n + (t_n - d) + (t_n - 2d) + \cdots + (a + 2d) + (a + d) + a$ _____
Divide both sides by 2:	
We can also modify the formula to replace t_n with $a + (n - 1)d$:	

Summary: Use the formula $\boldsymbol{S_n = \frac{n}{2}(a + t_n)}$ when you know

Use the formula $\boldsymbol{S_n = \frac{n}{2}(2a + (n - 1)d)}$ when you know

Ex. 1: Determine the sum of the first 10 terms of the arithmetic series that has $a = 2$ and $t_{10} = 38$.

Ex. 2: Determine the specified sum for each arithmetic series.

a) Sum of the first 15 terms of $-12 - 8 - 4 - \ldots$

b) Sum of $-11m - 2m + 7m + \ldots + 160m$

Ex. 3: Determine the common difference if the sum of the first 7 terms of an arithmetic series is -63 and the first term is 3.

Geometric Series

A geometric series is the **sum of the terms in a geometric sequence**.

e.g. -3, 6, -12, 24, … is a geometric sequence

-3 + 6 -12 + 24 - … is a geometric series.

How the sum formula is derived:

Write the sum formula:	$S_n = t_1 + t_2 + t_3 + \cdots + t_n$ $S_n = a + ar + ar^2 + \cdots + ar^{n-3} + ar^{n-2} + ar^{n-1}$
Multiply both sides by r:	$rS_n = \quad ar + ar^2 + \cdots + ar^{n-3} + ar^{n-2} + ar^{n-1} + ar^n$
Subtract the two sum equations:	$rS_n = \quad ar + ar^2 + \cdots + ar^{n-3} + ar^{n-2} + ar^{n-1} + ar^n$ $S_n = a + ar + ar^2 + \cdots + ar^{n-3} + ar^{n-2} + ar^{n-1}$ _____
Common factor both sides:	
Isolate for S_n:	

Summary: Use the formula $\boldsymbol{S_n = \frac{a(r^n-1)}{r-1}}$ when you know

Note: we do not use this formula if $r = 1$.

Ex. 4: Determine the sum of the first 10 terms of the geometric series with $a = 5,\ r = 4$

Ex. 5: Determine the sum of each geometric series.

a) $32 + 16 + 8 + \ldots + 1/8$

b) $1 - 3 + 9 - \ldots - 243$

Ex. 6: Determine the number of terms in a geometric series if the sum is -63, the ratio is -2 and the first term is 3.

Applications of Sequences and Series

When approaching a word problem, take these steps to help you determine the solution:

1) Identify whether the scenario describes an arithmetic or geometric sequence or series
   - If there is any mention of sum or total, then likely the question is about a **series**
   - If there is any mention of ratio or percentage of, then likely the question is about **geometric** sequences/series
2) Identify what quantities you are given, and what you need to determine
3) Write the appropriate formula and substitute all the values you already know to solve for the unknown variable value

Ex. 1: Wendy walks 2km on Monday. She walks an additional 0.5km every day for two weeks.

a) Determine how far Wendy walks on the last day.

b) Determine how far Wendy walks in total over the two weeks.

Ex. 2: Peter drops a ball from the top of a 50-ft tall building. It bounces back up to 80% of its previous height with every bounce. Approximately what total distance has the ball travelled after 5 bounces?

Additional Practice

1. The sum of the first 9 terms of an arithmetic series is 162 and the sum of the first 12 terms is 288. Find the general term of the series.

2. If $S_n = -441$ for the series $19 + 15 + 11 + ... + t_n$, find n.

3. In a geometric series $t_1 = 3$ and $S_3 = 21$. Find r and S_7.

Recursive Sequences

The general term of a sequence formula defines a relationship *between the term number and the term.*

e.g. $t_n = 2n - 1$ → the independent variable is _____ and the dependent variable is _____.

A sequence can be defined **recursively** if **each term can be calculated from the previous term or terms**. A recursion formula shows the relationship *between the terms of a sequence.*

Ex. 1: Write the first four terms of the sequence with the following recursion formulas:

a) $t_1 = 1,\ t_n = t_{n-1} + 1,\ n > 1$

b) $t_1 = 2,\ t_n = 3t_{n-1} + 5,\ n > 1$

Changing from the explicit formula to the recursive formula (and vice versa)

e.g. Express 1, 3, 5, 7, … using explicit and recursion formulae.

Explicit formula:
$t_n = 2n - 1$ or $f(n) = 2n - 1$

Recursion formula:
$t_1 = 1,\ t_n = t_{n-1} + 2,\ n > 1$

or

$f(1) = 1, f(n) = f(n-1) + 2, n > 1$

Given an **arithmetic sequence**…

n	1	2	3	4	…	n
t_n	a	$a + d$			…	Explicit formula:

Recursive formula:

Given a **geometric sequence**…

n	1	2	3	4	…	n
t_n	a	ar			…	Explicit formula:

Recursive formula:

Ex. 2: Determine the explicit and recursive formulae for each sequence.

a) -10, -7, -4, …

b) 2, -6, 18, -54, ...

Ex. 3: Determine the recursive formula for the sequence 1, 3, 6, 10, 15, 21, ...

Ex. 4: Determine the recursive formula for the Fibonacci sequence 1, 1, 2, 3, 5, 8, …

Ex. 4: A runner injures her knee in a race. Her doctor prescribes physiotherapy along with 500mg of an anti-inflammatory medicine every 4h for 3 days. The half-life of the anti-inflammatory medicine is approximately 4h. This means that after 4h, about half of the medicine is still in the body.

a) Describe what happens to the medicine in the runner's body over time.

b) Write the amount of medicine remaining after each 4h period as a sequence.

c) Write a recursion formula for the sequence.

Investigation: Binomial Expansion

Binomial	Expanded Form
$(a+b)^0$	1
$(a+b)^1$	$a+b$
$(a+b)^2$	$a^2+2ab+b^2$
$(a+b)^3$	$a^3+3a^2b+3ab^2+b^3$
$(a+b)^4$	$a^4+4a^3b+6a^2b^2+4ab^3+b^4$
$(a+b)^5$	$a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$
$(a+b)^6$	$a^6+6a^5b+15a^4b^2+20a^3b^3+15a^2b^4+6ab^5+b^6$

n	Binomial	Coefficients	Exponents on a	Exponents on b
0	$(a+b)^0$			
1	$(a+b)^1$			
2	$(a+b)^2$			
3	$(a+b)^3$			
4	$(a+b)^4$			
5	$(a+b)^5$			
6	$(a+b)^5$			

Looking at the results above, what patterns can you identify?

Lesson 7.5 – Binomial Expansion and Pascal's Triangle

The triangle of coefficients is often called **Pascal's Triangle** after the French mathematician Blaise Pascal who started using it extensively to study and publish about mathematical properties from the triangle. However, the existence of the triangle was known as early as the year 975 by Indian mathematician **Halayudha**, and its properties were studied by Arabian poet and mathematician **Omar Khayyam** in the 11th century as well as by Chinese mathematicians **Yang Hui** in the 12th century and **Zhu Shijie** in the 13th century.

There are many, many patterns in the triangle; one such pattern – the numbers in each row – allows us to quickly **expand a binomial raised to any whole number exponent**.

n = 0	1
n = 1	1 1
n = 2	1 2 1
n = 3	1 3 3 1
n = 4	1 4 6 4 1
n = 5	1 5 10 10 5 1
n = 6	
n = 7	

This pattern continues indefinitely!

Summary: **Binomial Expansion using Pascal's Triangle**

- The expansion of $(a + b)^k$ has _____ terms.
- The first term of the expansion of $(a + b)^k$ is $a^k b^0$ and the last term is _____.
- The exponent of a decreases by 1 going from _____ to _____.
- The exponent of b increases by ____ going from left to right.
- The exponent sum (or degree) of each term is _____.
- The coefficients of terms can be obtained from the n = _____th row of Pascal's Triangle.

To expand a binomial $(a + b)^k$ using Pascal's triangle:

1) Get the coefficients from row n = k in Pascal's Triangle

2) Follow the pattern that:
   - The exponents of a begin with k then decrease by 1
   - The exponents of b begin with 0 then increase by 1

3) Simplify fully as necessary.

Ex. 1: Expand $(x - 2)^5$ using the triangle.

Ex. 2: Expand $(2y + 3)^6$ using the triangle.

Another way to expand binomials is using **Omar Khayyam's method**, which does not require you to refer to the triangle.

$$(a+b)^n = a^n + na^{n-1}b + \frac{(n-1)n}{2}a^{n-2}b^2 + \frac{(n-2)\frac{n(n-1)}{2}}{3}a^{n-3}b^3 + \cdots$$

- Apply the exponent n to first term a; decrease exponent on a by 1 with each term
- Bring the previous exponent of a as the coefficient of the next term, multiply by the existing coefficient, and divide by the number of terms there have been so far
- Start with exponent 0 on second term b; increase exponent on b by 1 with each term

Ex 3: Expand $(3y+2)^5$ using Khayyam's method.

Ex 4: Expand $(3x-4y)^6$ using Khayyam's method.

Exam Review

Unit 1 Review – Rational and Complex Expressions

Factoring

- Greatest common factor → divide GCF from all terms
- Difference of squares → $a^2 - b^2 = (a - b)(a + b)$
- Simple trinomials $x^2 + bx + c = (x + m)(x + n)$ → $m \cdot n = c, m + n = b$
- Complex trinomials $ax^2 + bx + c$ → use cross-multiplication

1. $3x^2 - 30x - 72$

2. $6x^2 - 7x - 20$

3. $-8x^2 + 18y^4$

Multiplying/dividing rational expressions

- Common factors in any numerator-denominator pair can be canceled (divided out)
- Don't forget to state restrictions (denominator ≠ 0; numerator of divisor ≠ 0)

1. $\frac{25}{x} \times \frac{x^2y}{15}$

2. $\frac{50-2a^2}{a^2+3a-4} \times \frac{a^2-a}{a^2+6a+5}$

3. $\frac{200}{x+2y} \div \frac{150y}{x^2-4y^2}$

4. $\frac{4x^2-8x-12}{4x^2-14x+6} \div \frac{2x^2-3x-5}{6x^2-7x-20}$

Adding/subtracting rational expressions

- Find the LCD, convert all rational expressions to the same LCD, then add the numerators
- Don't forget to state restrictions (denominator ≠ 0)

1. $\frac{4}{a}+\frac{3}{b}$

2. $\frac{3x+6}{2x^2-5x+3}+\frac{4x+3}{2x^2-7x+6}$

3. $\frac{2x-3}{2x^2-x-3}-\frac{5x-7}{3x^2-x-4}$

Operations with radicals

- Radicand should be free of any perfect square factors, fractions
- Like radicals (same radicand) can be added/subtracted
- Rationalize radical denominators (multiply numerator and denominator by the radical)

1. $\sqrt{27}+\sqrt{12}$

2. $\left(\sqrt{3}-4\sqrt{6}\right)\left(-\sqrt{6}+2\sqrt{3}\right)$

3. $\left(\sqrt{50}-2\sqrt{48}\right)\left(\sqrt{75}+3\sqrt{18}\right)$

4. $\frac{6}{\sqrt{3}}$

5. $\frac{4+\sqrt{6}}{\sqrt{3}}$

6. $\frac{4-\sqrt{3}}{5+\sqrt{2}}$

MCR3U Unit 2-Quadratic Functions Exam Review

- Finding the **maximum/minimum of a quadratic function** (find the vertex (h,k); k = max/min value and h = where the max/min occurs)
  - Completing the square
  - Averaging the zeros
  - Partial factoring (then averaging the x-coordinates of the two points)

- **Quadratic optimization** problems: key words max, min, optimize, greatest, least, etc.
  - Number, area/perimeter, revenue/profit problems

  → Introduce variables using "let" statements

  → One possible solution

- **Solving for** x in the equation $0 = ax^2 + bx + c$
  - Isolating for x; factoring; using Quadratic Formula

- **Application word problems involving solving**

  → Introduce variables using "let" statements

  → Given value of dependent variable (sum of squares, area, revenue, etc.)

  → Two solutions possible; may need to omit an answer due to constraints

- **Determining the algebraic model of the family of quadratic relations**
  - Irrational zeros and one other point

- **Solving the linear-quadratic system** of equations (finding the Points of Intersections)
  - Equate the linear and quadratic (i.e. $y_1 = y_2$), simplify and solve to find the x-coordinates of the P.O.I.'s, then use y_1 or y_2 to find the y-coordinates

Discriminant uses – number of zeroes, nature of roots, number of solutions to system

	Quadratic Relation $y = ax^2 + bx + c$	**Quadratic Equation** $0 = ax^2 + bx + c$	**Quadratic/Linear Systems**
	If the discriminant is…		
Positive $b^2 - 4ac > 0$			
Zero $b^2 - 4ac = 0$			
Negative $b^2 - 4ac < 0$			

Practice Questions:

1. Find the max/min of the function and state where it occurs using:
   a) Completing the square
      i. $y = 3x^2 - 4x + 2$
      ii. $y = 2x^2 - 5x$

   b) Averaging the zeros
      i. $y = x^2 - 4x - 21$
      ii. $y = 3x^2 - 5x + 2$

   c) Partial factoring
      i. $y = -\frac{1}{2}x^2 - 3x - 5$
      ii. $0 = 2x^2 - 5x - 8$

2. A schoolyard is fenced in on three sides. The total length of fencing is 120m. Find the maximum area of the schoolyard and its dimensions.

3. Determine the nature of the roots of the quadratic equation $0 = 4x^2 - 4x + 1$

4. How many zeros do these functions have?
   a) $y = x^2 + 7x + 3$ b) $y = 2x^2 - 4x + 2$

5. Solve for x by
   a) Isolating for x
      i. $18 = 3(x + 1)^2 - 6$ ii. $0 = 16(x + 1)^2 - 36$

   b) Factoring
      i. $0 = x^2 - 4x - 21$ ii. $0 = -3x^2 - 6x$

   c) Quadratic Formula
      i. $0 = -\frac{1}{2}x^2 - 3x - 5$ iii. $0 = -5x^2 - 10x + 1$

6. Two numbers have a difference of 14. The sum of the squares is 100. Find the two numbers.

7. Determine the algebraic model of:
   a) the family of quadratic relations with zeros $-3 \pm \sqrt{2}$

   b) the quadratic relation that belongs to the family in part a) and passes through the point (2, 3).

8. Determine the number of solutions to the linear-quadratic system without solving.
   $4x + y = 7$
   $y = 3x^2 + x + 9$

9. Solve the following linear-quadratic system.
   $y = x - 2$
   $y = x^2 - 5x + 7$

Unit 3: Transformations of Functions Review

Function vs relation	**Function notation**
A function is a type of relation in which Vertical line test: Equations test:	

Transformed function, stating transformations				General Base Function: $y = f(x)$
$y = af[k(x - d)] + c$				
Vertical stretch/compress, factor of	Horizontal stretch/compress, factor of	Horizontal translation, right/left	Vertical translation, up/down	Mapping notation: $(x, y) \rightarrow \left(\frac{1}{k}x + d, ay + c\right)$ Default values:

Transformations of functions

	Base function	Base points	Transformed function
Quadratic	$y = x^2$		
Root	$y = \sqrt{x}$		
Rational	$y = \frac{1}{x}$		

Inverse relations

	Domain	Range	**Invariant points**
Quadratic		$a > 0$: $a < 0$:	After reflection on x-axis/ vertical scaling: After reflection on y-axis/ horizontal scaling: After reflection on both axis: After translation: After inverse:
Root	$k > 0$: $k < 0$:	$a > 0$: $a < 0$:	
Rational			
Inverse			

1. For each function below,

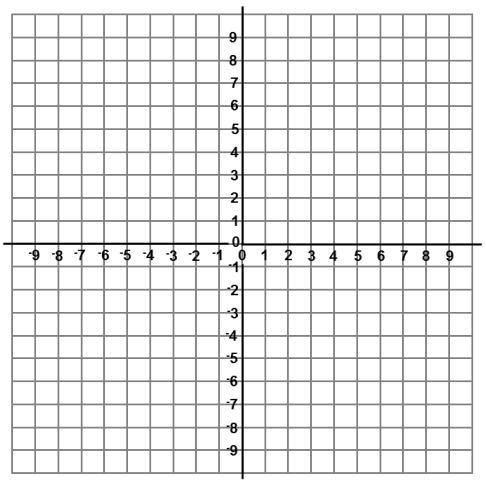
i. Graph the function.

ii. State the domain and range.

a) $f(x) = -\sqrt{\frac{1}{2}x - 3} + 4$

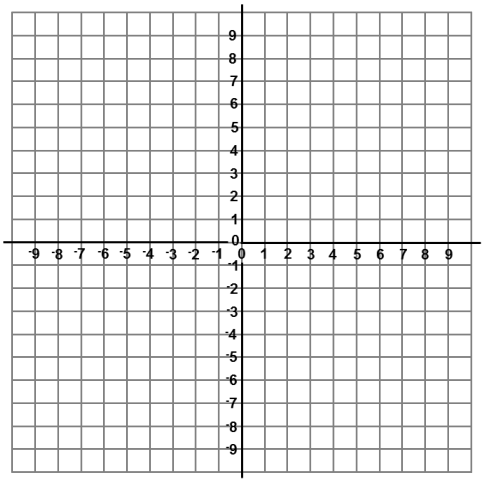
b) $y = 3\sqrt{-x - 4}$

c) $g(x) = \frac{2}{-x+5} + 3$



D:

R:



D:

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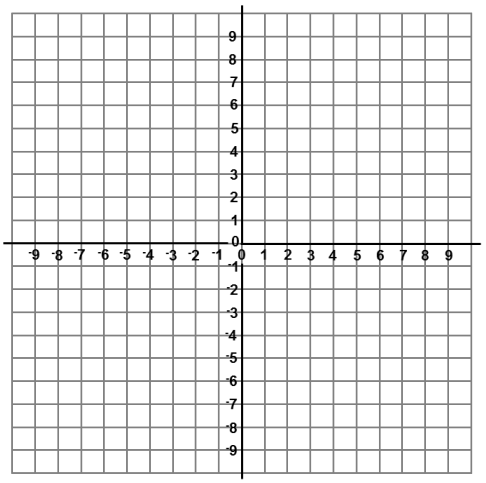


D:

R:

2. Fill in the table below.

a) $f(x) = -\sqrt{\frac{1}{2}x - 3} + 4$	b) $g(x) = 3\sqrt{-x - 4}$	c) $h(x) = 3x^2 - 6x + 2$
Function after reflection in x-axis:	Function after reflection in y-axis:	Inverse function:
Invariant point(s):	Invariant point(s):	Invariant point(s):

3. Given $g(x) = -\frac{3}{2x+6}$ and $h(x) = \sqrt{-x + 6}$, determine:

a) $h(-7)$

b) $h\big(g(3)\big)$

c) The solution(s) to $h(x) = -3$

Unit 4: Exponential Functions Review

Simplifying expressions with exponents

1) $\frac{(3x^{-2}y^5)^2}{(27x^3y^{-12})^{-\frac{1}{3}}}$

2) $\left(\sqrt[4]{81x^{12}y^{-8}}^3\right)\left(\sqrt[5]{\sqrt{y^{30}}}\right)$

Solving exponential equations

Solve for the unknown variable in exponent (simplify both sides into **equal bases**) → useful for finding in application questions	e.g. Solve: $185 = 2(10)^{-7n-1} - 15$
Solve for the unknown base (**raise** both sides by the **reciprocal of the exponent**) → useful for finding in application questions	e.g. Solve: $15n^{\frac{1}{4}} - 7 = 23$

Linear vs quadratic vs exponential functions

Function	Algebraic models	Numerical models	Graphical models
Linear			
Quadratic			
Exponential			

Transformation of exponential functions $y = ab^{k(x-d)} + c$ e.g. Graph $f(x) = 16^{-\frac{1}{4}x+1} - 5$.

Base function: Restrictions on b:

Base points:

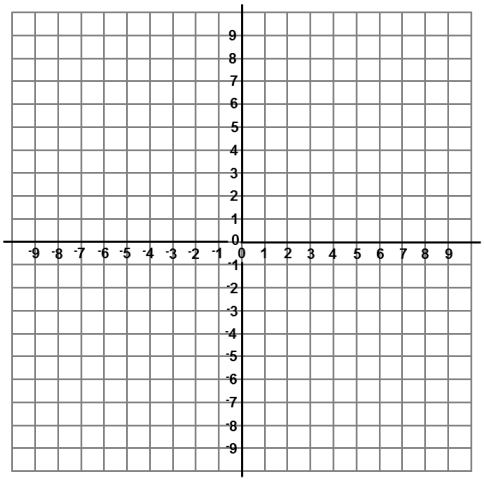
Mapping notation formula:

Horizontal asymptote:

Domain: Range:

y-int: x-int:

Increasing/decreasing:



Applications of exponential functions **Variations**

$A(t) = A_0 r^{\frac{t}{n}}$ $A(t) = A_0(1+i)^{\frac{t}{n}}$ $A(t) = A_0(1-i)^{\frac{t}{n}}$ $A(t) = A_0\left(\frac{1}{2}\right)^{\frac{t}{h}}$

e.g. The value of a car depreciates by i% every 3 years. Determine the depreciation rate if a $32,000 car depreciates to $7,000 after 10 years.

e.g. If it takes 1000 years for a radioactive substance to decay to 6.25% of its original mass, determine the half-life of the substance.

MCR3U Unit 5-Trigonometric Functions Exam Review

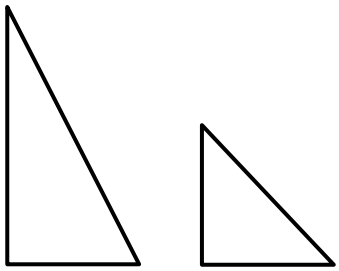
Primary Trig Ratios

SOH CAH TOA

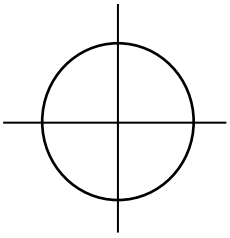
Reciprocal Trig Ratios

$\csc\theta =$ $\sec\theta =$ $\cot\theta =$

Special Triangles



Unit Circle



Coterminal Angles

CAST Rule and the Reference Angle

Reference Angle

Quadrant 2

Quadrant 1

Quadrant 3

Quadrant 4

Pythagorean theorem Cosine law Sine law

Sine law and ambiguous case

Given ∠A obtuse, side a, and side b		**Given ∠A acute, side a, and side b**	
Side a > side b	Side a ≤ side b	Side a > side b	Side a < side b

Proving Trigonometric Identities

$\csc\theta =$ $\sec\theta =$ $\cot\theta =$ $\tan\theta =$ $\sin^2\theta + \cos^2\theta = 1$

Steps:

- Completely separate LS from RS
- Work with more complex side; work on one side only
- Never move terms across = sign

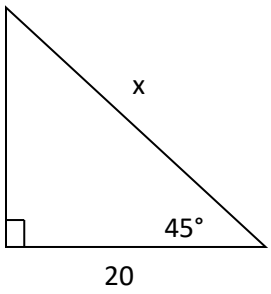
Techniques:

- Factoring
- Simplifying using LCD
- Conjugate binomial

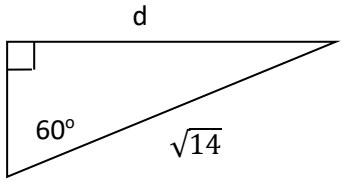
Practice Questions:

1. Solve for the indicated side length without using a calculator.

a)



b)



2. Determine the exact value of the following ratios:
   a) csc 225°
   b) tan 330°

3. Determine one other angle between 0° and 360° that has:
   a) The same sine ratio as sin300°
   b) the same cotangent ratio as cot170°

4. Determine the value(s) of θ for $0° \leq \theta \leq 360°$
   a) $sin\theta = -1$
   b) $cos\theta = \frac{\sqrt{3}}{2}$
   c) $sec\theta = -\sqrt{2}$

5. Ms. Kim is standing on a balcony that is 10 m above the ground. In one direction she can see a rabbit running around on the ground at an angle of depression of 14°. In another direction she can see Ms. Provvisionato stopping to smell some flowers at an angle of depression of 32°. If the angle between Ms. Kim's sightlines is 100°, find the distance between the rabbit and Ms. Provvisionato. Draw a diagram to help you.

6. Prove the following identities:

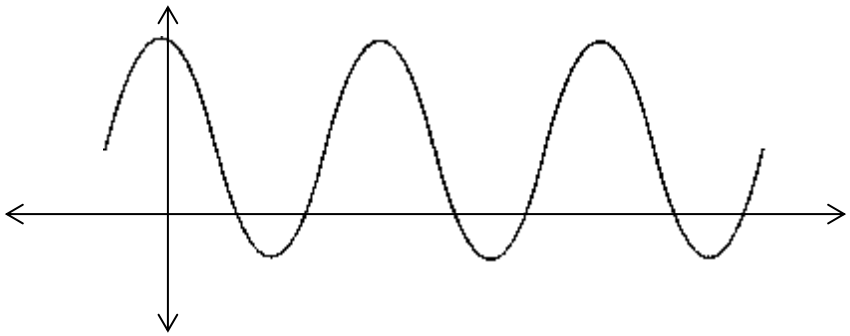
$a) \dfrac{sin\theta}{1+cos\theta} = \dfrac{1-cos\theta}{sin\theta}$

$b) \dfrac{sin\theta - 1}{cos\theta} = tan\theta - sec\theta$

MCR3U Unit 6-Sinusoidal Functions Exam Review

Key Features of Periodic Functions

- Central axis
- Amplitude
- Period
- Maximum/peak
- Minimum/trough



Helpful formulas:

Amplitude =

Central axis =

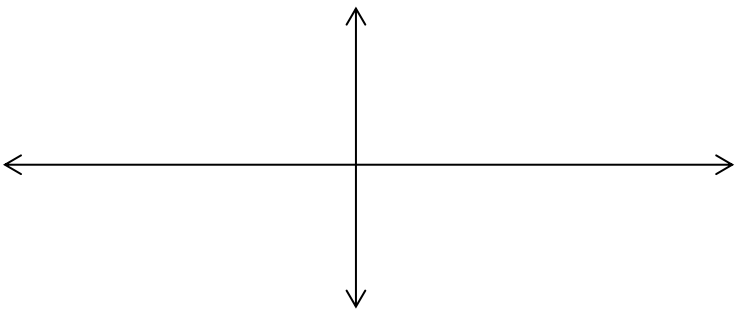
Periodic functions vs. sinusoidal functions

A **periodic** function is a function that has

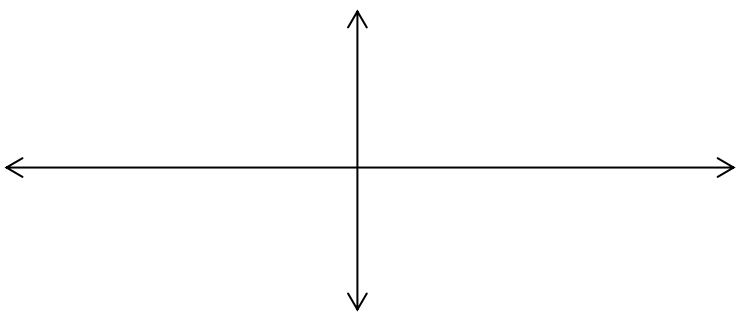
A **sinusoidal** function has all the same characteristics as a periodic function with the added feature that

Sine and Cosine curves

$\sin\theta$



$\cos\theta$



The sine curve is the same as the cosine curve shifted _____

OR the cosine curve is the same as the sine curve shifted _____

$\sin\theta = \cos$ or $\cos\theta = \sin$

$$y = a\sin k(\theta - d) + c \quad \textbf{or} \quad y = a\cos k(\theta - d) + c$$

Amplitude:	Central axis:	Period:	Phase shift:
Maximum/peak:	Minimum/trough:	Major point every:	
Range:		Best scale:	

Steps to graphing: First, determine key characteristics and best scale (see above), then…

For **SINE** functions	For **COSINE** functions
Regardless of a-value: 1) Plot the starting point (d, c) 2) Steps for $a > 0$: $\frac{1}{4}$ period right, $\lvert a\rvert$ units up $\frac{1}{4}$ period right, $\lvert a\rvert$ units down $\frac{1}{4}$ period right, $\lvert a\rvert$ units down $\frac{1}{4}$ period right, $\lvert a\rvert$ units up (If $a < 0$, change up/down steps)	1) If $a > 0$, start at max $(d, c + \lvert a\rvert)$ If $a < 0$, start at min $(d, c - \lvert a\rvert)$ 2) Steps for $a > 0$: $\frac{1}{4}$ period right, $\lvert a\rvert$ units down $\frac{1}{4}$ period right, $\lvert a\rvert$ units down $\frac{1}{4}$ period right, $\lvert a\rvert$ units up $\frac{1}{4}$ period right, $\lvert a\rvert$ units up (If $a < 0$, change up/down steps)
Mapping notation:	

Determining the algebraic model of a sinusoidal function

1) Determine the amplitude – same as
2) Determine the central axis – same as
3) Determine the period – use $\frac{360}{period}$ to determine
4) Decide on a starting point – the x-coordinate is the same as
   - For sine – choose starting point on the central axis
     (if increases, then $a > 0$; if decreases, then $a < 0$)
   - For cosine – choose starting point on the peak or trough
     (if peak, then $a > 0$; if trough, then $a < 0$)

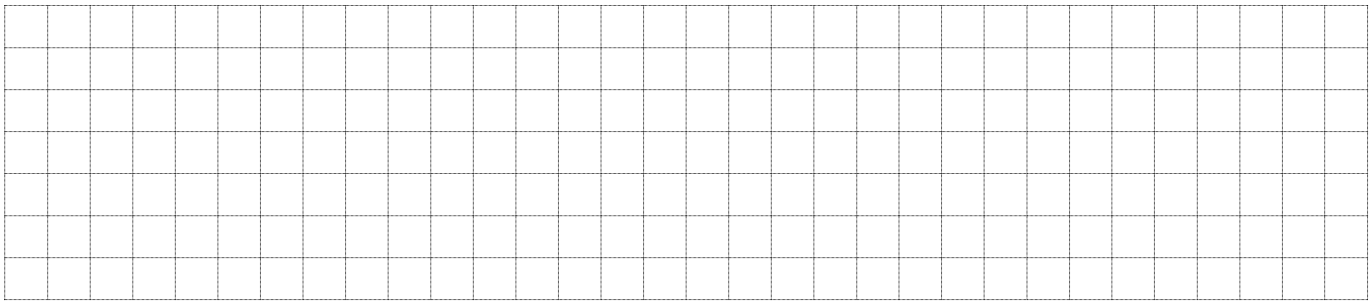
Practice Questions:

1. Fill in the table below:

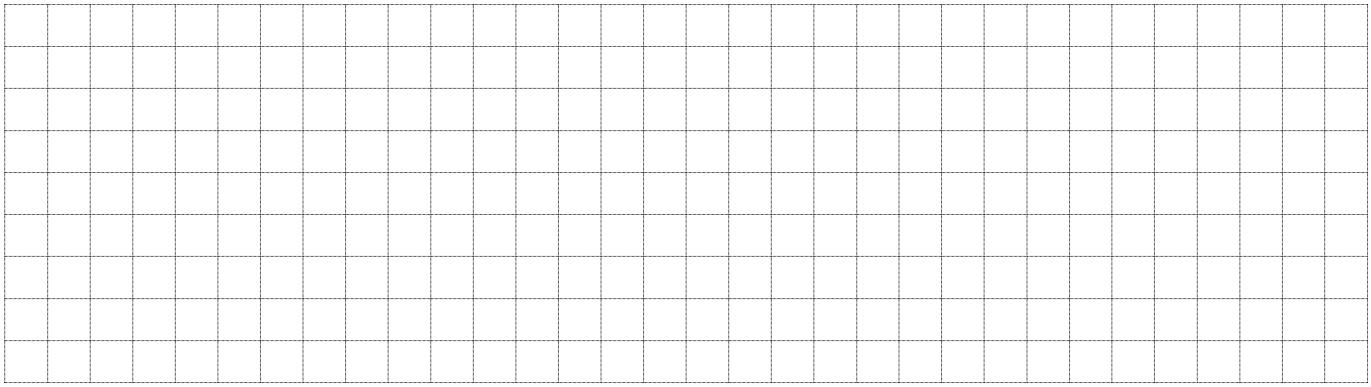
Sinusoidal function	Amplitude	Period	Phase Shift	Vertical Displacement	Scale
$y = -4\sin\frac{1}{2}(x - 45°) + 1$					
$y = 2\cos 3(x + 80°) - 6$					

2. Graph at least two complete cycles (if possible) of the following functions. Show all work.

a) $f(\theta) = -2\sin(2\theta + 60°) + 1$



b) $g(\theta) = 3\cos\frac{1}{3}(\theta - 45°) - 2$



3. A ferris wheel has a diameter of 18 m and completes one revolution every 40 seconds. A rider begins at the lowest point, 2 m from the ground.

   a) If the height, h, in metres over time, t, seconds is modeled by the function below, determine the values of b and c.

$$h = -9cosbt + c$$

   b) At what time(s) is the rider 15.5 m above the ground in the first 45 seconds?

4. A pendulum clock takes 6 seconds to swing from its centre position, to the right, back to centre, to the left and then back to centre again to complete one cycle. The pendulum swings a maximum horizontal distance of 3 cm from its centre position.

   a) Sketch the horizontal distance D, of the pendulum from its centre position, to time t, in seconds, for a 9 second interval.

   b) Determine the trigonometric function that models this relationship.

MCR3U Unit 7 - Sequences and Series Exam Review

- **Sequences** - discrete functions
  - **Domain** - **position of term;** set/subset of natural numbers
  - **Range** - **terms** of the sequence
- **Arithmetic sequence -** terms are separated by a common difference; linear function
- **Geometric sequence -**terms are separated by a common ratio; exponential function
- **General rule for the n^{th} term, t_n -** rule for finding the terms of a sequence; the value of a term depends on the value of n.

Sequence type	First term	Second term	Third term	Fourth term	General term, t_n	Give two terms t_k and t_c:
Arithmetic	a	$a+d$	$a+2d$	$a+3d$	$t_n = a+(n-1)d$	$d =$
Geometric	a	ar	ar^2	ar^3	$t_n = ar^{n-1}$	$r =$

Arithmetic Series **sum of the terms** in a sequence	$S_n = \frac{n}{2}(a+t_n)$	$S_n = \frac{n}{2}[2a+(n-1)d]$
Geometric Series **sum of the terms** in a sequence	$S_n = \frac{a(r^n-1)}{r-1}$, $r \neq 1$	

- **Pascal's triangle**

n = 0 1
n = 1 1 1
n = 2 1 2 1
n = 3 1 3 3 1
n = 4 1 4 6 4 1
n = 5 1 5 10 10 5 1
etc.

- **Binomial Expansion -** shortcut for expanding $(a+b)^k$ using Pascal's Triangle

  1) Get the coefficients from row n = k in Pascal's Triangle
  2) The exponents of a begin with k then decrease by 1
  3) The exponents of b begin with 0 then increase by 1

- **Khayyam's method for binomial expansion:**
  1) Apply the exponent n to first term a; decrease exponent on a by 1 with each term
  2) Bring the previous exponent of a as the coefficient of the next term, multiply by the existing coefficient, and divide by the number of terms there have been so far
  3) Start with exponent 0 on second term b; increase exponent on b by 1 with each term

Practice Questions:

1. If $t_3 = 17$ and $t_8 = 37$ in an arithmetic sequence,
   a) Determine the general term of the sequence.

   b) Determine the sum of the first 20 terms in the series.

2. If $t_3 = 24$ and $t_7 = \frac{3}{2}$ in a geometric sequence,
   a) Determine the general term of the sequence, if r is a negative value.

   b) Determine S_{10}.

3. Expand $(3x - 2y)^5$.